

Project Data Page:

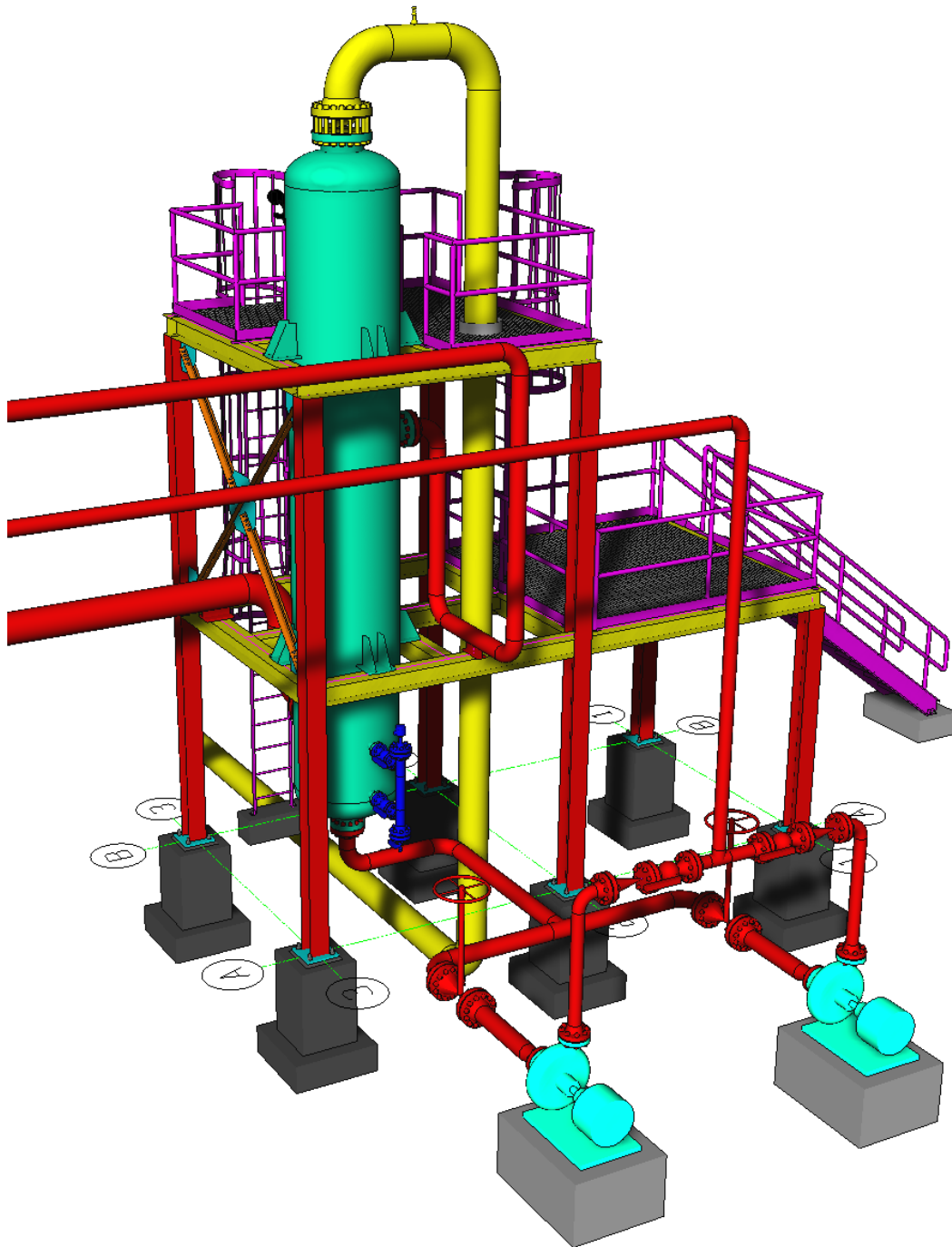




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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\121-DC02 ABSORBER\\Operating CaseR4 - up to HLL.pvdb

Date of Analysis : Apr 15, 2026 2:14pm

PV Elite 28, January 2026



Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 7200.000 x 4600.000 mm ID
 Vessel Design Length, Tangent to Tangent 43299.99 mm
 Distance of Bottom Tangent above Grade 9000.00 mm
 Distance of Base above Grade 0.00 mm
 Specified Datum Line Distance 0.00 mm
 Internal Design Temperature 130 °C
 Internal Design Pressure 4550.000 KPa.
 External Design Temperature 0 °C
 Maximum Allowable Working Pressure 4604 KPa.
 Shop Test Pressure 6425.212 KPa.
 Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -12.0 °C
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	+IT
Cone	SA-516 70	...	...	K02700	+IT
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-266 2	...	...	K03506	+IT
Nozzle	SA-765 I	...	...	K03046	+IT
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Btm skirt-1	...	...	...	1.6000	N/A	No
Btm skirt-2	...	...	...	1.6000	N/A	No
Btm skirt-3	...	...	...	1.6000	N/A	No
Top skirt	...	...	...	1.6000	N/A	No
BTM dish	4687.162	0.00	5221.000	6.0000	N/A	No
shell-1	4649.757	0.00	5020.000	6.0000	N/A	No
shell-2	4618.587	0.00	4967.000	6.0000	N/A	No
shell-3	4587.417	0.00	4998.000	6.0000	N/A	No
shell-4	4563.000	0.00	4939.000	6.0000	N/A	No
shell-5	4563.000	0.00	4939.000	6.0000	N/A	No
shell-6	4563.000	0.00	4939.000	6.0000	N/A	No
shell-7	4563.000	0.00	4855.000	6.0000	N/A	No
shell-8	4563.000	0.00	4855.000	6.0000	N/A	No
cone	4563.000	0.00	5041.000	6.0000	N/A	No
shell-9	4563.000	0.00	4867.000	3.0000	N/A	No
shell-10	4563.000	0.00	4867.000	3.0000	N/A	No
shell-11	4563.000	0.00	4736.000	3.0000	N/A	No
shell-12	4563.000	0.00	4736.000	3.0000	N/A	No



shell-13	4563.000	0.00	4736.000	3.0000	N/A	No
shell-14	4563.000	0.00	4736.000	3.0000	N/A	No
shell-15	4563.000	0.00	4604.000	3.0000	N/A	No
TOP dish	4563.000	0.00	4972.000	3.0000	N/A	No

Liquid Level: 11950.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.060

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	2000.0	2000.0	36.0	36.0	...	...	0.70	0.70
Skirt	5000.0	3000.0	34.0	34.0	...	...	0.70	0.70
Skirt	8000.0	3000.0	34.0	34.0	...	...	0.70	0.70
Skirt	9000.0	1000.0	34.0	34.0	...	...	0.70	0.55
Sphere	9000.0	0.0	78.0	68.3	60.4	...	1.00	1.00
Cylinder	12000.0	3000.0	126.0	126.0	114.8	...	1.00	1.00
Cylinder	15000.0	3000.0	124.0	124.0	114.1	...	1.00	1.00
Cylinder	18000.0	3000.0	124.0	124.0	113.3	...	1.00	1.00
Cylinder	21000.0	3000.0	122.0	122.0	112.8	...	1.00	1.00
Cylinder	24000.0	3000.0	122.0	122.0	112.8	...	1.00	1.00
Cylinder	27000.0	3000.0	122.0	122.0	112.8	...	1.00	1.00
Cylinder	29000.0	2000.0	120.0	120.0	112.8	...	1.00	1.00
Cylinder	30650.0	1650.0	120.0	120.0	112.8	...	1.00	1.00
Conical	33150.0	2500.0	160.0	139.5	126.3	...	1.00	1.00
Cylinder	36150.0	3000.0	76.0	76.0	71.2	...	1.00	1.00
Cylinder	39150.0	3000.0	76.0	76.0	71.2	...	1.00	1.00
Cylinder	42150.0	3000.0	74.0	74.0	71.2	...	1.00	1.00
Cylinder	45150.0	3000.0	74.0	74.0	71.2	...	1.00	1.00
Cylinder	48150.0	3000.0	74.0	74.0	71.2	...	1.00	1.00
Cylinder	51150.0	3000.0	74.0	74.0	71.2	...	1.00	1.00
Cylinder	52300.0	1150.0	72.0	72.0	71.2	...	1.00	1.00
Sphere	52300.0	0.0	46.0	40.0	36.8	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
C	WRC-107/537	0.885	n/a	Passed
A	WRC-297	0.914	Shell	Passed
P	WRC-297	0.946	Shell	Passed
D	WRC-297	0.972	Shell	Passed
B	WRC-107/537	0.926	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	22748.3	294722	584810	9045499
20	30	3500	21704.7	294608	540357	8456157
30	40	6500	20341.7	294204	477286	7572921
40	50	8500	19162.8	293454	418028	6691418
50	60	8999.98	18769.8	293133	399062	6398127
60	70	10500	18769.8	282504	399061	6398119
70	80	13500	17330.6	267814	344909	5572627
80	90	16500	15864.8	250266	295115	4795492
90	100	19500	14381.5	228679	249745	4077060
100	110	22500	13164.6	216060	208425	3409938
110	120	25500	11707	197370	171117	2789779
120	130	28000	10237.3	170696	138200	2237669
130	140	29825	9426.4	155536	118536	1911431
140	150	31900	8506.76	142606	103741	1665459
150	160	34650	7393.88	112404	83864.4	1346691
160	170	37650	6588.09	100793	62891.1	1026890

170	180	40650	5528.96	87378.6	44715.1	744627
180	190	43650	4463.82	75503	29725.7	500300
190	200	46650	3392.92	59974.4	17940.3	297080
200	210	49650	2587.79	45323.9	8969.08	139129
210	220	51725	1517.86	24701.8	2810.48	34088.9
220	230	52300	1209.48	12411.9	1242.27	12748.5

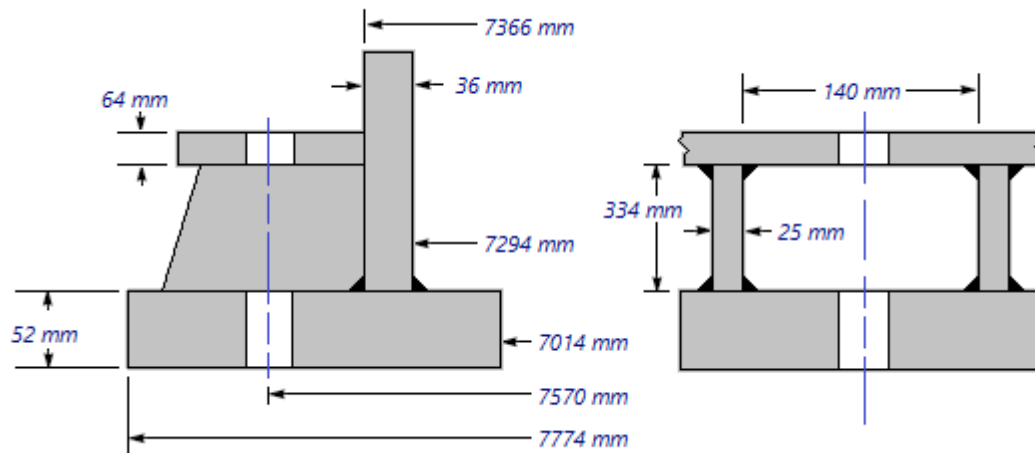
Abs Max of the all of the Stress Ratio's : 0.9315

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	52.0000	mm
Inside Diameter of Basing	7013.9995	mm
Outside Diameter of Basing	7774.0000	mm
Nominal Diameter of Bolts	76.0000	mm
Diameter of Bolt Circle	7570.0000	mm
Number of Bolts	52	

Thickness of Gusset Plates	25.0000	mm
Average Width of Gusset Plates	194.0000	mm
Height of Gussets	334.0000	mm
Distance between Gussets	140.0000	mm
Thickness of Top Plate or Ring	64.0000	mm
Radial Width of the Top Plate	204.0000	mm

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	28933.	Kgf
Total Earthquake Shear on Support	300907.	Kgf
Wind Moment on Support	1530106.	Kg-m.
Earthquake Moment on Support	9990794.	Kg-m.
Maximum Initial Bolt Load	66216.	Kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	934037.8	kg.
Shop Test - Fabricated + water (full)	2481463.2	kg.
Shipping - Fab. + rem. intls.+ shipping app.	1131822.6	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	1287459.5	kg.
Empty - Fab. + intls. + details + wgths.	1287459.4	kg.
Operating - Empty + operating liquid (no ca)	1888989.5	kg.



Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
C	5114	...	...	OK	OK	Passed
A	4972	...	...	OK	OK	Passed
MH-6	4952	...	...	...	OK	Passed
MH-5	4952	...	...	...	OK	Passed
MH-4	4868	...	...	...	OK	Passed
E	5327	...	...	OK	OK	Passed
P	4880	...	...	OK	OK	Passed
MH-3	4738	...	...	...	OK	Passed
MH-2	4738	...	...	...	OK	Passed
D	4749	...	...	OK	OK	Passed
MH-1	4617	...	...	...	OK	Passed
B	4868	...	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 4617 Nozzle : MH-1
 Minimum MAWP Shells/Flanges : 4604 Element : shell-15
 Minimum MAPnc Shells/Flanges : 5394 Element : shell-15

Computed Vessel M.A.W.P. : 4604 KPa.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

- *Error: Nozzle A, was too close to a gross structural discontinuity.*
- *Error: Nozzle MH-6, was too close to a gross structural discontinuity.*
- *Error: Nozzle E, was too close to a gross structural discontinuity.*
- *Error: Nozzle MH-1, was too close to a gross structural discontinuity.*

Please see ASME VIII-2 Section 4.5.13.1 for more information.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Inside Radius mm
50	C	0.000	0.000	463.200
80	A	8100.000	180.000	310.800
90	MH-6	9800.004	300.000	310.800
110	MH-5	16400.004	300.000	310.800
130	MH-4	20600.004	300.000	310.800
140	E	22623.805	180.000	463.200
150	P	24300.008	180.000	22.050
170	MH-3	31650.006	225.000	307.800
200	MH-2	40150.000	300.000	307.800
200	D	39650.000	180.000	206.200
210	MH-1	42449.996	300.000	307.800
220	B	0.000	0.000	307.800

The nozzle spacing is computed by the following:

$$= \sqrt{(l_l^2 + l_c^2)} \text{ where}$$

l_l - Arc length along the inside vessel surface in the long. direction.

l_c - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
P	1.500 in	Actual	WNF	2.445	12.000	...	...	276.44	600
D	16.000 in	Actual	WNF	19.780	48.000	...	...	490.28	600
A	24.000 in	Actual	WNF	31.402	94.000	...	...	547.06	600
MH-6	24.000 in	Actual	WNF	30.929	88.000	...	...	545.98	600
MH-5	24.000 in	Actual	WNF	30.929	88.000	...	...	545.98	600
MH-4	24.000 in	Actual	WNF	30.929	88.000	...	...	543.98	600
MH-3	24.000 in	Actual	WNF	29.512	70.000	...	...	509.19	600
MH-2	24.000 in	Actual	WNF	29.512	70.000	...	...	509.19	600
MH-1	24.000 in	Actual	WNF	29.512	70.000	...	...	509.71	600
B	24.000 in	Actual	WNF	27.150	40.000	...	...	473.27	600
C	36.000 in	Actual	WNF	43.717	98.000	...	...	614.55	600
E	36.000 in	Actual	WNF	38.047	26.000	...	...	716.19	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
P	SA-266 2	10.000	10.000	...	...	...
D	SA-266 2	20.000	20.000	...	...	...
A	SA-266 2	37.000	10.000	...	...	...
MH-6	SA-266 2	37.000	10.000	...	...	...
MH-5	SA-266 2	37.000	10.000	...	...	...
MH-4	SA-266 2	37.000	10.000	...	...	...
MH-3	SA-266 2	37.000	10.000	...	...	...
MH-2	SA-266 2	37.000	10.000	...	...	...
MH-1	SA-266 2	37.000	10.000	...	...	...
B	SA-266 2	37.000	10.000	...	...	...
C	SA-266 2	37.000	10.000	...	...	...
E	SA-765 I	10.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
P	33300.004	180.0	200.00	0.00	shell-9
D	48650.000	180.0	400.00	0.00	shell-14
A	17100.000	180.0	400.00	0.00	shell-3
MH-6	18800.002	300.0	400.00	0.00	shell-4
MH-5	25400.002	300.0	400.00	0.00	shell-6
MH-4	29600.004	300.0	400.00	0.00	shell-8
MH-3	40650.004	225.0	400.00	0.00	shell-11
MH-2	49149.996	300.0	400.00	0.00	shell-14
MH-1	51449.992	300.0	400.00	0.00	shell-15
B	...	0.0	400.00	0.00	TOP dish
C	...	0.0	500.00	0.00	BTM dish
E	31623.803	180.0	500.00	0.00	cone

PV Elite 28 Licensee: SL User
FileName : 121-D_C02 ABSORBER_Operating Case_R4 - up to HLL Page 8 of 225
Nozzle Schedule: Step: 54 2:14pm Apr 15,2026



HEXAGON

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 78.0mm THK X 7200.0mm ID X 0.0mm	SA-516 70
1	SPHERICAL HEAD: 46.0mm THK X 4600.0mm ID X 0.0mm	SA-516 70
13	PLATFORM: 1000mm X 1500mm WIDE	
1	PACKING: 3600.000mm	...
1	INSULATION: 3600mm X 80mm THK	insu
1	LADDER: 7000.0mm	...
1	PACKING: 1000.000mm	...
12	INSULATION: 3000mm X 80mm THK	insu
1	PACKING: 1350.000mm	...
1	PACKING: 2650.000mm	...
1	PACKING: 1250.000mm	...
1	PACKING: 2000.000mm	...
1	INSULATION: 2000mm X 80mm THK	insu
1	LADDER: 19600.0mm	...
1	PACKING: 750.003mm	...
1	INSULATION: 1650mm X 80mm THK	insu
1	INSULATION: 2500mm X 80mm THK	insu
1	PACKING: 2700.000mm	...
2	PACKING: 2999.990mm	...
1	PACKING: 150.007mm	...
1	PACKING: 749.991mm	...
1	PACKING: 2100.010mm	...
3	TRAYS # 3 SPACING 11.811024 mm	
1	INSULATION: 1150mm X 80mm THK	insu
1	PLATFORM: 1000mm X 8000mm WIDE	
1	LADDER: 26000.0mm	...
1	INSULATION: 2300mm X 80mm THK	insu
6	CLASS 600 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 7774.0mm OD X 7014.0 ID X 52.0mm	SA-516 70
52	BASERING BOLTS: 76.0mm DIA	SA-36
52	BASERING BOLT NUTS: 76.0mm DIA	...
1	BASERING TOP RING: 7774.0mm OD X 7366.0 ID X 64.0mm	...
104	BASERING GUSSET PLATES: 194.0mm X 194.0mm X 334.0mm X 25.0mm	...



4	CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 3118.9mm X 2000.0mm	SA-516 70
3	CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 3055.3mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 36.0mm THK X 21641.4mm X 2000.0mm	SA-516 70
10	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3056.9mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 21398.5mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3120.6mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 21653.2mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3120.6mm X 1000.0mm	SA-516 70
3	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 3056.9mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 34.0mm THK X 21653.2mm X 1000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 126.0mm THK X 8984.8mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 126.0mm THK X 4649.5mm X 3000.0mm	SA-516 70
12	CYLINDRICAL SEGMENT 124.0mm THK X 3048.0mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 124.0mm THK X 19571.1mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 124.0mm THK X 1283.4mm X 3000.0mm	SA-516 70
18	CYLINDRICAL SEGMENT 122.0mm THK X 3048.0mm X 3000.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 122.0mm THK X 19571.1mm X 3000.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 122.0mm THK X 1283.4mm X 3000.0mm	SA-516 70
6	CYLINDRICAL SEGMENT 120.0mm THK X 3048.0mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 120.0mm THK X 19571.1mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 120.0mm THK X 1283.4mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 120.0mm THK X 19571.1mm X 1650.0mm	SA-516 70
6	CYLINDRICAL SEGMENT 120.0mm THK X 3048.0mm X 1650.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 120.0mm THK X 1283.4mm X 1650.0mm	SA-516 70
7	CON. CONICAL SEGMENT 160.0mm THK X 3048.0mm X 2500.0mm X 1947.3m	SA-516 70
1	CON. CONICAL SEGMENT 160.0mm THK X 21335.7mm X 2500.0mm X 13631.	SA-516 70
7	CYLINDRICAL SEGMENT 76.0mm THK X 3048.0mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 76.0mm THK X 2259.3mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 76.0mm THK X 11403.1mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 76.0mm THK X 6095.9mm X 3000.0mm	SA-516 70
12	CYLINDRICAL SEGMENT 74.0mm THK X 3048.0mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 74.0mm THK X 11403.1mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 74.0mm THK X 6095.9mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 72.0mm THK X 11403.1mm X 1150.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 72.0mm THK X 3048.0mm X 1150.0mm	SA-516 70



1	CYLINDRICAL SEGMENT 72.0mm THK X 6095.9mm X 1150.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Notes							
BTM dish	[9]	!	-12	Yes			
shell-1	[8]	!	-12	Yes			
shell-2	[8]	!	-12	Yes			
shell-3	[8]	!	-12	Yes			
shell-4	[8]	!	-12	Yes			
shell-5	[8]	!	-12	Yes			
shell-6	[8]	!	-12	Yes			
shell-7	[8]	!	-12	Yes			
shell-8	[8]	!	-12	Yes			
cone	[8]	!	-12	Yes			
shell-9	[8]	!	-12	Yes			
shell-10	[8]	!	-12	Yes			
shell-11	[8]	!	-12	Yes			
shell-12	[8]	!	-12	Yes			
shell-13	[8]	!	-12	Yes			
shell-14	[8]	!	-12	Yes			
shell-15	[8]	!	-12	Yes			
TOP dish	[9]	!	-12	Yes			
C	[1]	!	-12	Yes			
Nozzle Flg	[5]	A	-18	-48	Yes		
A	[1]	!	-12	Yes			
Nozzle Flg	[5]	A	-18	-48	Yes		
MH-6	[1]	!	-12	Yes			
Nozzle Flg	[5]	A	-18	-48	Yes		
MH-5	[1]	!	-12	Yes			
Nozzle Flg	[5]	A	-18	-48	Yes		
MH-4	[1]	!	-12	Yes			
Nozzle Flg	[5]	A	-18	-48	Yes		
E	[1]	!	-12	Yes			
Nozzle Flg	[5]	!	-18	-57	Yes		
P	[1]	!	-12	Yes			
Nozzle Flg	[5]	A	-18	-48	Yes		
MH-3	[1]	!	-12	Yes			
Nozzle Flg	[5]	A	-18	-48	Yes		
MH-2	[1]	!	-12	Yes			
Nozzle Flg	[5]	A	-18	-48	Yes		
D	[1]	!	-12	Yes			
Nozzle Flg	[5]	A	-18	-48	Yes		
MH-1	[1]	!	-12	Yes			
Nozzle Flg	[5]	A	-18	-48	Yes		
B	[1]	!	-12	Yes			
Nozzle Flg	[5]	A	-18	-48	Yes		
Nozl. Bolting[22]			-48	No	1.000	64.000	1.00

Warmest MDMT: -12 -48

Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -12.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable



- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	4550	KPa.
Design Internal Temperature	130.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW



Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 97.2	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	0	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 mm
Distance Upwind of Crest	Lh	0 mm
Distance from Crest to the Vessel	x	0 mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration	[Ss] 0.147
Max. Eff. Ground Acceleration	[S] 0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/22
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Btm skirt-1
Distance "FROM" to "TO"	2000 mm
Skirt Outside Diameter	7366 mm
Diameter of Skirt at Base	7366 mm
Skirt Thickness	36 mm
Internal Corrosion Allowance	1.6 mm
Nominal Thickness	36 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2



Material Name	SA-516 70	
Allowable Stress, Ambient	175	N./mm ²
Allowable Stress, Operating	161.57	N./mm ²
Allowable Stress, Hydrotest	247	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	0	mm
Yield Stress, Operating	242.6	N./mm ²
MDMT Curve	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	Yes	
Element From Node	10	
Detail Type	Platform	
Detail ID	PLAT- SKIRT	
Dist. from "FROM" Node / Offset dist	1999	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	360.0	
Platform Wind Area	15000	cm ²
Platform Weight	1825.1	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	Kgf/mm
Platform Length (top head platform)	0	mm

Element#	2/22	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	Btm skirt-2	
Distance "FROM" to "TO"	3000	mm
Skirt Outside Diameter	7366	mm
Diameter of Skirt at Base	7366	mm
Skirt Thickness	34	mm
Internal Corrosion Allowance	1.6	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	Yes	

Element From Node	20	
Detail Type	Platform	
Detail ID	PLAT- SKIRT	
Dist. from "FROM" Node / Offset dist	3000	mm
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	15000	cm ²
Platform Weight	1826.7	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm



Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	Kgf/mm
Platform Length (top head platform)	0	mm

Element#	3/22	
Element From Node	30	
Element To Node	40	
Element Type	Skirt Support	
Description	Btm skirt-3	
Distance "FROM" to "TO"	3000	mm
Skirt Outside Diameter	7366	mm
Diameter of Skirt at Base	7366	mm
Skirt Thickness	34	mm
Internal Corrosion Allowance	1.6	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	82	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	Yes	

Element From Node	30	
Detail Type	Weight	
Detail ID	skirt manwy/ven	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	800	Kgf
Offset from Element Centerline	1000	mm

Element#	4/22	
Element From Node	40	
Element To Node	50	
Element Type	Skirt Support	
Description	Top skirt	
Distance "FROM" to "TO"	1000	mm
Skirt Outside Diameter	7366	mm
Diameter of Skirt at Base	7366	mm
Skirt Thickness	34	mm
Internal Corrosion Allowance	1.6	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	130	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.55	
Weld is pre-Heated	Yes	

Element#	5/22	
Element From Node	50	
Element To Node	60	
Element Type	Spherical	
Description	BTM dish	
Distance "FROM" to "TO"	0	mm
Inside Diameter	7200	mm
Element Thickness	68.3	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	78	mm



External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Packing	
Detail ID	Pack 1: Btm dis	
Dist. from "FROM" Node / Offset dist	-3600	mm
Height of Packed Section	3600	mm
Density	0.000163	kg./cm³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	-3600	mm
Height/Length of Liquid	3600	mm
Liquid Density	0.001058	kg./cm³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	-3600	mm
Height/Length of Insulation	3600	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	C	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	16824	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	+IT

Element#	6/22	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	shell-1	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	7200	mm
Element Thickness	126	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	126	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	



Element From Node	60
Detail Type	Platform
Detail ID	PLAT-1
Dist. from "FROM" Node / Offset dist	0 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	360.0
Platform Wind Area	30000 cm ²
Platform Weight	7650.7 Kg
Platform Railing Weight	0 Kg/mm
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	135 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	270.0
Ladder Start Elevation	0 mm
Ladder End Elevation	7000 mm
Unit Weight of Ladder	0.03702 Kg/mm
Platform Length (top head platform)	0 mm

Element From Node	60
Detail Type	Packing
Detail ID	pack#5
Dist. from "FROM" Node / Offset dist	0.005233 mm
Height of Packed Section	1000 mm
Density	0.000163 kg./cm ³
Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.0

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	3000 mm
Liquid Density	0.001058 kg./cm ³

Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm ³

Element From Node	60
Detail Type	Weight
Detail ID	VB
Dist. from "FROM" Node / Offset dist	0.005233 mm
Miscellaneous Weight	260 Kg
Offset from Element Centerline	0 mm

Element From Node	60
Detail Type	Weight
Detail ID	btm dish SF wt
Dist. from "FROM" Node / Offset dist	0.005233 mm
Miscellaneous Weight	2770 Kg
Offset from Element Centerline	0 mm

Element From Node	60
Detail Type	Weight
Detail ID	CPP0VH
Dist. from "FROM" Node / Offset dist	0.005233 mm
Miscellaneous Weight	5249 Kg
Offset from Element Centerline	5343 mm



Element To Node	80	
Element Type	Cylinder	
Description	shell-2	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	7200	mm
Element Thickness	124	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	124	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	70	
Detail Type	Platform	
Detail ID	PLAT-2	
Dist. from "FROM" Node / Offset dist	1000	mm
Platform Start Angle (degrees)	90.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	30000	cm ²
Platform Weight	3694.2	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	135	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	Kgf/mm
Platform Length (top head platform)	0	mm
Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	0.001058	kg./cm ³
Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element#	8/22	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	shell-3	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	7200	mm
Element Thickness	124	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	124	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.



Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	80	
Detail Type	Platform	
Detail ID	PLAT-3	
Dist. from "FROM" Node / Offset dist	2000	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	30000	cm ²
Platform Weight	3694.2	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	135	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	Kgf/mm
Platform Length (top head platform)	0	mm
Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2350	mm
Liquid Density	0.001058	kg./cm ³
Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³
Element From Node	80	
Detail Type	Nozzle	
Detail ID	A	
Dist. from "FROM" Node / Offset dist	2100	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	1204.6	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	+IT
Element From Node	80	
Detail Type	Weight	
Detail ID	FPA	
Dist. from "FROM" Node / Offset dist	2100	mm
Miscellaneous Weight	3300	Kgf
Offset from Element Centerline	0	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	FPABolting ba	
Dist. from "FROM" Node / Offset dist	2100	mm
Miscellaneous Weight	250	Kgf
Offset from Element Centerline	0	mm
Element From Node	80	



Detail Type	Weight
Detail ID	APPOVH
Dist. from "FROM" Node / Offset dist	2100 mm
Miscellaneous Weight	21250 Kg
Offset from Element Centerline	5267 mm

Element#	9/22
Element From Node	90
Element To Node	100
Element Type	Cylinder
Description	shell-4
Distance "FROM" to "TO"	3000 mm
Inside Diameter	7200 mm
Element Thickness	122 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	122 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4550 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	90
Detail Type	Packing
Detail ID	pack#4
Dist. from "FROM" Node / Offset dist	1650 mm
Height of Packed Section	1350 mm
Density	0.000163 kg./cm ³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	1.0599999

Element From Node	90
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm ³

Element From Node	90
Detail Type	Nozzle
Detail ID	MH-6
Dist. from "FROM" Node / Offset dist	800 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	300.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	1787.6 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	90
Detail Type	Weight
Detail ID	B#4btm TSR
Dist. from "FROM" Node / Offset dist	1650 mm
Miscellaneous Weight	300 Kg
Offset from Element Centerline	0 mm

Element From Node	90
Detail Type	Weight
Detail ID	B#4btm supp de
Dist. from "FROM" Node / Offset dist	1650 mm



Miscellaneous Weight 4400 Kgf
 Offset from Element Centerline 0 mm

Element# 10/22
 Element From Node 100
 Element To Node 110
 Element Type Cylinder
 Description shell-5
 Distance "FROM" to "TO" 3000 mm
 Inside Diameter 7200 mm
 Element Thickness 122 mm
 Internal Corrosion Allowance 6 mm
 Nominal Thickness 122 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 4550 KPa.
 Design Temperature Internal Pressure 130 °C
 Design External Pressure 0 KPa.
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 100
 Detail Type Platform
 Detail ID PLAT-4
 Dist. from "FROM" Node / Offset dist 0.001161 mm
 Platform Start Angle (degrees) 90.0
 Platform End Angle (degrees) 270.0
 Platform Wind Area 30000 cm²
 Platform Weight 3692.6 Kg
 Platform Railing Weight 0 Kg/mm
 Platform Grating Weight 0.017 Kgs/cm²
 Platform Width 1500 mm
 Platform Height 1000 mm
 Platform Clearance or End Offset 135 mm
 Platform Force Coefficient 1.0
 Ladder Layout Angle 0.0
 Ladder Start Elevation 0 mm
 Ladder End Elevation 0 mm
 Unit Weight of Ladder 0 Kg/mm
 Platform Length (top head platform) 0 mm

Element From Node 100
 Detail Type Packing
 Detail ID pack#4
 Dist. from "FROM" Node / Offset dist 0.02325 mm
 Height of Packed Section 2650 mm
 Density 0.000163 kg./cm³
 Percent Volume Holdup 25.0
 Specific Gravity of Packing Liquid 1.0599999

Element From Node 100
 Detail Type Insulation
 Detail ID Ins: 50
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 3000 mm
 Thickness of Insulation 80 mm
 Density 0.000352 kg./cm³

Element From Node 100
 Detail Type Weight
 Detail ID B#4top HDG
 Dist. from "FROM" Node / Offset dist 2650 mm
 Miscellaneous Weight 940 Kg
 Offset from Element Centerline 0 mm



Element From Node	100
Detail Type	Weight
Detail ID	B#4top dist. d
Dist. from "FROM" Node / Offset dist	2650 mm
Miscellaneous Weight	5010 Kg
Offset from Element Centerline	0 mm

Element From Node	100
Detail Type	Weight
Detail ID	B#4top dist. T
Dist. from "FROM" Node / Offset dist	2650 mm
Miscellaneous Weight	300 Kg
Offset from Element Centerline	0 mm

Element#	11/22
Element From Node	110
Element To Node	120
Element Type	Cylinder
Description	shell-6
Distance "FROM" to "TO"	3000 mm
Inside Diameter	7200 mm
Element Thickness	122 mm
Internal Corrosion Allowance	6 mm
Nominal Thickness	122 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4550 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	110
Detail Type	Platform
Detail ID	PLAT-5
Dist. from "FROM" Node / Offset dist	1000 mm
Platform Start Angle (degrees)	270.0
Platform End Angle (degrees)	90.0
Platform Wind Area	30000 cm ²
Platform Weight	3692.6 Kg
Platform Railing Weight	0 Kg/mm
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	135 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0 Kg/mm
Platform Length (top head platform)	0 mm

Element From Node	110
Detail Type	Packing
Detail ID	pack#3
Dist. from "FROM" Node / Offset dist	1750 mm
Height of Packed Section	1250 mm
Density	0.000163 kg./cm ³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	1.0599999

Element From Node	110
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm



Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element From Node	110	
Detail Type	Nozzle	
Detail ID	MH-5	
Dist. from "FROM" Node / Offset dist	1400	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	300.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	1783.8	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	

Element From Node	110	
Detail Type	Weight	
Detail ID	B#3btm TSR	
Dist. from "FROM" Node / Offset dist	1750	mm
Miscellaneous Weight	300	Kgf
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	B#3btm supp de	
Dist. from "FROM" Node / Offset dist	1750	mm
Miscellaneous Weight	4400	Kgf
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	Nozzle wt	
Dist. from "FROM" Node / Offset dist	1700	mm
Miscellaneous Weight	3250	Kgf
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	10T margin	
Dist. from "FROM" Node / Offset dist	1700	mm
Miscellaneous Weight	10000	Kgf
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	Cleat wt	
Dist. from "FROM" Node / Offset dist	1700	mm
Miscellaneous Weight	14300	Kgf
Offset from Element Centerline	0	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	Lifting wts	
Dist. from "FROM" Node / Offset dist	1700	mm
Miscellaneous Weight	27000	Kgf
Offset from Element Centerline	0	mm

Element#	12/22	
Element From Node	120	
Element To Node	130	
Element Type	Cylinder	
Description	shell-7	
Distance "FROM" to "TO"	2000	mm
Inside Diameter	7200	mm
Element Thickness	120	mm
Internal Corrosion Allowance	6	mm



Nominal Thickness	120	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	120	
Detail Type	Packing	
Detail ID	pack#3	
Dist. from "FROM" Node / Offset dist	0.02558	mm
Height of Packed Section	2000	mm
Density	0.000163	kg./cm ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	

Element From Node	120	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element#	13/22	
Element From Node	130	
Element To Node	140	
Element Type	Cylinder	
Description	shell-8	
Distance "FROM" to "TO"	1650	mm
Inside Diameter	7200	mm
Element Thickness	120	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	120	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	130	
Detail Type	Platform	
Detail ID	PLAT-6	
Dist. from "FROM" Node / Offset dist	0	mm
Platform Start Angle (degrees)	90.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	30000	cm ²
Platform Weight	4416.5	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	135	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	270.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	19600	mm
Unit Weight of Ladder	0.03702	Kgf/mm



Platform Length (top head platform)	0	mm
Element From Node	130	
Detail Type	Packing	
Detail ID	pack#3	
Dist. from "FROM" Node / Offset dist	0.02325	mm
Height of Packed Section	750	mm
Density	0.000163	kg./cm ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	
Element From Node	130	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1650	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³
Element From Node	130	
Detail Type	Nozzle	
Detail ID	MH-4	
Dist. from "FROM" Node / Offset dist	600	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	300.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	1783.8	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	
Element From Node	130	
Detail Type	Weight	
Detail ID	B#3top HDG	
Dist. from "FROM" Node / Offset dist	749.98	mm
Miscellaneous Weight	940	Kgf
Offset from Element Centerline	0	mm
Element From Node	130	
Detail Type	Weight	
Detail ID	B#3top distri.0	
Dist. from "FROM" Node / Offset dist	749.98	mm
Miscellaneous Weight	6000	Kgf
Offset from Element Centerline	0	mm
Element From Node	130	
Detail Type	Weight	
Detail ID	B#3top dist. T	
Dist. from "FROM" Node / Offset dist	749.98	mm
Miscellaneous Weight	300	Kgf
Offset from Element Centerline	0	mm

Element#	14/22	
Element From Node	140	
Element To Node	150	
Element Type	Conical	
Description	cone	
Distance "FROM" to "TO"	2500	mm
Inside Diameter	7200	mm
Element Thickness	139.5	mm
Internal Corrosion Allowance	6	mm
Nominal Thickness	160	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C

Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Cone Diameter at "To" End	4600	mm
Design Length of Cone	2500	mm
Half Apex Angle of Cone	27.474428	degrees
Toriconical (Y/N)	N	
Weld is pre-Heated	No	

Element From Node	140	
Detail Type	Platform	
Detail ID	PLAT-7	
Dist. from "FROM" Node / Offset dist	2350	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	30000	cm ²
Platform Weight	3706.6	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	135	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	Kgf/mm
Platform Length (top head platform)	0	mm

Element From Node	140	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2500	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element From Node	140	
Detail Type	Nozzle	
Detail ID	E	
Dist. from "FROM" Node / Offset dist	973.8	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	2407.9	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 I	+IT

Element From Node	140	
Detail Type	Weight	
Detail ID	EPQVH	
Dist. from "FROM" Node / Offset dist	973.78	mm
Miscellaneous Weight	56821	Kgf
Offset from Element Centerline	5343	mm

Element From Node	140	
Detail Type	Weight	
Detail ID	FPE	
Dist. from "FROM" Node / Offset dist	973.78	mm
Miscellaneous Weight	3150	Kgf
Offset from Element Centerline	0	mm

Element From Node	140	
Detail Type	Weight	
Detail ID	FPEBolting ba	
Dist. from "FROM" Node / Offset dist	973.78	mm
Miscellaneous Weight	250	Kgf



Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	FPEinsert plt	
Dist. from "FROM" Node / Offset dist	973.78	mm
Miscellaneous Weight	1720	Kgf
Offset from Element Centerline	0	mm

Element#	15/22	
Element From Node	150	
Element To Node	160	
Element Type	Cylinder	
Description	shell-9	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	4600	mm
Element Thickness	76	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	76	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	150	
Detail Type	Packing	
Detail ID	pack#2	
Dist. from "FROM" Node / Offset dist	299.98	mm
Height of Packed Section	2700	mm
Density	0.000153	kg./cm³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	

Element From Node	150	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm³

Element From Node	150	
Detail Type	Nozzle	
Detail ID	P	
Dist. from "FROM" Node / Offset dist	150	mm
Nozzle Diameter	1.5	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	11.311	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	

Element From Node	150	
Detail Type	Weight	
Detail ID	B#2btm TSR	
Dist. from "FROM" Node / Offset dist	299.98	mm
Miscellaneous Weight	200	Kgf
Offset from Element Centerline	0	mm

Element From Node	150	
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Detail Type	Weight
Detail ID	B#2btm supp de
Dist. from "FROM" Node / Offset dist	299.98 mm
Miscellaneous Weight	1800 Kg
Offset from Element Centerline	0 mm

Element#	16/22
Element From Node	160
Element To Node	170
Element Type	Cylinder
Description	shell-10
Distance "FROM" to "TO"	3000 mm
Inside Diameter	4600 mm
Element Thickness	76 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	76 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4550 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	160
Detail Type	Platform
Detail ID	PLAT-8
Dist. from "FROM" Node / Offset dist	850 mm
Platform Start Angle (degrees)	90.0
Platform End Angle (degrees)	270.0
Platform Wind Area	30000 cm²
Platform Weight	2613.8 Kg
Platform Railing Weight	0 Kg/mm
Platform Grating Weight	0.017 Kgs/cm²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	135 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	0 mm
Unit Weight of Ladder	0 Kg/mm
Platform Length (top head platform)	0 mm

Element From Node	160
Detail Type	Packing
Detail ID	pack#2
Dist. from "FROM" Node / Offset dist	0.02558 mm
Height of Packed Section	3000 mm
Density	0.000153 kg./cm³
Percent Volume Holdup	25.0
Specific Gravity of Packing Liquid	1.0599999

Element From Node	160
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm³

Element#	17/22
Element From Node	170



Element To Node	180	
Element Type	Cylinder	
Description	shell-11	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	4600	mm
Element Thickness	74	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	74	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	170	
Detail Type	Platform	
Detail ID	PLAT-9	
Dist. from "FROM" Node / Offset dist	1850	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	30000	cm ²
Platform Weight	2612.1	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	135	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	Kgf/mm
Platform Length (top head platform)	0	mm
Element From Node	170	
Detail Type	Packing	
Detail ID	pack#2	
Dist. from "FROM" Node / Offset dist	0.02325	mm
Height of Packed Section	150.01	mm
Density	0.000153	kg./cm ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	
Element From Node	170	
Detail Type	Packing	
Detail ID	pack#1	
Dist. from "FROM" Node / Offset dist	2250	mm
Height of Packed Section	749.99	mm
Density	0.000153	kg./cm ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	
Element From Node	170	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³
Element From Node	170	
Detail Type	Nozzle	
Detail ID	MH-3	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	24	in.



Nozzle Schedule	None
Nozzle Class	600
Layout Angle	225.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	1543.7 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	170
Detail Type	Weight
Detail ID	B#2top HDG
Dist. from "FROM" Node / Offset dist	149.98 mm
Miscellaneous Weight	300 Kg
Offset from Element Centerline	0 mm

Element From Node	170
Detail Type	Weight
Detail ID	B#2top distri.0
Dist. from "FROM" Node / Offset dist	149.98 mm
Miscellaneous Weight	1850 Kg
Offset from Element Centerline	0 mm

Element From Node	170
Detail Type	Weight
Detail ID	B#2top dis. TA
Dist. from "FROM" Node / Offset dist	149.98 mm
Miscellaneous Weight	200 Kg
Offset from Element Centerline	0 mm

Element From Node	170
Detail Type	Weight
Detail ID	B#1btm TSR
Dist. from "FROM" Node / Offset dist	2250 mm
Miscellaneous Weight	200 Kg
Offset from Element Centerline	0 mm

Element From Node	170
Detail Type	Weight
Detail ID	B#1btm supp de
Dist. from "FROM" Node / Offset dist	2250 mm
Miscellaneous Weight	1800 Kg
Offset from Element Centerline	0 mm

Element#	18/22
Element From Node	180
Element To Node	190
Element Type	Cylinder
Description	shell-12
Distance "FROM" to "TO"	3000 mm
Inside Diameter	4600 mm
Element Thickness	74 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	74 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4550 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	180
Detail Type	Platform
Detail ID	PLAT-10
Dist. from "FROM" Node / Offset dist	2850 mm



Platform Start Angle (degrees)	90.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	30000	cm ²
Platform Weight	2612.1	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	135	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	Kgf/mm
Platform Length (top head platform)	0	mm

Element From Node	180	
Detail Type	Packing	
Detail ID	pack#1	
Dist. from "FROM" Node / Offset dist	0.0186	mm
Height of Packed Section	3000	mm
Density	0.000153	kg./cm ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	

Element From Node	180	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element#	19/22	
Element From Node	190	
Element To Node	200	
Element Type	Cylinder	
Description	shell-13	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	4600	mm
Element Thickness	74	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	74	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	190	
Detail Type	Packing	
Detail ID	pack#1	
Dist. from "FROM" Node / Offset dist	0.0186	mm
Height of Packed Section	2100	mm
Density	0.000153	kg./cm ³
Percent Volume Holdup	25.0	
Specific Gravity of Packing Liquid	1.0599999	

Element From Node	190	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm



Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³
Element From Node	190	
Detail Type	Weight	
Detail ID	B#1top HDG	
Dist. from "FROM" Node / Offset dist	2100	mm
Miscellaneous Weight	300	Kgf
Offset from Element Centerline	0	mm
Element From Node	190	
Detail Type	Weight	
Detail ID	B#1top distri.0	
Dist. from "FROM" Node / Offset dist	2100	mm
Miscellaneous Weight	2260	Kgf
Offset from Element Centerline	0	mm
Element From Node	190	
Detail Type	Weight	
Detail ID	B#1top dis. TA	
Dist. from "FROM" Node / Offset dist	2100	mm
Miscellaneous Weight	200	Kgf
Offset from Element Centerline	0	mm

Element#	20/22	
Element From Node	200	
Element To Node	210	
Element Type	Cylinder	
Description	shell-14	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	4600	mm
Element Thickness	74	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	74	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	200	
Detail Type	Platform	
Detail ID	PLAT-11	
Dist. from "FROM" Node / Offset dist	2850	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	30000	cm ²
Platform Weight	2612.1	Kgf
Platform Railing Weight	0	Kgf/mm
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	135	mm
Platform Force Coefficient	1.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	0	mm
Unit Weight of Ladder	0	Kgf/mm
Platform Length (top head platform)	0	mm

Element From Node	200
Detail Type	Insulation
Detail ID	Ins: 50



Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element From Node	200	
Detail Type	Nozzle	
Detail ID	MH-2	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	300.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	1543.7	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	

Element From Node	200	
Detail Type	Nozzle	
Detail ID	D	
Dist. from "FROM" Node / Offset dist	500.01	mm
Nozzle Diameter	16	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	463.88	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2 +IT	

Element From Node	200	
Detail Type	Tray	
Detail ID	Tray#1-2-3	
Dist. from "FROM" Node / Offset dist	1900	mm
Tray Spacing	300	mm
Number of Trays	3	
Tray Weight per Unit Area	0.015	Kgs/cm ²
Height of Liquid on Tray	0	mm
Density of Liquid on Tray	0	kg./cm ³
Support Ring and Bolting Bar Weight	270	Kgf

Element From Node	200	
Detail Type	Weight	
Detail ID	FPD	
Dist. from "FROM" Node / Offset dist	699.98	mm
Miscellaneous Weight	1000	Kgf
Offset from Element Centerline	0	mm

Element From Node	200	
Detail Type	Weight	
Detail ID	FPDBolting ba	
Dist. from "FROM" Node / Offset dist	699.98	mm
Miscellaneous Weight	150	Kgf
Offset from Element Centerline	0	mm

Element From Node	200	
Detail Type	Weight	
Detail ID	DPPQVH	
Dist. from "FROM" Node / Offset dist	499.98	mm
Miscellaneous Weight	17147	Kgf
Offset from Element Centerline	3779	mm

Element#	21/22
Element From Node	210
Element To Node	220
Element Type	Cylinder
Description	shell-15



Distance "FROM" to "TO"	1150	mm
Inside Diameter	4600	mm
Element Thickness	72	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	72	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.23	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	210	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1150	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³
Element From Node	210	
Detail Type	Nozzle	
Detail ID	MH-1	
Dist. from "FROM" Node / Offset dist	300	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	300.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	1546.6	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	+IT
Element From Node	210	
Detail Type	Weight	
Detail ID	Demister dead	
Dist. from "FROM" Node / Offset dist	849.98	mm
Miscellaneous Weight	410	Kgf
Offset from Element Centerline	0	mm
Element From Node	210	
Detail Type	Weight	
Detail ID	Demister TA	
Dist. from "FROM" Node / Offset dist	849.98	mm
Miscellaneous Weight	1400	Kgf
Offset from Element Centerline	0	mm
Element From Node	210	
Detail Type	Weight	
Detail ID	Tray Truss	
Dist. from "FROM" Node / Offset dist	849.98	mm
Miscellaneous Weight	1210	Kgf
Offset from Element Centerline	0	mm
Element From Node	210	
Detail Type	Weight	
Detail ID	davit	
Dist. from "FROM" Node / Offset dist	499.98	mm
Miscellaneous Weight	900	Kgf
Offset from Element Centerline	2300	mm
Element From Node	210	
Detail Type	Weight	
Detail ID	BPPQVH	
Dist. from "FROM" Node / Offset dist	574.98	mm
Miscellaneous Weight	17550	Kgf



Offset from Element Centerline 3841 mm

Element#	22/22
Element From Node	220
Element To Node	230
Element Type	Spherical
Description	TOP dish
Distance "FROM" to "TO"	0 mm
Inside Diameter	4600 mm
Element Thickness	40 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	46 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4550 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.23
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	220
Detail Type	Platform
Detail ID	TOP PLT
Dist. from "FROM" Node / Offset dist	2500 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	80000 cm ²
Platform Weight	11848 Kgf
Platform Railing Weight	0 Kgf/mm
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	8000 mm
Platform Height	1000 mm
Platform Clearance or End Offset	4000 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	270.0
Ladder Start Elevation	0 mm
Ladder End Elevation	26000 mm
Unit Weight of Ladder	0.03702 Kgf/mm
Platform Length (top head platform)	8000 mm

Element From Node	220
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2300 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm ³

Element From Node	220
Detail Type	Nozzle
Detail ID	B
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	744 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	220
Detail Type	Weight
Detail ID	top dish SF wt
Dist. from "FROM" Node / Offset dist	0 mm

Miscellaneous Weight	1050	Kgf
Offset from Element Centerline	0	mm
Element From Node	220	
Detail Type	Weight	
Detail ID	PSV-1	
Dist. from "FROM" Node / Offset dist	2300	mm
Miscellaneous Weight	3000	Kgf
Offset from Element Centerline	4000	mm
Element From Node	220	
Detail Type	Weight	
Detail ID	PSV-2	
Dist. from "FROM" Node / Offset dist	2300	mm
Miscellaneous Weight	3000	Kgf
Offset from Element Centerline	4000	mm
Element From Node	220	
Detail Type	For./Mom.	
Detail ID	BFxFz	
Dist. from "FROM" Node / Offset dist	2300	mm
Force in X Direction	4373.2	Kgf
Force in Y Direction	-6054.4	Kgf
Force in Z Direction	4373.2	Kgf
Moment about X Axis	8074	Kg-m.
Moment about Y Axis	0	Kg-m.
Moment about Z Axis	8074	Kg-m.
Force/Moment Combination Method	SRSS	



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Btm skirt-1	...	...	2000	...	2000
Btm skirt-2	...	...	5000	...	3000
Btm skirt-3	...	...	8000	...	3000
Top skirt	...	...	9000	...	1000
BTM dish	...	...	9000	...	...
shell-1	...	...	12000	...	3000
shell-2	...	...	15000	...	3000
shell-3	...	...	18000	...	3000
shell-4	...	...	21000	...	3000
shell-5	...	...	24000	...	3000
shell-6	...	...	27000	...	3000
shell-7	...	...	29000	...	2000
shell-8	...	...	30650	...	1650
cone	...	...	33150	...	2500
shell-9	...	...	36150	...	3000
shell-10	...	...	39150	...	3000
shell-11	...	...	42150	...	3000
shell-12	...	...	45150	...	3000
shell-13	...	...	48150	...	3000
shell-14	...	...	51150	...	3000
shell-15	...	...	52300	...	1150
TOP dish	...	...	52300	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
Btm skirt-1	...	...	36	1.6	7294	...	1
Btm skirt-2	...	...	34	1.6	7298	...	2
Btm skirt-3	...	...	34	1.6	7298	...	3
Top skirt	...	...	34	1.6	7298	...	4
BTM dish		4687.2	78	6	7200	156.4	5
shell-1		4649.8	126	6	7200	156.4	6
shell-2		4618.6	124	6	7200	156.4	7
shell-3		4587.4	124	6	7200	156.4	8
shell-4		4563	122	6	7200	156.4	9
shell-5		4563	122	6	7200	156.4	10
shell-6		4563	122	6	7200	156.4	11
shell-7		4563	120	6	7200	156.4	12
shell-8		4563	120	6	7200	156.4	13
cone		4563	160	6	7200	156.4	14
shell-9		4563	76	3	4600	156.4	15
shell-10		4563	76	3	4600	156.4	16
shell-11		4563	74	3	4600	156.4	17
shell-12		4563	74	3	4600	156.4	18
shell-13		4563	74	3	4600	156.4	19
shell-14		4563	74	3	4600	156.4	20
shell-15		4563	72	3	4600	156.4	21
TOP dish		4563	46	3	4600	156.4	22

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
Btm skirt-1	...	...	No Calc	No Calc	36	No Calc	1
Btm skirt-2	...	...	No Calc	No Calc	34	No Calc	2
Btm skirt-3	...	...	No Calc	No Calc	34	No Calc	3
Top skirt	...	...	No Calc	No Calc	34	No Calc	4
BTM dish		4550	5220.55	6577.71	68.3	60.4442	5
shell-1		4550	5019.88	6019.89	126	114.822	6
shell-2		4550	4967.09	5925.95	124	114.081	7
shell-3		4550	4998.26	5925.95	124	113.342	8
shell-4		4550	4938.66	5831.93	122	112.762	9
shell-5		4550	4938.66	5831.93	122	112.762	10
shell-6		4550	4938.66	5831.93	122	112.762	11
shell-7		4550	4854.59	5737.88	120	112.762	12
shell-8		4550	4854.59	5737.88	120	112.762	13
cone		4550	5041.29	5915	139.5	126.333	14
shell-9		4550	4867.29	5688.8	76	71.1842	15
shell-10		4550	4867.29	5688.8	76	71.1842	16
shell-11		4550	4735.6	5541.43	74	71.1842	17
shell-12		4550	4735.6	5541.43	74	71.1842	18
shell-13		4550	4735.6	5541.43	74	71.1842	19
shell-14		4550	4735.6	5541.43	74	71.1842	20
shell-15		4550	4603.8	5393.95	72	71.1842	21
TOP dish		4550	4972.24	6034.29	40	36.8435	22

Minimum 4604 5394

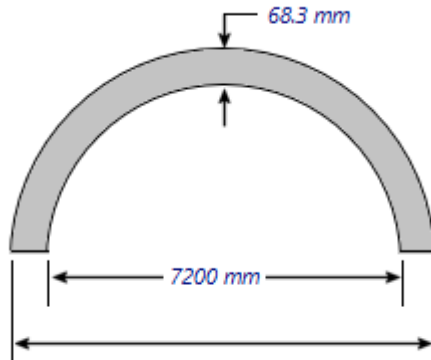
MAWP: 4604 KPa., limited by: shell-15.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: BTM dish

Spherical Head Design Information:					
Design Pressure	P	4687.162 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		0.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7200.000 mm	Actual Stress		134.5 N./mm ²
Spec. Min. Thk t _e , or s _a		68.3000 mm	Spec. Nominal Thk.		78.000 mm
Surface Area		849969.5 cm ²	SG of Contents		1.060
Element Volume		97733608.0 cm ³	Weight of Contents		103534.0 Kg _f
Empty Weight		50299.3 Kg _f	Operating Weight		153833.3 Kg _f
			Comp. All. Ope. Stress		105.1 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 90.000 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 45.000 mm

Material treatment options: +IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2 \cdot (\exp(0.5P/(S \cdot E)) - 1) + c_i + c_o$
 $= 7212.0/2 \cdot (\exp(0.5 \cdot 4687.162/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 54.4442 + 6.0 + 0.0 = 60.4442 \text{ mm}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$
 $= 2 \cdot 156.4 \cdot 1.0 \cdot \ln((2 \cdot 62.3 + 7212.0)/7212.0) - 137.162$
 $= 5357.708 - 137.162 = 5220.5459 \text{ KPa.}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D)$
 $= 2 \cdot 175.0 \cdot 1.0 \cdot \ln((2 \cdot 68.3 + 7200.0)/7200.0)$
 $= 6577.71 \text{ KPa.}$



Computed Stress Intensity at Design Pressure [Si]:

$$\begin{aligned} &= P \cdot D^2 / (E (D_o^2 - D^2)) \\ &= 4687.162 \cdot 7212.0^2 / (1.0 (7336.6^2 - 7212.0^2)) \\ &= 134.495 \text{ N./mm}^2 \end{aligned}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$\begin{aligned} &= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D)) \\ &= 0.5 \cdot 4687.162 / (1.0 \cdot \ln((2 \cdot 62.3 + 7212.0) / 7212.0)) \\ &= 136.826 \text{ N./mm}^2 \end{aligned}$$

Percent Fiber Elongation per Table 6.1

$$\begin{aligned} \text{For Heads formed from Segments } (75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) & \quad 1.608 \% \\ \text{For one Piece doubly Curved Heads } 100 \cdot \ln(D_b / (D_f - 2t)) & \quad 36.801 \% \end{aligned}$$

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 849970 / 3.14159) \\ &= 10402.955 \text{ mm} \end{aligned}$$

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

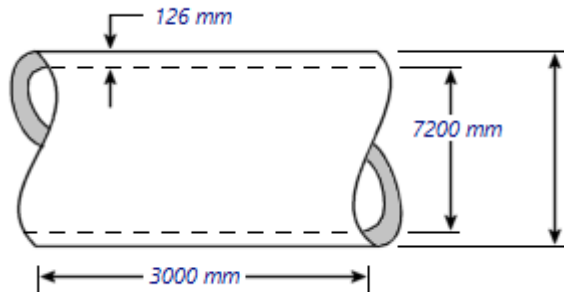
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 90.000 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 45.000 mm

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	4649.757 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7200.000 mm	Actual Stress		139.7 N./mm ²
Spec. Min. Thk t,e, or sa		126.0000 mm	Spec. Nominal Thk.		126.000 mm
Surface Area		702334.5 cm ²	SG of Contents		1.060
Element Volume		122167016.0 cm ³	Weight of Contents		129417.5 Kg
Empty Weight		67424.2 Kg	Operating Weight		196841.8 Kg
			Comp. All. Ope. Stress		116.6 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7212.0 \cdot (\exp(4649.757/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 108.8221 + 6.0 + 0.0 = 114.8221 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 120.0 + 7212.0) / 7212.0) - 99.757$$

$$= 5119.636 - 99.757 = 5019.8779 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 126.0 + 7200.0) / 7200.0)$$

$$= 6019.8945 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4649.757 \cdot 7212.0 / (1.0 (7452.0 - 7212.0))$$

$$= 139.7333 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4649.757 / (1.0 \cdot \ln((2 \cdot 120.0 + 7212.0) / 7212.0))$$
$$= 142.0457 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.720 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

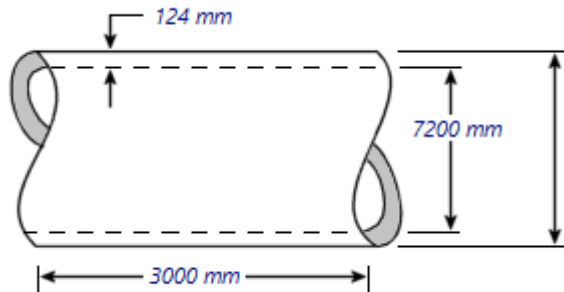
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	4618.587 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7200.000 mm	Actual Stress	141.1 N./mm ²	
Spec. Min. Thk t,e, or sa		124.0000 mm	Spec. Nominal Thk.	124.000 mm	
Surface Area		701957.5 cm ²	SG of Contents	1.060	
Element Volume		122167016.0 cm ³	Weight of Contents	129417.5 Kg	
Empty Weight		66336.0 Kg	Operating Weight	195753.6 Kg	
			Comp. All. Ope. Stress	116.4 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7212.0 \cdot (\exp(4618.587/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 108.0815 + 6.0 + 0.0 = 114.0815 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 118.0 + 7212.0) / 7212.0) - 68.587$$

$$= 5035.678 - 68.587 = 4967.0908 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 124.0 + 7200.0) / 7200.0)$$

$$= 5925.9473 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4618.587 \cdot 7212.0 / (1.0 (7448.0 - 7212.0))$$

$$= 141.1489 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4618.587 / (1.0 \cdot \ln((2 \cdot 118.0 + 7212.0) / 7212.0))$$
$$= 143.4458 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.693 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

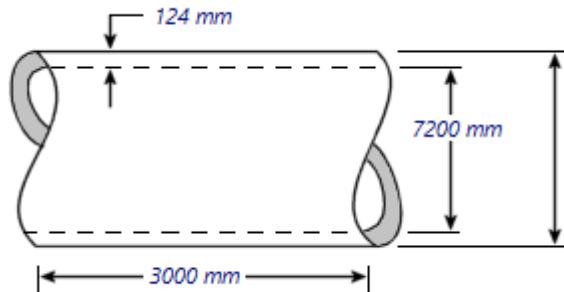
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	4587.417 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7200.000 mm	Actual Stress	140.2 N./mm ²	
Spec. Min. Thk t _e , or s _a		124.0000 mm	Spec. Nominal Thk.	124.000 mm	
Surface Area		701957.5 cm ²	SG of Contents	1.060	
Element Volume		122167016.0 cm ³	Weight of Contents	101377.1 Kg	
Empty Weight		66336.0 Kg	Operating Weight	167713.1 Kg	
			Comp. All. Ope. Stress	116.4 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7212.0 \cdot (\exp(4587.417/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 107.3417 + 6.0 + 0.0 = 113.3417 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 118.0 + 7212.0) / 7212.0) - 37.417$$

$$= 5035.678 - 37.417 = 4998.2612 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 124.0 + 7200.0) / 7200.0)$$

$$= 5925.9473 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4587.417 \cdot 7212.0 / (1.0 (7448.0 - 7212.0))$$

$$= 140.1963 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4587.417 / (1.0 \cdot \ln((2 \cdot 118.0 + 7212.0) / 7212.0))$$
$$= 142.4777 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.693 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

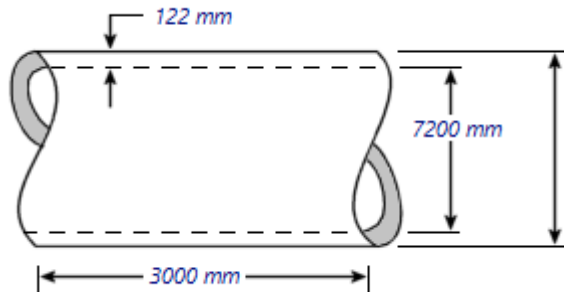
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	4563.000 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7200.000 mm	Actual Stress	141.9 N./mm ²	
Spec. Min. Thk t _e , or s _a		122.0000 mm	Spec. Nominal Thk.	122.000 mm	
Surface Area		701580.5 cm ²	SG of Contents	0.000	
Element Volume		122167016.0 cm ³	Weight of Contents	0.0 Kgf	
Empty Weight		65248.0 Kgf	Operating Weight	65248.0 Kgf	
			Comp. All. Ope. Stress	116.1 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7212.0 \cdot (\exp(4563.0/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 106.7618 + 6.0 + 0.0 = 112.7618 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 116.0 + 7212.0) / 7212.0) - 13.0$$

$$= 4951.657 - 13.0 = 4938.6567 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 122.0 + 7200.0) / 7200.0)$$

$$= 5831.9292 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4563.0 \cdot 7212.0 / (1.0 (7444.0 - 7212.0))$$

$$= 141.8547 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 4563.0 / (1.0 \cdot \ln((2 \cdot 116.0 + 7212.0) / 7212.0))$$
$$= 144.1241 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.666 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

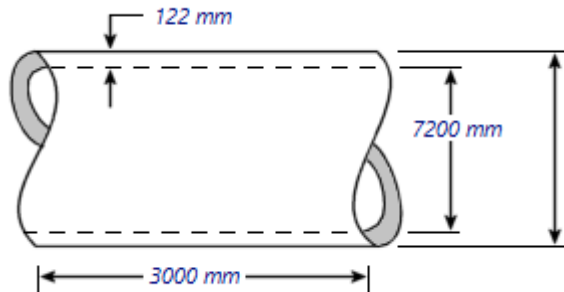
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	4563.000 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7200.000 mm	Actual Stress		141.9 N./mm ²
Spec. Min. Thk t _e , or s _a		122.0000 mm	Spec. Nominal Thk.		122.000 mm
Surface Area		701580.5 cm ²	SG of Contents		0.000
Element Volume		122167016.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		65248.0 Kgf	Operating Weight		65248.0 Kgf
			Comp. All. Ope. Stress		116.1 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7212.0 \cdot (\exp(4563.0/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 106.7618 + 6.0 + 0.0 = 112.7618 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 116.0 + 7212.0) / 7212.0) - 13.0$$

$$= 4951.657 - 13.0 = 4938.6567 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 122.0 + 7200.0) / 7200.0)$$

$$= 5831.9292 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4563.0 \cdot 7212.0 / (1.0 (7444.0 - 7212.0))$$

$$= 141.8547 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4563.0 / (1.0 \cdot \ln((2 \cdot 116.0 + 7212.0) / 7212.0))$$
$$= 144.1241 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.666 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

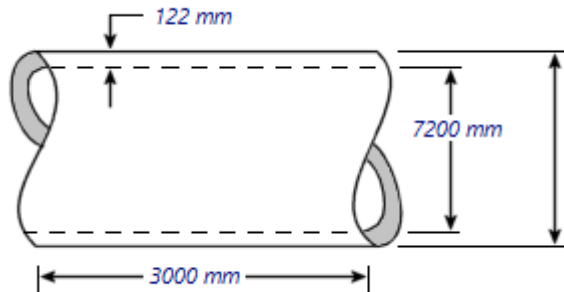
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	4563.000 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7200.000 mm	Actual Stress	141.9 N./mm ²	
Spec. Min. Thk t _e , or s _a		122.0000 mm	Spec. Nominal Thk.	122.000 mm	
Surface Area		701580.5 cm ²	SG of Contents	0.000	
Element Volume		122167016.0 cm ³	Weight of Contents	0.0 Kgf	
Empty Weight		65248.0 Kgf	Operating Weight	65248.0 Kgf	
			Comp. All. Ope. Stress	116.1 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7212.0 \cdot (\exp(4563.0/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 106.7618 + 6.0 + 0.0 = 112.7618 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 116.0 + 7212.0) / 7212.0) - 13.0$$

$$= 4951.657 - 13.0 = 4938.6567 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 122.0 + 7200.0) / 7200.0)$$

$$= 5831.9292 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4563.0 \cdot 7212.0 / (1.0 (7444.0 - 7212.0))$$

$$= 141.8547 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4563.0 / (1.0 \cdot \ln((2 \cdot 116.0 + 7212.0) / 7212.0))$$
$$= 144.1241 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.666 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

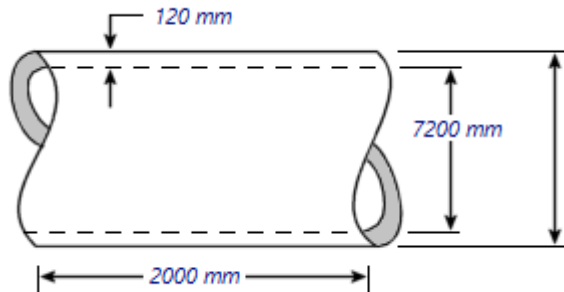
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: shell-7

Cylindrical Shell Design Information:					
Design Pressure	P	4563.000 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7200.000 mm	Actual Stress		144.3 N./mm ²
Spec. Min. Thk t _e , or s _a		120.0000 mm	Spec. Nominal Thk.		120.000 mm
Surface Area		467469.0 cm ²	SG of Contents		0.000
Element Volume		81444680.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		42774.0 Kgf	Operating Weight		42774.0 Kgf
			Comp. All. Ope. Stress		115.9 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7212.0 \cdot (\exp(4563.0/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 106.7618 + 6.0 + 0.0 = 112.7618 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 114.0 + 7212.0) / 7212.0) - 13.0$$

$$= 4867.591 - 13.0 = 4854.5908 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 120.0 + 7200.0) / 7200.0)$$

$$= 5737.8809 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4563.0 \cdot 7212.0 / (1.0 (7440.0 - 7212.0))$$

$$= 144.3433 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 4563.0 / (1.0 \cdot \ln((2 \cdot 114.0 + 7212.0) / 7212.0))$$
$$= 146.6132 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.639 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

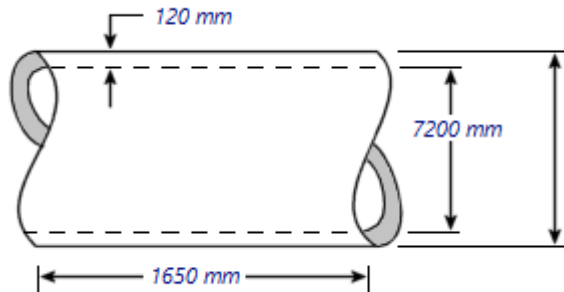
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: shell-8

Cylindrical Shell Design Information:					
Design Pressure	P	4563.000 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1650.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7200.000 mm	Actual Stress		144.3 N./mm ²
Spec. Min. Thk t _e , or s _a		120.0000 mm	Spec. Nominal Thk.		120.000 mm
Surface Area		385662.0 cm ²	SG of Contents		0.000
Element Volume		67191856.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		35288.6 Kgf	Operating Weight		35288.6 Kgf
			Comp. All. Ope. Stress		115.9 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7212.0 \cdot (\exp(4563.0/(156.4 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 106.7618 + 6.0 + 0.0 = 112.7618 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 114.0 + 7212.0) / 7212.0) - 13.0$$

$$= 4867.591 - 13.0 = 4854.5908 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 120.0 + 7200.0) / 7200.0)$$

$$= 5737.8809 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4563.0 \cdot 7212.0 / (1.0 (7440.0 - 7212.0))$$

$$= 144.3433 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4563.0 / (1.0 \cdot \ln((2 \cdot 114.0 + 7212.0) / 7212.0))$$
$$= 146.6132 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.639 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

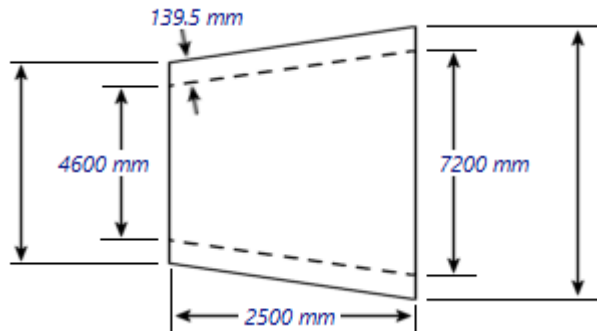
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: cone

Conical Head/Section Design Information:					
Design Pressure	P	4563.000 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2500.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	6.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	7200.000 mm	Actual Stress		138.9 N./mm ²
Spec. Min. Thk t,e, or sa		139.5000 mm	Spec. Nominal Thk.		160.000 mm
Surface Area		554219.2 cm ²	SG of Contents		0.000
Element Volume		69467832.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		66744.5 Kgf	Operating Weight		66744.5 Kgf
			Comp. All. Ope. Stress		119.4 N./mm ²
			To End Diameter		4600.000 mm

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Thickness due to Internal Pressure: Equation 4.3.4.1 [tr]:

$$= 0.5 \cdot D_c / \cos(\alpha) \cdot (\exp(P / (S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 7212.0 / \cos(27.474) \cdot (\exp(4563.0 / (156.4 \cdot 1.0)) - 1) + c_i + c_o$$

$$= 120.3333 + 6.0 + 0.0 = 126.3333 \text{ mm}$$

Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln(2 \cdot t_{cor} \cdot \cos(\alpha) + D_{cor}) / D_{cor}$$

$$= 156.4 \cdot 1.0 \cdot \ln(2 \cdot 133.5 \cdot \cos(27.474) + 7212.0) / 7212.0$$

$$= 5054.2891 \text{ KPa.}$$

Maximum Allowable Pressure New and Cold [MAPnc]:

$$= S_a \cdot E \cdot \ln(2 \cdot t \cdot \cos(\alpha) + D) / D$$

$$= 175.0 \cdot 1.0 \cdot \ln(2 \cdot 139.5 \cdot \cos(27.474) + 7200.0) / 7200.0$$

$$= 5914.9971 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D) \cos(\alpha))$$

$$= 4563.0 \cdot 7212.0 / (1.0(7479.0 - 7212.0) \cos(27.474))$$

$$= 138.928 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t_{cor} \cdot \cos(\alpha) + D_{cor}) / D_{cor}))$$

$$= 4563.0 / (1.0 \cdot \ln((2 \cdot 133.5 \cdot \cos(27.474) + 7212.0) / 7212.0))$$

$$= 141.198 \text{ N./mm}^2$$



Percent Fiber Elongation per Table 6.1

Cones formed from Plate $(50 * t_{nom} / R_f) * (1 - R_f / R_o)$ 3.347 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

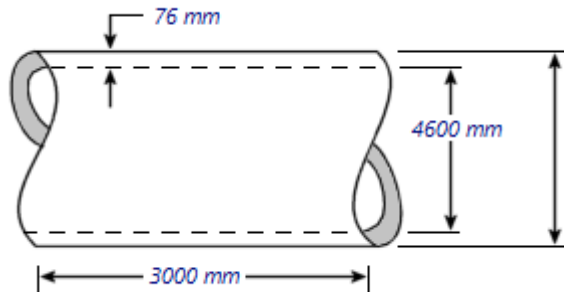
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: shell-9

Cylindrical Shell Design Information:					
Design Pressure	P	4563.000 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	4600.000 mm	Actual Stress	144.0 N./mm ²	
Spec. Min. Thk t _e , or s _a		76.0000 mm	Spec. Nominal Thk.	76.000 mm	
Surface Area		447865.5 cm ²	SG of Contents	0.000	
Element Volume		49866016.0 cm ³	Weight of Contents	0.0 Kgf	
Empty Weight		25957.7 Kgf	Operating Weight	25957.7 Kgf	
			Comp. All. Ope. Stress	115.9 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 4606.0 \cdot (\exp(4563.0/(156.4 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 68.1842 + 3.0 + 0.0 = 71.1842 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 73.0 + 4606.0) / 4606.0) - 13.0$$

$$= 4880.295 - 13.0 = 4867.2944 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 76.0 + 4600.0) / 4600.0)$$

$$= 5688.7983 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4563.0 \cdot 4606.0 / (1.0 (4752.0 - 4606.0))$$

$$= 143.9616 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4563.0 / (1.0 \cdot \ln((2 \cdot 73.0 + 4606.0) / 4606.0))$$
$$= 146.2316 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.625 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

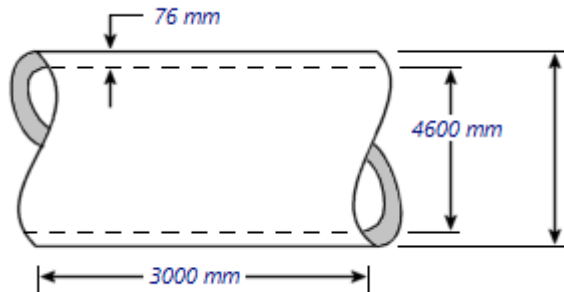
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: shell-10

Cylindrical Shell Design Information:					
Design Pressure	P	4563.000 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	4600.000 mm	Actual Stress		144.0 N./mm ²
Spec. Min. Thk t _e , or s _a		76.0000 mm	Spec. Nominal Thk.		76.000 mm
Surface Area		447865.5 cm ²	SG of Contents		0.000
Element Volume		49866016.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		25957.7 Kgf	Operating Weight		25957.7 Kgf
			Comp. All. Ope. Stress		115.9 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 4606.0 \cdot (\exp(4563.0/(156.4 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 68.1842 + 3.0 + 0.0 = 71.1842 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 73.0 + 4606.0) / 4606.0) - 13.0$$

$$= 4880.295 - 13.0 = 4867.2944 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 76.0 + 4600.0) / 4600.0)$$

$$= 5688.7983 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4563.0 \cdot 4606.0 / (1.0 (4752.0 - 4606.0))$$

$$= 143.9616 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4563.0 / (1.0 \cdot \ln((2 \cdot 73.0 + 4606.0) / 4606.0))$$
$$= 146.2316 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.625 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

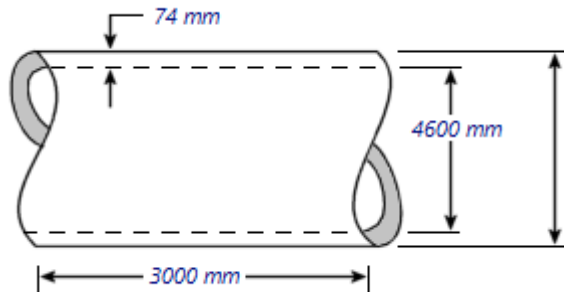
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: shell-11

Cylindrical Shell Design Information:					
Design Pressure	P	4563.000 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	4600.000 mm	Actual Stress		148.0 N./mm ²
Spec. Min. Thk t _e , or sa		74.0000 mm	Spec. Nominal Thk.		74.000 mm
Surface Area		447488.5 cm ²	SG of Contents		0.000
Element Volume		49866016.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		25263.8 Kgf	Operating Weight		25263.8 Kgf
			Comp. All. Ope. Stress		115.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 4606.0 \cdot (\exp(4563.0/(156.4 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 68.1842 + 3.0 + 0.0 = 71.1842 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 71.0 + 4606.0) / 4606.0) - 13.0$$

$$= 4748.596 - 13.0 = 4735.5962 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 74.0 + 4600.0) / 4600.0)$$

$$= 5541.4268 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4563.0 \cdot 4606.0 / (1.0 (4748.0 - 4606.0))$$

$$= 148.0170 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4563.0 / (1.0 \cdot \ln((2 \cdot 71.0 + 4606.0) / 4606.0))$$
$$= 150.2872 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.583 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

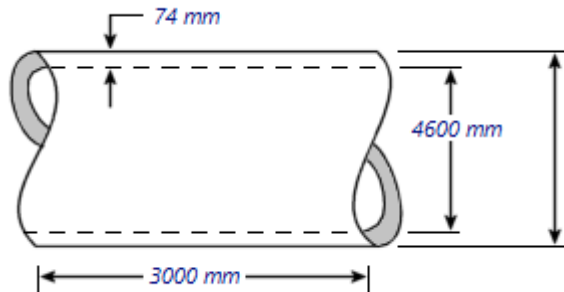
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: shell-12

Cylindrical Shell Design Information:					
Design Pressure	P	4563.000 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	4600.000 mm	Actual Stress	148.0 N./mm ²	
Spec. Min. Thk t _e , or s _a		74.0000 mm	Spec. Nominal Thk.	74.000 mm	
Surface Area		447488.5 cm ²	SG of Contents	0.000	
Element Volume		49866016.0 cm ³	Weight of Contents	0.0 Kgf	
Empty Weight		25263.8 Kgf	Operating Weight	25263.8 Kgf	
			Comp. All. Ope. Stress	115.5 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 4606.0 \cdot (\exp(4563.0/(156.4 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 68.1842 + 3.0 + 0.0 = 71.1842 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 71.0 + 4606.0) / 4606.0) - 13.0$$

$$= 4748.596 - 13.0 = 4735.5962 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 74.0 + 4600.0) / 4600.0)$$

$$= 5541.4268 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4563.0 \cdot 4606.0 / (1.0 (4748.0 - 4606.0))$$

$$= 148.0170 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 4563.0 / (1.0 \cdot \ln((2 \cdot 71.0 + 4606.0) / 4606.0))$$
$$= 150.2872 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.583 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

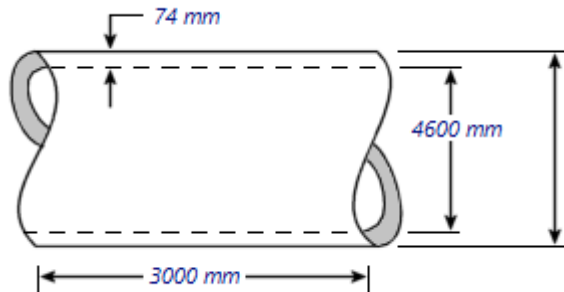
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: shell-13

Cylindrical Shell Design Information:					
Design Pressure	P	4563.000 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	4600.000 mm	Actual Stress		148.0 N./mm ²
Spec. Min. Thk t _e , or s _a		74.0000 mm	Spec. Nominal Thk.		74.000 mm
Surface Area		447488.5 cm ²	SG of Contents		0.000
Element Volume		49866016.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		25263.8 Kgf	Operating Weight		25263.8 Kgf
			Comp. All. Ope. Stress		115.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 4606.0 \cdot (\exp(4563.0/(156.4 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 68.1842 + 3.0 + 0.0 = 71.1842 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 71.0 + 4606.0) / 4606.0) - 13.0$$

$$= 4748.596 - 13.0 = 4735.5962 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 74.0 + 4600.0) / 4600.0)$$

$$= 5541.4268 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4563.0 \cdot 4606.0 / (1.0 (4748.0 - 4606.0))$$

$$= 148.0170 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4563.0 / (1.0 \cdot \ln((2 \cdot 71.0 + 4606.0) / 4606.0))$$
$$= 150.2872 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.583 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

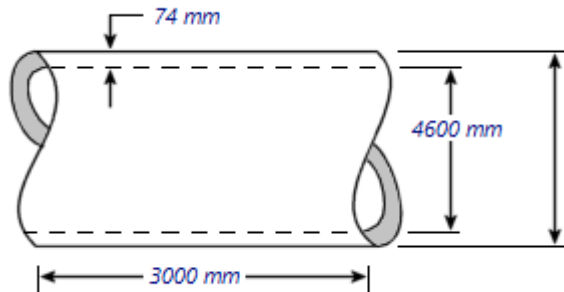
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

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Internal Pressure Results for: shell-14

Cylindrical Shell Design Information:					
Design Pressure	P	4563.000 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	4600.000 mm	Actual Stress	148.0 N./mm ²	
Spec. Min. Thk t,e, or sa		74.0000 mm	Spec. Nominal Thk.	74.000 mm	
Surface Area		447488.5 cm ²	SG of Contents	0.000	
Element Volume		49866016.0 cm ³	Weight of Contents	0.0 Kgf	
Empty Weight		25263.8 Kgf	Operating Weight	25263.8 Kgf	
			Comp. All. Ope. Stress	115.5 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 4606.0 \cdot (\exp(4563.0/(156.4 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 68.1842 + 3.0 + 0.0 = 71.1842 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 71.0 + 4606.0) / 4606.0) - 13.0$$

$$= 4748.596 - 13.0 = 4735.5962 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 74.0 + 4600.0) / 4600.0)$$

$$= 5541.4268 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4563.0 \cdot 4606.0 / (1.0 (4748.0 - 4606.0))$$

$$= 148.0170 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 4563.0 / (1.0 \cdot \ln((2 \cdot 71.0 + 4606.0) / 4606.0))$$
$$= 150.2872 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.583 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

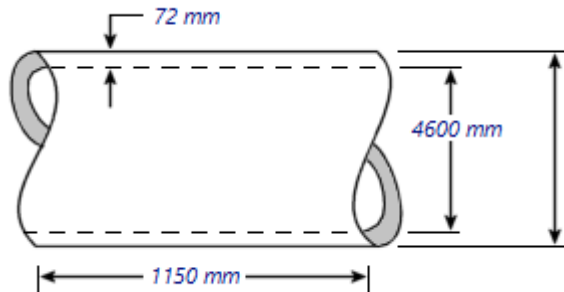
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell-15

Cylindrical Shell Design Information:					
Design Pressure	P	4563.000 KPa.	Int. Design Temperature	130 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		1150.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	4600.000 mm	Actual Stress	152.3 N./mm ²	
Spec. Min. Thk t _e , or s _a		72.0000 mm	Spec. Nominal Thk.	72.000 mm	
Surface Area		171392.8 cm ²	SG of Contents	0.000	
Element Volume		19115306.0 cm ³	Weight of Contents	0.0 Kgf	
Empty Weight		9418.7 Kgf	Operating Weight	9418.7 Kgf	
			Comp. All. Ope. Stress	115.1 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 4606.0 \cdot (\exp(4563.0/(156.4 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 68.1842 + 3.0 + 0.0 = 71.1842 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 156.4 \cdot 1.0 \cdot \ln((2 \cdot 69.0 + 4606.0) / 4606.0) - 13.0$$

$$= 4616.805 - 13.0 = 4603.8042 \text{ KPa.}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 72.0 + 4600.0) / 4600.0)$$

$$= 5393.9517 \text{ KPa.}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 4563.0 \cdot 4606.0 / (1.0 (4744.0 - 4606.0))$$

$$= 152.3071 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 4563.0 / (1.0 \cdot \ln((2 \cdot 69.0 + 4606.0) / 4606.0))$$
$$= 154.5773 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.541 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

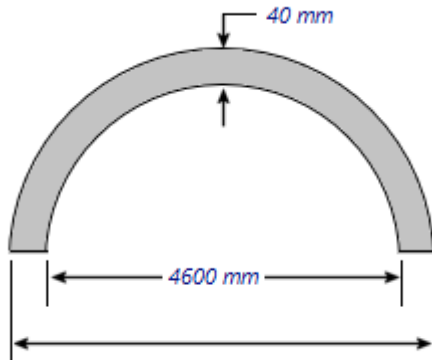
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: TOP dish

Spherical Head Design Information:					
Design Pressure	P	4563.000 KPa.	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		0.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	156.4 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	4600.000 mm	Actual Stress		140.9 N./mm ²
Spec. Min. Thk t _e , or s _a		40.0000 mm	Spec. Nominal Thk.		46.000 mm
Surface Area		345808.7 cm ²	SG of Contents		0.000
Element Volume		25487076.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		12088.1 Kgf	Operating Weight		12088.1 Kgf
			Comp. All. Ope. Stress		103.5 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 57.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 28.750 mm

Material treatment options: +IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2 \cdot (\exp(0.5P/(S \cdot E)) - 1) + c_i + c_o$
 $= 4606.0/2 \cdot (\exp(0.5 \cdot 4563.0/(156.4 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 33.8435 + 3.0 + 0.0 = 36.8435 \text{ mm}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$
 $= 2 \cdot 156.4 \cdot 1.0 \cdot \ln((2 \cdot 37.0 + 4606.0)/4606.0) - 13.0$
 $= 4985.236 - 13.0 = 4972.2358 \text{ KPa.}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D)$
 $= 2 \cdot 175.0 \cdot 1.0 \cdot \ln((2 \cdot 40.0 + 4600.0)/4600.0)$
 $= 6034.2891 \text{ KPa.}$

Computed Stress Intensity at Design Pressure [Si]:

$$\begin{aligned} &= P \cdot D^2 / (E (D_o^2 - D^2)) \\ &= 4563.0 \cdot 4606.0^2 / (1.0 (4680.0^2 - 4606.0^2)) \\ &= 140.884 \text{ N./mm}^2 \end{aligned}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$\begin{aligned} &= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D)) \\ &= 0.5 \cdot 4563.0 / (1.0 \cdot \ln((2 \cdot 37.0 + 4606.0) / 4606.0)) \\ &= 143.153 \text{ N./mm}^2 \end{aligned}$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$	1.485 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$	36.638 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 345808.7 / 3.14159) \\ &= 6635.490 \text{ mm} \end{aligned}$$

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 57.500 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 28.750 mm

Material treatment options: +IT

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Hydrostatic Test Pressure Results:

Test Pressure at high point in vessel per Equation 8.1
 $= \gamma_{St/s} \cdot MAWP(St/S)$
 $= 1.25 \cdot 4603.8042(1.117)$
 $= 6425.212 \text{ KPa.}$

Horizontal Test performed per: $\gamma_{St/s} \cdot MAWP(St/S)$

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
BTM dish	172.8	247.0	0.700	6495.79
shell-1	188.8	247.0	0.765	6495.79
shell-2	191.8	247.0	0.777	6495.79
shell-3	191.8	247.0	0.777	6495.79
shell-4	194.9	247.0	0.789	6495.79
shell-5	194.9	247.0	0.789	6495.79
shell-6	194.9	247.0	0.789	6495.79
shell-7	198.1	247.0	0.802	6495.79
shell-8	198.1	247.0	0.802	6495.79
cone	192.2	247.0	0.778	6495.79
shell-9	199.0	247.0	0.806	6470.30
shell-10	199.0	247.0	0.806	6470.30
shell-11	204.3	247.0	0.827	6470.30
shell-12	204.3	247.0	0.827	6470.30
shell-13	204.3	247.0	0.827	6470.30
shell-14	204.3	247.0	0.827	6470.30
shell-15	209.9	247.0	0.850	6470.30
TOP dish	187.6	247.0	0.760	6470.30

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
C	Nozzle	165.00	147.60	1.118
A	Nozzle	165.00	147.60	1.118
MH-6	Nozzle	165.00	147.60	1.118
MH-5	Nozzle	165.00	147.60	1.118
MH-4	Nozzle	165.00	147.60	1.118
E	Nozzle	138.00	123.60	1.117
P	Nozzle	165.00	147.60	1.118
MH-3	Nozzle	165.00	147.60	1.118
MH-2	Nozzle	165.00	147.60	1.118
D	Nozzle	165.00	147.60	1.118
MH-1	Nozzle	165.00	147.60	1.118
B	Nozzle	165.00	147.60	1.118

Minimum 1.117

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
BTM dish	175.00	156.40	1.119
shell-1	175.00	156.40	1.119
shell-2	175.00	156.40	1.119
shell-3	175.00	156.40	1.119
shell-4	175.00	156.40	1.119
shell-5	175.00	156.40	1.119
shell-6	175.00	156.40	1.119
shell-7	175.00	156.40	1.119
shell-8	175.00	156.40	1.119



cone	175.00	156.40	1.119
shell-9	175.00	156.40	1.119
shell-10	175.00	156.40	1.119
shell-11	175.00	156.40	1.119
shell-12	175.00	156.40	1.119
shell-13	175.00	156.40	1.119
shell-14	175.00	156.40	1.119
shell-15	175.00	156.40	1.119
TOP dish	175.00	156.40	1.119

Minimum 1.119

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
C	33.45	237.50	0.141
A	24.17	237.50	0.102
MH-6	25.61	237.50	0.108
MH-5	25.61	237.50	0.108
MH-4	25.61	237.50	0.108
E	117.45	194.70	0.603
P	13.25	237.50	0.056
MH-3	31.30	237.50	0.132
MH-2	31.30	237.50	0.132
D	30.51	237.50	0.128
MH-1	31.30	237.50	0.132
B	52.48	237.50	0.221

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	9000	...	...	No Calc	No Calc
20	30	9000	...	...	No Calc	No Calc
30	40	9000	...	...	No Calc	No Calc
40	50	9000	...	...	No Calc	No Calc
50	60	No Calc	7336.6	62.3	No Calc	No Calc
60	70	22850	7452	120	No Calc	No Calc
70	80	22850	7448	118	No Calc	No Calc
80	90	22850	7448	118	No Calc	No Calc
90	100	22850	7444	116	No Calc	No Calc
100	110	22850	7444	116	No Calc	No Calc
110	120	22850	7444	116	No Calc	No Calc
120	130	22850	7440	114	No Calc	No Calc
130	140	22850	7440	114	No Calc	No Calc
140	150	2500	7514.47	133.5	No Calc	No Calc
150	160	19916.7	4752	73	No Calc	No Calc
160	170	19916.7	4752	73	No Calc	No Calc
170	180	19916.7	4748	71	No Calc	No Calc
180	190	19916.7	4748	71	No Calc	No Calc
190	200	19916.7	4748	71	No Calc	No Calc
200	210	19916.7	4748	71	No Calc	No Calc
210	220	19916.7	4744	69	No Calc	No Calc
220	230	No Calc	4680	37	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	...	No Calc	...	No Calc
50	60	68.3	No Calc	...	2890.98
60	70	126	No Calc	...	2039.87
70	80	124	No Calc	...	1974.69
80	90	124	No Calc	...	1974.69
90	100	122	No Calc	...	1909.38
100	110	122	No Calc	...	1909.38
110	120	122	No Calc	...	1909.38
120	130	120	No Calc	...	1844.01
130	140	120	No Calc	...	1844.01
140	150	139.5	No Calc	...	4162.26
150	160	76	No Calc	...	1423.27
160	170	76	No Calc	...	1423.27
170	180	74	No Calc	...	1332.26
180	190	74	No Calc	...	1332.26
190	200	74	No Calc	...	1332.26
200	210	74	No Calc	...	1332.26
210	220	72	No Calc	...	1243.87
220	230	40	No Calc	...	2620.01

Minimum

1244

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners	Allowable Length Bet. Stiffeners	Ring Inertia Required	Ring Inertia Available
------	----	----------------------------------	-------------------------------------	--------------------------	---------------------------



		mm	mm	cm^4	cm^4
10	20	9000	No Calc	No Calc	No Calc
20	30	9000	No Calc	No Calc	No Calc
30	40	9000	No Calc	No Calc	No Calc
40	50	9000	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	22850	No Calc	No Calc	No Calc
70	80	22850	No Calc	No Calc	No Calc
80	90	22850	No Calc	No Calc	No Calc
90	100	22850	No Calc	No Calc	No Calc
100	110	22850	No Calc	No Calc	No Calc
110	120	22850	No Calc	No Calc	No Calc
120	130	22850	No Calc	No Calc	No Calc
130	140	22850	No Calc	No Calc	No Calc
140	150	2500	2500	No Calc	No Calc
150	160	19916.7	No Calc	No Calc	No Calc
160	170	19916.7	No Calc	No Calc	No Calc
170	180	19916.7	No Calc	No Calc	No Calc
180	190	19916.7	No Calc	No Calc	No Calc
190	200	19916.7	No Calc	No Calc	No Calc
200	210	19916.7	No Calc	No Calc	No Calc
210	220	19916.7	No Calc	No Calc	No Calc
220	230	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	20120.1	...	19551.7	...	...
20	30	18209.6	...	17356.5	...	...
30	40	18209.6	...	17356.5	...	...
40	50	6069.56	...	5785.17	...	...
50	60	50299.3	97733608	46506.5	98223088	...
60	70	67424.2	122167016	64266.2	122574576	...
70	80	66336	122167016	63177.9	122574576	...
80	90	66336	122167016	63177.9	122574576	...
90	100	65248	122167016	62089.9	122574576	...
100	110	65248	122167016	62089.9	122574576	...
110	120	65248	122167016	62089.9	122574576	...
120	130	42774	81444680	40668.7	81716384	...
130	140	35288.6	67191856	33551.6	67416016	...
140	150	66744.5	69467832	64313	69781616	...
150	160	25957.7	49866016	24949.1	49996180	...
160	170	25957.7	49866016	24949.1	49996180	...
170	180	25263.8	49866016	24255.2	49996180	...
180	190	25263.8	49866016	24255.2	49996180	...
190	200	25263.8	49866016	24255.2	49996180	...
200	210	25263.8	49866016	24255.2	49996180	...
210	220	9418.7	19115306	9032.06	19165204	...
220	230	12088.1	25487076	11314.3	25586936	...
Total		828033	0.139E+10	789246	0.140E+10	0

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Plat	1825.06	2879.96	1999	2879.96	PLAT- SKIRT
20	Plat	1826.66	-2881.22	3000	2881.22	PLAT- SKIRT
30	Wght	800	1000	...	...	skirt manwy/ven
50	Pack	15946.4	...	-1800	...	Pack 1: Btm dis
50	Liqd	103534	...	-1800	...	Liquid: 40
50	Insl	2733.67	...	-1800	...	Ins: 50
50	Nozl	16823.8	...	-3815.85	...	C
60	Plat	7650.71	0.40084E-04	...	...	PLAT-1
60	Pack	6644.32	...	500.005	...	pack#5
60	Liqd	129418	...	1500	...	Liquid: 50
60	Insl	2001.35	...	1500	...	Ins: 50
60	Wght	260	...	0.0052334	...	VB
60	Wght	2770	...	0.0052334	...	btm dish SF wt
60	Wght	5249	-5343	0.0052334	...	C_PP_OVH
70	Plat	3694.2	-2881.85	1000	...	PLAT-2
70	Liqd	129418	...	1500	...	Liquid: 60
70	Insl	2000.29	...	1500	...	Ins: 50
80	Plat	3694.2	2881.85	2000	...	PLAT-3
80	Liqd	101377	...	1175	...	Liquid: 70
80	Insl	2000.29	...	1500	...	Ins: 50
80	Nozl	1204.56	-3800	2100	...	A
80	Wght	3300	...	2099.98	...	FP_A
80	Wght	250	...	2099.98	...	FP_A_Bolting ba
80	Wght	21250	-5267	2099.98	...	A_PP_OVH
90	Pack	8969.83	...	2324.97	...	pack#4
90	Insl	1999.22	...	1500	...	Ins: 50
90	Nozl	1787.64	1900	800.002	3290.9	MH-6
90	Wght	300	...	1649.97	...	B#4_btm TSR
90	Wght	4400	...	1649.97	...	B#4_btm supp de
100	Plat	3692.59	-2881.85	0.0011613	...	PLAT-4
100	Pack	17607.5	...	1325.02	...	pack#4
100	Insl	1999.22	...	1500	...	Ins: 50



100	Wght	940	...	2649.98	...	B#4_top HDG
100	Wght	5010	...	2649.98	...	B#4_top dist. d
100	Wght	300	...	2649.98	...	B#4_top dist. T
110	Plat	3692.59	2881.85	1000	...	PLAT-5
110	Pack	8305.4	...	2374.97	...	pack#3
110	Insl	1999.22	...	1500	...	Ins: 50
110	Nozl	1783.79	1900	1400	3290.9	MH-5
110	Wght	300	...	1749.97	...	B#3_btm TSR
110	Wght	4400	...	1749.97	...	B#3_btm supp de
110	Wght	3250	...	1699.98	...	Nozzle wt
110	Wght	10000	...	1699.98	...	10T margin
110	Wght	14300	...	1699.98	...	Cleat wt
110	Wght	27000	...	1699.98	...	Lifting wts
120	Pack	13288.6	...	1000.03	...	pack#3
120	Insl	1332.11	...	1000	...	Ins: 50
130	Plat	4416.49	-2881.85	...	...	PLAT-6
130	Pack	4983.26	...	375.025	...	pack#3
130	Insl	1098.99	...	825	...	Ins: 50
130	Nozl	1783.79	1900	600.003	3290.9	MH-4
130	Wght	940	...	749.975	...	B#3_top HDG
130	Wght	6000	...	749.975	...	B#3_top distri.
130	Wght	300	...	749.975	...	B#3_top dist. T
140	Plat	3706.62	2881.85	2350	...	PLAT-7
140	Insl	1792.55	...	1250	...	Ins: 50
140	Nozl	2407.87	-3218.62	973.803	...	E
140	Wght	56821	-5343	973.778	...	E_PP_OVH
140	Wght	3150	...	973.778	...	FP_E
140	Wght	250	...	973.778	...	FP_E_Bolting ba
140	Wght	1720	...	973.778	...	FP_E_insert plt
150	Pack	6873.36	...	1649.98	...	pack#2
150	Insl	1283.92	...	1500	...	Ins: 50
150	Nozl	11.3106	-2400	150.005	...	P
150	Wght	200	...	299.977	...	B#2_btm TSR
150	Wght	1800	...	299.977	...	B#2_btm supp de
160	Plat	2613.75	-2065.11	850.002	...	PLAT-8
160	Pack	7637.04	...	1500.02	...	pack#2
160	Insl	1283.92	...	1500	...	Ins: 50
170	Plat	2612.15	2065.11	1850	...	PLAT-9
170	Pack	381.871	...	75.0268	...	pack#2
170	Pack	1909.24	...	2624.97	...	pack#1
170	Insl	1282.86	...	1500	...	Ins: 50
170	Nozl	1543.71	-1767.77	1500.01	1767.77	MH-3
170	Wght	300	...	149.977	...	B#2_top HDG
170	Wght	1850	...	149.977	...	B#2_top distri.
170	Wght	200	...	149.977	...	B#2_top dis. TA
170	Wght	200	...	2249.98	...	B#1_btm TSR
170	Wght	1800	...	2249.98	...	B#1_btm supp de
180	Plat	2612.15	-2065.11	2850	...	PLAT-10
180	Pack	7637.04	...	1500.01	...	pack#1
180	Insl	1282.86	...	1500	...	Ins: 50
190	Pack	5345.97	...	1050.02	...	pack#1
190	Insl	1282.86	...	1500	...	Ins: 50
190	Wght	300	...	2099.98	...	B#1_top HDG
190	Wght	2260	...	2099.98	...	B#1_top distri.
190	Wght	200	...	2099.98	...	B#1_top dis. TA
200	Plat	2612.15	2065.11	2850	...	PLAT-11
200	Insl	1282.86	...	1500	...	Ins: 50
200	Nozl	1543.71	1250	1000	2165.06	MH-2
200	Nozl	463.882	-2500	500.007	...	D
200	Tray	8292.41	...	2199.98	...	Tray#1-2-3
200	Wght	1000	...	699.976	...	FP_D
200	Wght	150	...	699.976	...	FP_D_Bolting ba
200	Wght	17147	-3779	499.979	...	D_PP_OVH
210	Insl	491.355	...	575	...	Ins: 50
210	Nozl	1546.59	1250	300	2165.06	MH-1
210	Wght	410	...	849.981	...	Demister dead
210	Wght	1400	...	849.981	...	Demister TA
210	Wght	1210	...	849.981	...	Tray Truss
210	Wght	900	2300	499.979	...	davit
210	Wght	17550	-3841	574.979	...	B_PP_OVH



220	Plat	11848	...	2500.01	...
220	Insl	1127.77	...	1150	...
220	Nozl	744	...	2480	...
220	Wght	1050	...	...	...
220	Wght	3000	-4000	2300	...
220	Wght	3000	-3758.77	2300	1368.08
220	Forc	...	...	2300	...
90	Pliq	14559.53	...	2324.975	...
100	Pliq	28579.81	...	1325.023	...
110	Pliq	13481.04	...	2374.974	...
120	Pliq	21569.67	...	1000.026	...
130	Pliq	8088.66	...	375.025	...
150	Pliq	11885.79	...	1649.977	...
160	Pliq	13206.39	...	1500.021	...
170	Pliq	660.35	...	75.027	...
170	Pliq	3301.57	...	2624.974	...
180	Pliq	13206.39	...	1500.014	...
190	Pliq	9244.55	...	1050.024	...

TOP PLT
 Ins: 50
 B
 top dish SF wt
 PSV-1
 PSV-2
 B_FxFz

Note: The individual tray liquid weights are listed below, but these weights are included in the tray weights above.

From	Tliq	Weight	X Offset	Y Offset
200	Tliq	0.00	0.000	2199.977

Total Weight of Each Detail Type:

Platforms	56497.3
Packing	105529.8
Packing Liquid	137783.8
Liquid	463746.2
Insulation	28275.3
Nozzles	31644.7
Trays	7482.4
Tray Bolting Weight	810.0
Weights	229187.0

Sum of the Detail Weights 1060956.4 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	828033.1	828033.1	828033.1	828033.1	828033.1	828033.1
Nozzles	31644.7	31644.7	31644.7	31644.7	31644.7	31644.7
Tray Bar	810.0	810.0	810.0	810.0	810.0	810.0
Wld Weights	73550.0	73550.0	73550.0	73550.0	73550.0	73550.0
Platforms	...	...	56497.3	56497.3	56497.3	56497.3
Packing	...	...	105529.8	105529.8	105529.8	105529.8
Insulation	...	...	28275.3	28275.3	28275.3	28275.3
Trays	...	...	7482.4	7482.4	7482.4	7482.4
Empty Weights	...	...	...	155637.0	155637.0	...
Test Weights	...	155637.0	...	...	...	...
Ope Weights	...	...	...	...	...	155637.0
Ope. Liquid	...	...	...	...	...	463746.2
Packing Liq	...	...	...	...	...	137783.8
Test Liquid	...	1391788.4	...	...	...	...
Totals	934037.8	2481463.2	1131822.6	1287459.5	1287459.4	1888989.5

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	934037.8 kg.
Shop Test Wt.	- Fabricated weight + water (full)	2481463.2 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	1131822.6 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	1287459.5 kg.



Op. Wt. no Liq - Fab. weight + internals. + details + weights 1287459.4 kg.
 Operating Wt. - Empty weight + operating liq. uncorroded 1888989.5 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating) 323378.8 kg.
 Mass of the Upper 1/3 of the Vertical Vessel (filled w/water) 588758.6 kg.
 Mass of the Upper 1/3 of the Vertical Vessel (empty) 284745.2 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	462819	458296
20	30	694229	687820
30	40	694229	687820
40	50	231410	229273
50	60	849970	814301
60	70	702334	678584
70	80	701958	678584
80	90	701958	678584
90	100	701580	678584
100	110	701580	678584
110	120	701580	678584
120	130	467469	452389
130	140	385662	373221
140	150	554219	522291
150	160	447866	433540
160	170	447866	433540
170	180	447489	433540
180	190	447489	433540
190	200	447489	433540
200	210	447489	433540
210	220	171393	166190
220	230	345809	332381

Total 11753885.0 11396724.0 cm²
 Total 12651.8 12267.3 ft²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	21945.1	21945.1	21945.1	5256.2	21945.1
20	30	20036.3	20036.3	20036.3	5263.12	20036.3
30	40	19009.6	19009.6	19009.6	800.016	19009.6
40	50	6069.56	6069.56	6069.56	...	6069.56
50	60	85803.1	189337	167531	...	85803.1
60	70	91999.6	221417	207448	28046	91999.6
70	80	72030.5	201448	194123	10646.3	72030.5
80	90	98035.1	199412	220128	105857	98035.1
90	100	82704.7	97264.2	195827	3396.58	82704.7
100	110	94797.3	123377	199282	10641.7	94797.3
110	120	140279	153760	254066	14031	140279
120	130	57394.8	78964.4	125501	...	57394.8
130	140	54811.1	62899.7	116979	9338.65	54811.1
140	150	136593	136593	206018	300669	136593
150	160	36126.3	48012.1	79088.5	27.146	36126.3
160	170	37492.4	50698.8	79691	5397.79	37492.4
170	180	37343.6	41305.5	84888.1	2665.51	37343.6
180	190	36795.8	50002.2	78994.4	5394.48	36795.8
190	200	34652.6	43897.2	79142.2	...	34652.6
200	210	57755.8	57755.8	107591	58635.4	57755.8
210	220	32926.6	32926.6	52030.3	63407.6	32926.6
220	230	32857.8	32857.8	58329.4	23276.8	32857.8



Empty Support Force + the Sum of the Y forces 1155730.25 Kgf
 Operating Support Force + the Sum of the Y forces 1895043.62 Kgf
 Hydro Support Force + the Sum of the Y forces 2579772.50 Kgf

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	1287459	1888989	2573718
20	30	1265514	1867044	2551773
30	40	1245478	1847007	2531736
40	50	1226468	1827998	2512727
50	60	1220398	1821928	2506657
60	70	1134595	1632591	2339127
70	80	1042596	1411174	2131679
80	90	970566	1209726	1937556
90	100	872530	1010314	1717428
100	110	789826	913050	1521601
110	120	695028	789673	1322319
120	130	554750	635913	1068253
130	140	497355	556948	942752
140	150	442544	494049	825773
150	160	305951	357456	619755
160	170	269825	309444	540667
170	180	232332	258745	460976
180	190	194989	217440	376088
190	200	158193	167437	297093
200	210	123540	123540	217951
210	220	65784.5	65784.5	110360
220	230	32857.8	32857.8	58329.4

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	652750	652750	652750
20	30	647494	647494	647494
30	40	642230	642230	642230
40	50	641431	641431	641431
50	60	641431	641431	641431
60	70	641431	641431	641431
70	80	613385	613385	613385
80	90	602738	602738	602738
90	100	496881	496881	496881
100	110	493485	493485	493485
110	120	482843	482843	482843
120	130	468812	468812	468812
130	140	468812	468812	468812
140	150	459473	459473	459473
150	160	158805	158805	158805
160	170	158777	158777	158777
170	180	153380	153380	153380
180	190	150714	150714	150714
190	200	145320	145320	145320
200	210	145320	145320	145320
210	220	86684.3	86684.3	86684.3
220	230	23276.8	23276.8	23276.8

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
C	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
A	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
MH-6	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
MH-5	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
MH-4	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
E	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
P	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
MH-3	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
MH-2	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
D	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
MH-1	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
B	9142.1	10200.0	130	600	GR 1.1	...	...	...	...
Min Rating	9142.1	10200.0, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-2, 4.16.12 for more information.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 1.61307 Hz.

The Natural Frequency for the Vessel (Ope...) is 1.5002 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. Kgf	X,Z Moment and For Res Kg-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	...	...
190	200	...	...
200	210	...	...
210	220	...	...
220	230	6184.62	25643.3

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. Kgf	X,Z Moment and For Res Kg-m.
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	...	...
190	200	...	...
200	210	...	...
210	220	...	...
220	230	6184.62	25643.3

User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
220	2300.00	4373.	-6054.	4373.	8074.		8074.

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.165 \cdot 0.879) / (1 + 1.7 \cdot 3.4 \cdot 0.165)))$
 $= \min(0.85, 0.87)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	1.500 Hz
Natural Frequency of Vessel (Empty)	1.613 Hz
Natural Frequency of Vessel (Test)	1.196 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.609
Structure Height to Diameter ratio	8.555
Height to top of Structure	54719.992 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Zg = 274320.0$ mm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1000.0 + 0.0 = 1000.0$ mm
 $= 3.281$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (3.281 ft.) < 15 ft.
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15 / 274320.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:



K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [78.12] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.609 \cdot 190.289$$

$$= 1575.1 \text{ lbs. } [714.5] \text{ Kgf}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Btm skirt-1	1000.0	0.000	0.000	0.000	0.849	1.000	78.120
Btm skirt-2	3500.0	0.000	0.000	0.000	0.849	1.000	78.120
Btm skirt-3	6500.0	0.000	0.000	0.000	0.914	1.000	78.120
Top skirt	8500.0	0.000	0.000	0.000	0.967	1.000	78.120
BTM dish	9000.0	0.000	0.000	0.000	0.979	1.000	78.120
shell-1	10500.0	0.000	0.000	0.000	1.011	1.000	78.120
shell-2	13500.0	0.000	0.000	0.000	1.066	1.000	78.120
shell-3	16500.0	0.000	0.000	0.000	1.112	1.000	78.120
shell-4	19500.0	0.000	0.000	0.000	1.152	1.000	78.120
shell-5	22500.0	0.000	0.000	0.000	1.187	1.000	78.120
shell-6	25500.0	0.000	0.000	0.000	1.219	1.000	78.120
shell-7	28000.0	0.000	0.000	0.000	1.243	1.000	78.120
shell-8	29825.0	0.000	0.000	0.000	1.260	1.000	78.120
cone	31808.2	0.000	0.000	0.000	1.277	1.000	78.120
shell-9	34650.0	0.000	0.000	0.000	1.300	1.000	78.120
shell-10	37650.0	0.000	0.000	0.000	1.323	1.000	78.120
shell-11	40650.0	0.000	0.000	0.000	1.345	1.000	78.120
shell-12	43650.0	0.000	0.000	0.000	1.365	1.000	78.120
shell-13	46650.0	0.000	0.000	0.000	1.384	1.000	78.120
shell-14	49650.0	0.000	0.000	0.000	1.403	1.000	78.120
shell-15	51725.0	0.000	0.000	0.000	1.415	1.000	78.120
TOP dish	53327.1	0.000	0.000	0.000	1.424	1.000	78.120

Note:

The height/(avg. diameter) ratio [8.0], was less than or equal to 15,
 therefore no wind vibration calculations have been performed.

Platform Load Calculations

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf	Cf
PLAT- SKIRT	15000.00	1999.00	78.12	117.18	1.00
PLAT- SKIRT	15000.00	5000.00	78.12	117.18	1.00
PLAT-1	30000.00	9000.00	78.12	234.36	1.00
PLAT-2	30000.00	13000.00	78.12	234.36	1.00
PLAT-3	30000.00	17000.00	78.12	234.36	1.00
PLAT-4	30000.00	21000.00	78.12	234.36	1.00
PLAT-5	30000.00	25000.00	78.12	234.36	1.00
PLAT-6	30000.00	29000.00	78.12	234.36	1.00
PLAT-7	30000.00	33000.00	78.12	234.36	1.00
PLAT-8	30000.00	37000.00	78.12	234.36	1.00
PLAT-9	30000.00	41000.00	78.12	234.36	1.00
PLAT-10	30000.00	45000.00	78.12	234.36	1.00
PLAT-11	30000.00	51000.00	78.12	234.36	1.00
TOP PLT	80000.00	54799.97	78.12	624.96	1.00

Wind Loads on Masses/Equipment/Piping



ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
skirt manwy/v	0.00	5000.00	78.12	0.00
VB	0.00	9000.00	78.12	0.00
btm dish SF w	0.00	9000.00	78.12	0.00
CPPOVH	0.00	9000.00	78.12	0.00
FPA	0.00	17099.97	78.12	0.00
FPABolting	0.00	17099.97	78.12	0.00
APPOVH	0.00	17099.98	78.12	0.00
B#4btm TSR	0.00	19649.98	78.12	0.00
B#4btm supp	0.00	19649.98	78.12	0.00
B#4top HDG	0.00	23649.98	78.12	0.00
B#4top dist.	0.00	23649.98	78.12	0.00
B#4top dist.	0.00	23649.98	78.12	0.00
B#3btm TSR	0.00	25749.98	78.12	0.00
B#3btm supp	0.00	25749.98	78.12	0.00
Nozzle wt	0.00	25699.98	78.12	0.00
10T margin	0.00	25699.98	78.12	0.00
Cleat wt	0.00	25699.98	78.12	0.00
Lifting wts	0.00	25699.98	78.12	0.00
B#3top HDG	0.00	29749.98	78.12	0.00
B#3top distr	0.00	29749.98	78.12	0.00
B#3top dist.	0.00	29749.98	78.12	0.00
EPPOVH	0.00	31623.78	78.12	0.00
FPE	0.00	31623.78	78.12	0.00
FPEBolting	0.00	31623.78	78.12	0.00
FPEinsert p	0.00	31623.78	78.12	0.00
B#2btm TSR	0.00	33449.98	78.12	0.00
B#2btm supp	0.00	33449.98	78.12	0.00
B#2top HDG	0.00	39299.98	78.12	0.00
B#2top distr	0.00	39299.98	78.12	0.00
B#2top dis.	0.00	39299.98	78.12	0.00
B#1btm TSR	0.00	41399.98	78.12	0.00
B#1btm supp	0.00	41399.98	78.12	0.00
B#1top HDG	0.00	47249.97	78.12	0.00
B#1top distr	0.00	47249.97	78.12	0.00
B#1top dis.	0.00	47249.97	78.12	0.00
FPD	0.00	48849.97	78.12	0.00
FPDBolting	0.00	48849.97	78.12	0.00
DPPOVH	0.00	48649.97	78.12	0.00
Demister dead	0.00	51999.97	78.12	0.00
Demister TA	0.00	51999.97	78.12	0.00
Tray Truss	0.00	51999.97	78.12	0.00
davit	0.00	51649.97	78.12	0.00
BPPOVH	0.00	51724.97	78.12	0.00
top dish SF w	0.00	52299.96	78.12	0.00
PSV-1	0.00	54599.96	78.12	0.00
PSV-2	0.00	54599.96	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 1.5002 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1000	8839.2	176784	78.12	1043.59
20	30	3500	8839.2	265176	78.12	1363.02
30	40	6500	8839.2	265176	78.12	1178.88
40	50	8500	8839.2	88392	78.12	392.959
50	60	8999.98	8995.92	2.74196	78.12	0.01219
60	70	10500	9134.4	274032	78.12	1439.22
70	80	13500	9129.6	273888	78.12	1465.86
80	90	16500	9129.6	273888	78.12	1483.22
90	100	19500	9124.8	273744	78.12	1216.97
100	110	22500	9124.8	273744	78.12	1457.58
110	120	25500	9124.8	273744	78.12	1469.71



120	130	28000	9120	182400	78.12	810.884
130	140	29825	9120	150480	78.12	919.645
140	150	31808.2	7606.8	190170	78.12	1112.88
150	160	34650	6041.76	181253	78.12	805.784
160	170	37650	6041.76	181253	78.12	1059.13
170	180	40650	6036.84	181105	78.12	1065.14
180	190	43650	6036.84	181105	78.12	1070.9
190	200	46650	6036.84	181105	78.12	805.128
200	210	49650	6036.84	181105	78.12	1069.93
210	220	51725	6031.92	69367.1	78.12	308.381
220	230	53327.1	5953.2	113152	78.12	1209.48

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\
 SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(179.13380.75)$
 = 0.979 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/1.5)
 = 0.667

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.979$, 0.667) per 12.8.2
 = 0.667

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 = $Sds / (R / I)$
 = $0.157 / (1.25 / 1.25)$
 = 0.157

Maximum value of C_s per equation 12.8-3 [Cs]:
 = $Sd1 / (T(R / I_e))$
 = $0.104 / (0.667(1.25 / 1.25))$
 = 0.156

Minimum value of C_s per equation 15.4-1 [Cs]:
 = $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
 = $\max((0.044 \cdot 0.1568 \cdot 1.25), 0.030)$
 = 0.030

Total Base Shear [V]:
 = $C_s \cdot W$ (Equation 12.8-1):
 = $0.156 \cdot 1888989$
 = 294722.094 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.157 \cdot 1888989$



= 59238.7 Kgf

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x k / (\text{Sum of } W_i * h_i k)$) and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.0833

The Natural Frequency for the Vessel (Ope...) is 1.5002 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	1000	21945.1	114	108.799	53184.3
20	30	3500	20036.3	404.36	374.8	52496.1
30	40	6500	19009.6	750.173	685.367	51867.8
40	50	8500	6069.56	320.298	290.803	51271.6
50	60	8999.98	189337	10629.7	4367.74	51081.3
60	70	10500	221417	14690	5514.44	45143.7
70	80	13500	201448	17547.2	5635.36	38200
80	90	16500	199412	21587.7	9487.75	31882.6
90	100	19500	97264.2	12618.3	9554.62	25629.1
100	110	22500	123377	18689.9	12745.4	22578.8
110	120	25500	153760	26674.7	21536.1	18709.7
120	130	28000	78964.4	15159.6	9729.74	13887.8
130	140	29825	62899.7	12930.4	9934.92	11411.5
140	150	31900	136593	30201.9	26587.9	9438.97
150	160	34650	48012.1	11610.8	7676.22	5155.43
160	170	37650	50698.8	13414.5	8699.45	3649.77
170	180	40650	41305.5	11875.5	9398.46	2059.85
180	190	43650	50002.2	15528.6	9986.59	764.508
190	200	46650	43897.2	14650.5	10091.4	-803.561
200	210	49650	57755.8	20622.1	17968.2	-2180.18
210	220	51725	32926.6	12289.9	10698	-3991.4
220	230	52300	32857.8	12411.9	10801.5	-5023.98



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas Kgf	Cumulative Shr Eqk Cas Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	6184.62	6184.62	349105	349105
20	30	3500	6184.62	6184.62	336736	336736
30	40	6500	6184.62	6184.62	318182	318182
40	50	8500	6184.62	6184.62	299627	299627
50	60	8999.98	6184.62	6184.62	293443	293443
60	70	10500	6184.62	6184.62	293443	293443
70	80	13500	6184.62	6184.62	274888	274888
80	90	16500	6184.62	6184.62	256334	256334
90	100	19500	6184.62	6184.62	237780	237780
100	110	22500	6184.62	6184.62	219226	219226
110	120	25500	6184.62	6184.62	200671	200671
120	130	28000	6184.62	6184.62	182117	182117
130	140	29825	6184.62	6184.62	169748	169748
140	150	31900	6184.62	6184.62	159543	159543
150	160	34650	6184.62	6184.62	144081	144081
160	170	37650	6184.62	6184.62	125527	125527
170	180	40650	6184.62	6184.62	106972	106972
180	190	43650	6184.62	6184.62	88418.2	88418.2
190	200	46650	6184.62	6184.62	69864	69864
200	210	49650	6184.62	6184.62	51309.8	51309.8
210	220	51725	6184.62	6184.62	32755.5	32755.5
220	230	52300	6184.62	6184.62	25643.3	25643.3

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	1000	22748.3	294722	584810	9045499
20	30	3500	21704.7	294608	540357	8456157
30	40	6500	20341.7	294204	477286	7572921
40	50	8500	19162.8	293454	418028	6691418
50	60	8999.98	18769.8	293133	399062	6398127
60	70	10500	18769.8	282504	399061	6398119
70	80	13500	17330.6	267814	344909	5572627
80	90	16500	15864.8	250266	295115	4795492
90	100	19500	14381.5	228679	249745	4077060
100	110	22500	13164.6	216060	208425	3409938
110	120	25500	11707	197370	171117	2789779
120	130	28000	10237.3	170696	138200	2237669
130	140	29825	9426.4	155536	118536	1911431
140	150	31900	8506.76	142606	103741	1665459
150	160	34650	7393.88	112404	83864.4	1346691
160	170	37650	6588.09	100793	62891.1	1026890
170	180	40650	5528.96	87378.6	44715.1	744627
180	190	43650	4463.82	75503	29725.7	500300
190	200	46650	3392.92	59974.4	17940.3	297080
200	210	49650	2587.79	45323.9	8969.08	139129
210	220	51725	1517.86	24701.8	2810.48	34088.9
220	230	52300	1209.48	12411.9	1242.27	12748.5

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear Kg	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	6184.62	0.0015487	0.0061581	0.61213E-05
20	30	6184.62	0.018889	0.038587	0.15410E-04
30	40	6184.62	0.065053	0.098091	0.24172E-04
40	50	6184.62	0.11053	0.12367	0.26976E-04
50	60	6184.62	0.12367	0.12367	0.26976E-04
60	70	6184.62	0.165	0.208	0.29222E-04
70	80	6184.62	0.25266	0.29891	0.31358E-04
80	90	6184.62	0.34671	0.396	0.33345E-04
90	100	6184.62	0.44674	0.49888	0.35216E-04
100	110	6184.62	0.55237	0.60715	0.36935E-04
110	120	6184.62	0.66316	0.72034	0.38503E-04
120	130	6184.62	0.7591	0.79834	0.39481E-04
130	140	6184.62	0.83107	0.86411	0.40236E-04
140	150	6184.62	0.91492	0.96674	0.41839E-04
150	160	6184.62	1.03212	1.10252	0.48531E-04
160	170	6184.62	1.1776	1.257	0.54302E-04
170	180	6184.62	1.34045	1.42763	0.59295E-04
180	190	6184.62	1.51821	1.61182	0.63340E-04
190	200	6184.62	1.70811	1.80672	0.66436E-04
200	210	6184.62	1.9073	2.00949	0.68584E-04
210	220	6184.62	2.04902	2.08871	0.69174E-04
220	230	6184.62	2.08872	2.08872	0.69174E-04

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 261.500 Actual deflection : 2.089 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kg	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	22748.3	0.0026089	0.010301	0.10167E-04
20	30	21704.7	0.031256	0.063165	0.24774E-04
30	40	20341.7	0.10536	0.15719	0.37625E-04
40	50	19162.8	0.1765	0.19681	0.41595E-04
50	60	18769.8	0.19681	0.19681	0.41595E-04
60	70	18769.8	0.26039	0.32624	0.44614E-04
70	80	17330.6	0.3942	0.46415	0.47257E-04
80	90	15864.8	0.53592	0.60938	0.49506E-04
90	100	14381.5	0.68441	0.76088	0.51432E-04
100	110	13164.6	0.83866	0.91764	0.53026E-04
110	120	11707	0.99771	1.07874	0.54325E-04
120	130	10237.3	1.13326	1.18814	0.55058E-04
130	140	9426.4	1.23368	1.27943	0.55581E-04
140	150	8506.76	1.34925	1.4197	0.56597E-04
150	160	7393.88	1.50613	1.59536	0.60331E-04
160	170	6588.09	1.68699	1.78068	0.63062E-04
170	180	5528.96	1.8761	1.97297	0.65004E-04
180	190	4463.82	2.07102	2.17	0.66241E-04
190	200	3392.92	2.26968	2.36988	0.66938E-04
200	210	2587.79	2.47044	2.57122	0.67233E-04
210	220	1517.86	2.60989	2.64856	0.67274E-04
220	230	1209.48	2.64856	2.64857	0.67274E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	238.048	1487.8
20	30	238.048	1487.8
30	40	238.048	1487.8
40	50	238.048	1487.8
50	60	242.268	1514.18
60	70	245.998	1537.48
70	80	245.868	1536.68
80	90	245.868	1536.68
90	100	245.739	1535.87
100	110	245.739	1535.87
110	120	245.739	1535.87
120	130	245.61	1535.06
130	140	245.61	1535.06
140	150	204.858	1280.36
150	160	162.71	1016.94
160	170	162.71	1016.94
170	180	162.578	1016.11
180	190	162.578	1016.11
190	200	162.578	1016.11
200	210	162.578	1016.11
210	220	162.445	1015.28
220	230	160.325	1002.03

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 261.500 Actual deflection : 2.649 mm

Total Deflection in the Operating Condition + Applied Forces:
 Allowable deflection : 261.500 Actual deflection : 4.737 mm

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	8290.05	7923.36	1511763072	1445523200
20	30	7831.6	7464.67	1428935424	1362578944
30	40	7831.6	7464.67	1428935424	1362578944
40	50	7831.6	7464.67	1428935424	1362578944
50	60	15595.7	14237.4	2.808E+09	2.567E+09
60	70	28999.3	27641	5.223E+09	4.986E+09
70	80	28531.2	27172.9	5.139E+09	4.902E+09
80	90	28531.2	27172.9	5.139E+09	4.902E+09
90	100	28063.3	26705	5.054E+09	4.817E+09
100	110	28063.3	26705	5.054E+09	4.817E+09
110	120	28063.3	26705	5.054E+09	4.817E+09
120	130	27595.8	26237.5	4.970E+09	4.733E+09
130	140	27595.8	26237.5	4.970E+09	4.733E+09
140	150	36342	34810.9	6.548E+09	6.283E+09
150	160	11164.5	10730.7	1284593024	1236237056
160	170	11164.5	10730.7	1284593024	1236237056
170	180	10866	10432.2	1250217088	1201820672
180	190	10866	10432.2	1250217088	1201820672
190	200	10866	10432.2	1250217088	1201820672
200	210	10866	10432.2	1250217088	1201820672
210	220	10567.8	10134	1215876992	1167439744
220	230	5830.8	5396.99	670641792	621541952

Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	113.099	246.996	-89.9194	-91.2212
20	30	113.099	246.996	-88.1668	-89.5805
30	40	113.099	246.996	-88.1668	-89.5805
40	50	86.02	247	-88.1668	-89.5805
50	60	156.4	247	-105.078	-107.032
60	70	156.4	247	-116.595	-117.273
70	80	156.4	247	-116.362	-117.061
80	90	156.4	247	-116.362	-117.061
90	100	156.4	247	-116.122	-116.843
100	110	156.4	247	-116.122	-116.843
110	120	156.4	247	-116.122	-116.843
120	130	156.4	247	-115.875	-116.618
130	140	156.4	247	-115.875	-116.618
140	150	156.4	247	-119.406	-119.905
150	160	156.4	247	-115.913	-116.497
160	170	156.4	247	-115.913	-116.497
170	180	156.4	247	-115.511	-116.125
180	190	156.4	247	-115.511	-116.125
190	200	156.4	247	-115.511	-116.125
200	210	156.4	247	-115.511	-116.125
210	220	156.4	247	-115.088	-115.733
220	230	156.4	247	-103.489	-105.219



Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have been computed in the new and cold condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	...	...	...
50	60	130.932	...	167.75
60	70	67.4409	...	90.2153
70	80	68.6026	...	91.6953
80	90	68.6026	...	91.6953
90	100	69.8046	...	93.2242
100	110	69.8046	...	93.2242
110	120	69.8046	...	93.2242
120	130	71.0485	...	94.8037
130	140	71.0485	...	94.8037
140	150	60.3832	...	81.1389
150	160	70.8577	...	95.649
160	170	70.8577	...	95.649
170	180	72.8849	...	98.2762
180	190	72.8849	...	98.2762
190	200	72.8849	...	98.2762
200	210	72.8849	...	98.2762
210	220	75.0295	...	101.049
220	230	140.884	...	183.143

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-15.935	-23.3802	...	4.42835	4.42835
20	30	-16.6259	-24.5286	...	4.66009	4.66009
30	40	-16.3627	-24.2654	...	4.62221	4.62221
40	50	-16.1129	-24.0156	...	4.61645	4.61645
50	60	-8.40622	-9.35528	...	2.45008	2.45008
60	70	-4.02546	-4.51431	...	1.2615	1.2615
70	80	-3.76277	-4.26004	...	1.22714	1.22714
80	90	-3.50281	-4.00008	...	1.20584	1.20584
90	100	-3.20418	-3.71016	...	1.0115	1.0115
100	110	-2.90046	-3.35297	...	1.00459	1.00459
110	120	-2.55234	-2.8999	...	0.98292	0.98292
120	130	-2.0735	-2.37686	...	0.97138	0.97138
130	140	-1.85897	-2.08172	...	0.97138	0.97138
140	150	-1.24672	-1.39182	...	0.71716	0.71716
150	160	-2.79611	-3.26682	...	1.25974	1.25974
160	170	-2.46595	-2.82803	...	1.25953	1.25953
170	180	-2.18405	-2.43235	...	1.25155	1.25155
180	190	-1.833	-2.04405	...	1.2298	1.2298
190	200	-1.4871	-1.574	...	1.18578	1.18578
200	210	-1.16135	-1.16135	...	1.18578	1.18578
210	220	-0.63661	-0.63661	...	0.72816	0.72816
220	230	-0.59706	-0.59706	...	0.36726	0.36726

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:



From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	...	3.96744	61.366	...	...
20	30	...	3.88901	60.8599	...	...
30	40	...	3.43508	54.5032	...	...
40	50	...	3.00859	48.1589	...	...
50	60	...	1.5243	24.439	...	...
60	70	...	0.78483	12.5832	...	...
70	80	...	0.69003	11.1487	...	...
80	90	...	0.59041	9.59391	...	...
90	100	...	0.50841	8.29967	...	...
100	110	...	0.42429	6.94161	...	...
110	120	...	0.34834	5.67915	...	...
120	130	...	0.28635	4.63648	...	...
130	140	...	0.24561	3.96051	...	...
140	150	...	0.16192	2.59949	...	...
150	160	...	0.66527	10.6828	...	...
160	170	...	0.49889	8.14596	...	...
170	180	...	0.36487	6.07602	...	...
180	190	...	0.24256	4.08236	...	...
190	200	...	0.14639	2.42412	...	...
200	210	...	0.073186	1.13527	...	...
210	220	...	0.023608	0.28635	...	...
220	230	...	0.0196	0.20114	...	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	45.3262
20	30	...	...	...	45.1803
30	40	...	...	...	40.8278
40	50	...	...	...	36.4869
50	60	...	...	...	18.5985
60	70	...	...	...	9.57603
70	80	...	...	...	8.58105
80	90	...	...	...	7.45439
90	100	...	...	...	6.48491
100	110	...	...	...	5.44282
110	120	...	...	...	4.46884
120	130	...	...	...	3.66374
130	140	...	...	...	3.13865
140	150	...	...	...	2.06331
150	160	...	...	...	8.53075
160	170	...	...	...	6.59161
170	180	...	...	...	4.98615
180	190	...	...	...	3.41344
190	200	...	...	...	2.078
200	210	...	...	...	0.98832
210	220	...	...	...	0.24921
220	230	...	...	...	0.17505

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	-0.074936	0.65827
20	30	-0.079541	0.68968
30	40	-0.079541	0.68142
40	50	-0.079541	0.67359
50	60	-0.041703	0.35185



60	70	-0.021481	0.16017
70	80	-0.021851	0.13787
80	90	-0.021851	0.11507
90	100	-0.022233	0.094117
100	110	-0.022233	0.082916
110	120	-0.022233	0.068707
120	130	-0.02263	0.051909
130	140	-0.02263	0.042653
140	150	-0.017056	0.026591
150	160	-0.055332	0.047116
160	170	-0.055332	0.033355
170	180	-0.056915	0.019364
180	190	-0.056915	0.0071868
190	200	-0.056915	-0.0075539
200	210	-0.056915	-0.020495
210	220	-0.058589	-0.038625
220	230	-0.11001	-0.09129

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	2.36838	2.36838	2.26461	2.26461
20	30	2.42353	2.42353	2.31098	2.31098
30	40	2.28999	2.28999	2.18365	2.18365
40	50	2.15645	2.15645	2.05631	2.05631
50	60	1.12087	1.12087	1.02492	1.02492
60	70	0.57711	0.57711	0.55097	0.55097
70	80	0.54994	0.54994	0.52461	0.52461
80	90	0.51282	0.51282	0.4892	0.4892
90	100	0.48405	0.48405	0.46136	0.46136
100	110	0.44628	0.44628	0.42536	0.42536
110	120	0.40851	0.40851	0.38936	0.38936
120	130	0.37735	0.37735	0.35935	0.35935
130	140	0.35172	0.35172	0.33495	0.33495
140	150	0.24902	0.24902	0.23896	0.23896
150	160	1.14294	1.14294	1.09992	1.09992
160	170	0.99576	0.99576	0.95828	0.95828
170	180	0.87288	0.87288	0.83909	0.83909
180	190	0.72148	0.72148	0.69355	0.69355
190	200	0.57008	0.57008	0.54801	0.54801
200	210	0.41868	0.41868	0.40247	0.40247
210	220	0.27515	0.27515	0.26419	0.26419
220	230	0.4046	0.4046	0.37498	0.37498

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...
140	150	...
150	160	...

160	170	...
170	180	...
180	190	...
190	200	...
200	210	...
210	220	...
220	230	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Un-Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

For Division 2, the overall stress intensity is computed in accordance with the Section VIII Division II Example problem manual PTB-3, example 4.3.6, stress due to combined loads.
 The stresses are combined per equation 4.3.44 to determine the overall stress intensity.
 The unity check is from Section 4.4.12 equations 4.4.85 and 4.4.90.

An asterisk (*) in the final column denotes overstress.

Note: Performing Combined Stress Analysis per VIII-2 paragraph 4.3.10.2.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	26.77	113.10	-26.77	100.63	0.2367	0.2661	0.1697
20	27.68	113.10	-27.68	98.54	0.2447	0.2809	0.1807
30	26.79	113.10	-26.79	98.54	0.2369	0.2719	0.1778
40	25.97	86.02	-25.97	94.58	0.3020	0.2746	0.1822
50	13.54	156.40	-13.54	73.10	0.0866	0.1853	0.3257
60	6.67	156.40	-6.67	165.53	0.0426	0.0403	0.0268
70	6.25	156.40	-6.25	164.86	0.0400	0.0379	0.0252
80	5.83	156.40	-5.83	164.86	0.0373	0.0354	0.0234
90	5.23	156.40	-5.23	164.18	0.0334	0.0319	0.0215
100	4.80	156.40	-4.80	164.18	0.0307	0.0292	0.0195
110	4.31	156.40	-4.31	164.18	0.0276	0.0263	0.0172
120	3.73	156.40	-3.73	163.47	0.0239	0.0228	0.0141
130	3.45	156.40	-3.45	163.47	0.0221	0.0211	0.0126
150	5.92	156.40	-5.92	163.80	0.0378	0.0361	0.0211
160	5.28	156.40	-5.28	163.80	0.0337	0.0322	0.0187
170	4.73	156.40	-4.73	162.63	0.0302	0.0291	0.0167
180	4.08	156.40	-4.08	162.63	0.0261	0.0251	0.0141



190	3.45	156.40	-3.45	162.63	0.0220	0.0212	0.0117
200	2.90	156.40	-2.90	162.63	0.0185	0.0178	0.0103
210	1.72	156.40	-1.72	161.40	0.0110	0.0107	0.0064
220	1.50	156.40	-1.50	71.15	0.0096	0.0211	0.0191

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	67.40	113.10	-67.40	100.63	0.5959	0.6697	0.4829
20	68.20	113.10	-68.20	98.54	0.6030	0.6921	0.4912
30	63.42	113.10	-63.42	98.54	0.5608	0.6436	0.4487
40	58.70	86.02	-58.70	94.58	0.6824	0.6206	0.4224
50	30.22	156.40	-30.22	138.97	0.1932	0.2175	0.4068
60	15.28	156.40	-15.28	165.53	0.0977	0.0923	0.0690
70	13.98	156.40	-13.98	164.86	0.0894	0.0848	0.0628
80	12.56	156.40	-12.56	164.86	0.0803	0.0762	0.0556
90	11.09	156.40	-11.09	164.18	0.0709	0.0676	0.0486
100	9.71	156.40	-9.71	164.18	0.0621	0.0592	0.0420
110	8.34	156.40	-8.34	164.18	0.0533	0.0508	0.0357
120	7.03	156.40	-7.03	163.47	0.0450	0.0430	0.0307
130	6.28	156.40	-6.28	163.47	0.0401	0.0384	0.0273
150	13.68	156.40	-13.68	163.80	0.0875	0.0835	0.0667
160	11.28	156.40	-11.28	163.80	0.0721	0.0689	0.0540
170	9.28	156.40	-9.28	162.63	0.0593	0.0570	0.0464
180	7.19	156.40	-7.19	162.63	0.0460	0.0442	0.0496
190	5.33	156.40	-5.33	162.63	0.0341	0.0328	0.0527
200	3.77	156.40	-3.77	162.63	0.0241	0.0232	0.0569
210	1.93	156.40	-1.93	161.40	0.0123	0.0119	0.0618
220	1.64	156.40	-1.64	136.08	0.0105	0.0120	0.0115

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	34.22	113.10	-34.22	100.63	0.3026	0.3400	0.2486
20	35.58	113.10	-35.58	98.54	0.3146	0.3611	0.2661
30	34.69	113.10	-34.69	98.54	0.3067	0.3521	0.2633
40	33.88	86.02	-33.88	94.58	0.3938	0.3582	0.2711
50	14.49	156.40	-14.49	73.10	0.0927	0.1982	0.4338
60	7.16	156.40	-7.16	165.53	0.0458	0.0433	0.0301
70	6.75	156.40	-6.75	164.86	0.0432	0.0409	0.0285
80	6.33	156.40	-6.33	164.86	0.0405	0.0384	0.0268
90	5.74	156.40	-5.74	164.18	0.0367	0.0349	0.0249
100	5.25	156.40	-5.25	164.18	0.0336	0.0320	0.0225
110	4.66	156.40	-4.66	164.18	0.0298	0.0284	0.0195
120	4.03	156.40	-4.03	163.47	0.0258	0.0247	0.0161
130	3.67	156.40	-3.67	163.47	0.0235	0.0225	0.0141
150	6.39	156.40	-6.39	163.80	0.0409	0.0390	0.0246
160	5.64	156.40	-5.64	163.80	0.0360	0.0344	0.0214
170	4.98	156.40	-4.98	162.63	0.0318	0.0306	0.0186
180	4.29	156.40	-4.29	162.63	0.0275	0.0264	0.0157
190	3.53	156.40	-3.53	162.63	0.0226	0.0217	0.0122
200	2.90	156.40	-2.90	162.63	0.0185	0.0178	0.0103
210	1.72	156.40	-1.72	161.40	0.0110	0.0107	0.0064
220	1.50	156.40	-1.50	71.15	0.0096	0.0211	0.0191

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	90.88	113.10	-90.88	100.63	0.8036	0.9031	0.6315
20	91.78	113.10	-91.78	98.54	0.8115	0.9315	0.6386
30	85.00	113.10	-85.00	98.54	0.7515	0.8626	0.5773
40	78.27	86.02	-78.27	94.58	0.9099	0.8276	0.5364
50	37.01	156.40	-37.01	139.04	0.2367	0.2662	0.5474
60	18.78	156.40	-18.78	165.53	0.1201	0.1134	0.0871
70	17.05	156.40	-17.05	164.86	0.1090	0.1034	0.0784
80	15.20	156.40	-15.20	164.86	0.0972	0.0922	0.0686
90	13.41	156.40	-13.41	164.18	0.0857	0.0817	0.0597



100	11.66	156.40	-11.66	164.18	0.0746	0.0710	0.0511
110	9.90	156.40	-9.90	164.18	0.0633	0.0603	0.0431
120	8.31	156.40	-8.31	163.47	0.0531	0.0508	0.0366
130	7.32	156.40	-7.32	163.47	0.0468	0.0448	0.0323
150	16.31	156.40	-16.31	163.80	0.1043	0.0995	0.0799
160	13.20	156.40	-13.20	163.80	0.0844	0.0806	0.0635
170	10.61	156.40	-10.61	162.63	0.0679	0.0653	0.0504
180	8.07	156.40	-8.07	162.63	0.0516	0.0496	0.0496
190	5.76	156.40	-5.76	162.63	0.0368	0.0354	0.0527
200	3.92	156.40	-3.92	162.63	0.0251	0.0241	0.0569
210	1.96	156.40	-1.96	161.40	0.0126	0.0122	0.0618
220	1.66	156.40	-1.66	136.08	0.0106	0.0122	0.0118

Analysis of Load Case 5 : NP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	247.00	0.00	91.22	0.0000	-0.0000	0.0023
20	0.00	247.00	0.00	89.58	0.0000	-0.0000	0.0030
30	0.00	247.00	0.00	89.58	0.0000	-0.0000	0.0034
40	0.00	247.00	0.00	89.58	0.0000	-0.0000	0.0037
50	0.00	247.00	0.00	107.03	0.0000	-0.0000	...
60	0.00	247.00	0.00	117.27	0.0000	-0.0000	...
70	0.00	247.00	0.00	117.06	0.0000	-0.0000	...
80	0.00	247.00	0.00	117.06	0.0000	-0.0000	0.0010
90	0.00	247.00	0.00	116.84	0.0000	-0.0000	0.0011
100	0.00	247.00	0.00	116.84	0.0000	-0.0000	0.0012
110	0.00	247.00	0.00	116.84	0.0000	-0.0000	0.0013
120	0.00	247.00	0.00	116.62	0.0000	-0.0000	0.0014
130	0.00	247.00	0.00	116.62	0.0000	-0.0000	0.0015
150	0.00	247.00	0.00	116.50	0.0000	-0.0000	0.0042
160	0.00	247.00	0.00	116.50	0.0000	-0.0000	0.0044
170	0.00	247.00	0.00	116.12	0.0000	-0.0000	0.0048
180	0.00	247.00	0.00	116.12	0.0000	-0.0000	0.0050
190	0.00	247.00	0.00	116.12	0.0000	-0.0000	0.0051
200	0.00	247.00	0.00	116.12	0.0000	-0.0000	0.0053
210	0.00	247.00	0.00	115.73	0.0000	-0.0000	0.0056
220	0.00	247.00	0.00	105.22	0.0000	-0.0000	...

Analysis of Load Case 6 : NP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	247.00	0.00	91.22	0.0000	-0.0000	0.0020
20	0.00	247.00	0.00	89.58	0.0000	-0.0000	0.0021
30	0.00	247.00	0.00	89.58	0.0000	-0.0000	0.0021
40	0.00	247.00	0.00	89.58	0.0000	-0.0000	0.0022
50	0.00	247.00	0.00	107.03	0.0000	-0.0000	...
60	0.00	247.00	0.00	117.27	0.0000	-0.0000	...
70	0.00	247.00	0.00	117.06	0.0000	-0.0000	...
80	0.00	247.00	0.00	117.06	0.0000	-0.0000	...
90	0.00	247.00	0.00	116.84	0.0000	-0.0000	...
100	0.00	247.00	0.00	116.84	0.0000	-0.0000	...
110	0.00	247.00	0.00	116.84	0.0000	-0.0000	...
120	0.00	247.00	0.00	116.62	0.0000	-0.0000	...
130	0.00	247.00	0.00	116.62	0.0000	-0.0000	...
150	0.00	247.00	0.00	116.50	0.0000	-0.0000	0.0012
160	0.00	247.00	0.00	116.50	0.0000	-0.0000	0.0012
170	0.00	247.00	0.00	116.12	0.0000	-0.0000	0.0012
180	0.00	247.00	0.00	116.12	0.0000	-0.0000	0.0012
190	0.00	247.00	0.00	116.12	0.0000	-0.0000	0.0012
200	0.00	247.00	0.00	116.12	0.0000	-0.0000	0.0012
210	0.00	247.00	0.00	115.73	0.0000	-0.0000	0.0012
220	0.00	247.00	0.00	105.22	0.0000	-0.0000	...

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	34.22	113.10	-34.22	100.63	0.3026	0.3400	0.2486



20	35.58	113.10	-35.58	98.54	0.3146	0.3611	0.2661
30	34.69	113.10	-34.69	98.54	0.3067	0.3521	0.2633
40	33.88	86.02	-33.88	94.58	0.3938	0.3582	0.2711
50	130.74	156.40		73.10	0.8359		0.4338
60	121.39	156.40		165.53	0.7762		0.0301
70	122.55	156.40		164.86	0.7836		0.0285
80	121.67	156.40		164.86	0.7779		0.0268
90	123.04	156.40		164.18	0.7867		0.0249
100	123.02	156.40		164.18	0.7865		0.0225
110	122.99	156.40		164.18	0.7864		0.0195
120	125.11	156.40		163.47	0.7999		0.0161
130	125.10	156.40		163.47	0.7999		0.0141
150	124.90	156.40		163.80	0.7986		0.0246
160	124.86	156.40		163.80	0.7983		0.0214
170	128.33	156.40		162.63	0.8205		0.0186
180	128.30	156.40		162.63	0.8203		0.0157
190	128.27	156.40		162.63	0.8201		0.0122
200	128.25	156.40		162.63	0.8200		0.0103
210	131.93	156.40		161.40	0.8436		0.0064
220	140.93	156.40		103.49	0.9011		...

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	90.88	113.10	-90.88	100.63	0.8036	0.9031	0.6315
20	91.78	113.10	-91.78	98.54	0.8115	0.9315	0.6386
30	85.00	113.10	-85.00	98.54	0.7515	0.8626	0.5773
40	78.27	86.02	-78.27	94.58	0.9099	0.8276	0.5364
50	142.84	156.40		139.04	0.9133		0.5474
60	122.86	156.40		165.53	0.7855		0.0871
70	123.71	156.40		164.86	0.7910		0.0784
80	122.56	156.40		164.86	0.7836		0.0686
90	123.71	156.40		164.18	0.7910		0.0597
100	123.51	156.40		164.18	0.7897		0.0511
110	123.34	156.40		164.18	0.7886		0.0431
120	125.36	156.40		163.47	0.8015		0.0366
130	125.29	156.40		163.47	0.8011		0.0323
150	125.89	156.40		163.80	0.8049		0.0799
160	125.49	156.40		163.80	0.8024		0.0635
170	128.72	156.40		162.63	0.8230		0.0504
180	128.52	156.40		162.63	0.8217		0.0496
190	128.37	156.40		162.63	0.8208		0.0527
200	128.29	156.40		162.63	0.8202		0.0569
210	131.94	156.40		161.40	0.8436		0.0618
220	141.03	156.40		103.49	0.9017		...

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	34.22	113.10	-34.22	100.81	0.3026	0.3394	0.2478
20	35.58	113.10	-35.58	98.73	0.3146	0.3604	0.2653
30	34.69	113.10	-34.69	98.73	0.3067	0.3514	0.2624
40	33.88	86.02	-33.88	94.88	0.3938	0.3571	0.2697
50	14.49	156.40	-14.49	73.69	0.0927	0.1967	0.4303
60	7.16	156.40	-7.16	166.73	0.0458	0.0429	0.0298
70	6.75	156.40	-6.75	166.08	0.0432	0.0406	0.0282
80	6.33	156.40	-6.33	166.08	0.0405	0.0381	0.0265
90	5.74	156.40	-5.74	165.41	0.0367	0.0347	0.0247
100	5.25	156.40	-5.25	165.41	0.0336	0.0317	0.0223
110	4.66	156.40	-4.66	165.41	0.0298	0.0282	0.0193
120	4.03	156.40	-4.03	164.72	0.0258	0.0245	0.0159
130	3.67	156.40	-3.67	164.72	0.0235	0.0223	0.0140
150	6.39	156.40	-6.39	165.04	0.0409	0.0387	0.0243
160	5.64	156.40	-5.64	165.04	0.0360	0.0342	0.0211
170	4.98	156.40	-4.98	163.90	0.0318	0.0304	0.0183
180	4.29	156.40	-4.29	163.90	0.0275	0.0262	0.0155
190	3.53	156.40	-3.53	163.90	0.0226	0.0216	0.0120
200	2.90	156.40	-2.90	163.90	0.0185	0.0177	0.0102



210	1.72	156.40	-1.72	162.70	0.0110	0.0106	0.0063
220	1.50	156.40	-1.50	71.78	0.0096	0.0209	0.0190

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	90.88	113.10	-90.88	100.81	0.8036	0.9015	0.6306
20	91.78	113.10	-91.78	98.73	0.8115	0.9296	0.6377
30	85.00	113.10	-85.00	98.73	0.7515	0.8609	0.5764
40	78.27	86.02	-78.27	94.88	0.9099	0.8250	0.5351
50	37.01	156.40	-37.01	140.02	0.2367	0.2644	0.5436
60	18.78	156.40	-18.78	166.73	0.1201	0.1126	0.0865
70	17.05	156.40	-17.05	166.08	0.1090	0.1026	0.0778
80	15.20	156.40	-15.20	166.08	0.0972	0.0915	0.0681
90	13.41	156.40	-13.41	165.41	0.0857	0.0811	0.0592
100	11.66	156.40	-11.66	165.41	0.0746	0.0705	0.0507
110	9.90	156.40	-9.90	165.41	0.0633	0.0599	0.0427
120	8.31	156.40	-8.31	164.72	0.0531	0.0504	0.0363
130	7.32	156.40	-7.32	164.72	0.0468	0.0445	0.0321
150	16.31	156.40	-16.31	165.04	0.1043	0.0988	0.0793
160	13.20	156.40	-13.20	165.04	0.0844	0.0800	0.0630
170	10.61	156.40	-10.61	163.90	0.0679	0.0648	0.0500
180	8.07	156.40	-8.07	163.90	0.0516	0.0492	0.0495
190	5.76	156.40	-5.76	163.90	0.0368	0.0352	0.0525
200	3.92	156.40	-3.92	163.90	0.0251	0.0239	0.0567
210	1.96	156.40	-1.96	162.70	0.0126	0.0121	0.0616
220	1.66	156.40	-1.66	137.14	0.0106	0.0121	0.0117

Analysis of Load Case 11 : HP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	247.00	0.00	91.22	0.0000	-0.0000	0.0023
20	0.00	247.00	0.00	89.58	0.0000	-0.0000	0.0030
30	0.00	247.00	0.00	89.58	0.0000	-0.0000	0.0034
40	0.00	247.00	0.00	89.58	0.0000	-0.0000	0.0037
50	170.34	247.00		107.03	0.6896		...
60	163.59	247.00		117.27	0.6623		...
70	166.18	247.00		117.06	0.6728		...
80	166.18	247.00		117.06	0.6728		0.0010
90	168.86	247.00		116.84	0.6836		0.0011
100	168.86	247.00		116.84	0.6836		0.0012
110	168.86	247.00		116.84	0.6836		0.0013
120	171.63	247.00		116.62	0.6948		0.0014
130	171.63	247.00		116.62	0.6948		0.0015
150	172.42	247.00		116.50	0.6980		0.0042
160	172.42	247.00		116.50	0.6980		0.0044
170	177.00	247.00		116.12	0.7166		0.0048
180	177.00	247.00		116.12	0.7166		0.0050
190	177.00	247.00		116.12	0.7166		0.0051
200	177.00	247.00		116.12	0.7166		0.0053
210	181.84	247.00		115.73	0.7362		0.0056
220	185.43	247.00		105.22	0.7507		...

Analysis of Load Case 12 : HP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	247.00	0.00	91.22	0.0000	-0.0000	0.0020
20	0.00	247.00	0.00	89.58	0.0000	-0.0000	0.0021
30	0.00	247.00	0.00	89.58	0.0000	-0.0000	0.0021
40	0.00	247.00	0.00	89.58	0.0000	-0.0000	0.0022
50	170.34	247.00		107.03	0.6896		...
60	163.59	247.00		117.27	0.6623		...
70	166.18	247.00		117.06	0.6728		...
80	166.18	247.00		117.06	0.6728		...
90	168.86	247.00		116.84	0.6836		...
100	168.86	247.00		116.84	0.6836		...
110	168.86	247.00		116.84	0.6836		...



120	171.63	247.00	116.62	0.6948	...
130	171.63	247.00	116.62	0.6948	...
150	172.42	247.00	116.50	0.6980	0.0012
160	172.42	247.00	116.50	0.6980	0.0012
170	177.00	247.00	116.12	0.7166	0.0012
180	177.00	247.00	116.12	0.7166	0.0012
190	177.00	247.00	116.12	0.7166	0.0012
200	177.00	247.00	116.12	0.7166	0.0012
210	181.84	247.00	115.73	0.7362	0.0012
220	185.43	247.00	105.22	0.7507	...

Analysis of Load Case 13 : IP+WE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	20.36	113.10	-20.36	100.63	0.1800	0.2024	0.1689
20	21.29	113.10	-21.29	98.54	0.1882	0.2160	0.1798
30	20.98	113.10	-20.98	98.54	0.1855	0.2130	0.1769
40	20.73	86.02	-20.73	94.58	0.2410	0.2192	0.1813
50	130.00	156.40		73.10	0.8312		0.2539
60	121.26	156.40		165.53	0.7753		0.0267
70	122.44	156.40		164.86	0.7828		0.0250
80	121.57	156.40		164.86	0.7773		0.0233
90	122.97	156.40		164.18	0.7862		0.0214
100	122.95	156.40		164.18	0.7861		0.0194
110	122.94	156.40		164.18	0.7860		0.0170
120	125.07	156.40		163.47	0.7997		0.0139
130	125.07	156.40		163.47	0.7997		0.0125
150	124.78	156.40		163.80	0.7978		0.0207
160	124.77	156.40		163.80	0.7978		0.0183
170	128.27	156.40		162.63	0.8201		0.0163
180	128.25	156.40		162.63	0.8200		0.0137
190	128.24	156.40		162.63	0.8200		0.0111
200	128.23	156.40		162.63	0.8199		0.0087
210	131.93	156.40		161.40	0.8435		0.0048
220	140.77	156.40		103.49	0.9001		...

Analysis of Load Case 14 : IP+WF+CW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	22.35	113.10	-22.35	100.63	0.1976	0.2221	0.2368
20	23.38	113.10	-23.38	98.54	0.2067	0.2373	0.2528
30	23.13	113.10	-23.13	98.54	0.2045	0.2347	0.2501
40	22.89	86.02	-22.89	94.58	0.2661	0.2420	0.2575
50	128.87	156.40		73.10	0.8240		0.3131
60	121.20	156.40		165.53	0.7749		0.0285
70	122.39	156.40		164.86	0.7825		0.0270
80	121.53	156.40		164.86	0.7770		0.0253
90	122.94	156.40		164.18	0.7860		0.0236
100	122.93	156.40		164.18	0.7860		0.0213
110	122.91	156.40		164.18	0.7859		0.0184
120	125.05	156.40		163.47	0.7996		0.0152
130	125.04	156.40		163.47	0.7995		0.0133
150	124.75	156.40		163.80	0.7976		0.0233
160	124.73	156.40		163.80	0.7975		0.0201
170	128.23	156.40		162.63	0.8199		0.0174
180	128.22	156.40		162.63	0.8198		0.0146
190	128.21	156.40		162.63	0.8198		0.0113
200	128.21	156.40		162.63	0.8197		0.0083
210	131.91	156.40		161.40	0.8434		0.0046
220	140.61	156.40		103.49	0.8990		...

Analysis of Load Case 15 : IP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	27.81	113.10	-27.81	100.63	0.2459	0.2763	0.2478
20	29.19	113.10	-29.19	98.54	0.2581	0.2962	0.2653
30	28.89	113.10	-28.89	98.54	0.2554	0.2932	0.2624



40	28.63	86.02	-28.63	94.58	0.3329	0.3027	0.2702
50	129.58	156.40		73.10	0.8285		0.3620
60	121.29	156.40		165.53	0.7755		0.0299
70	122.47	156.40		164.86	0.7830		0.0283
80	121.60	156.40		164.86	0.7775		0.0266
90	122.99	156.40		164.18	0.7864		0.0248
100	122.97	156.40		164.18	0.7863		0.0224
110	122.95	156.40		164.18	0.7861		0.0194
120	125.08	156.40		163.47	0.7998		0.0159
130	125.07	156.40		163.47	0.7997		0.0140
150	124.80	156.40		163.80	0.7980		0.0242
160	124.78	156.40		163.80	0.7978		0.0209
170	128.28	156.40		162.63	0.8202		0.0181
180	128.26	156.40		162.63	0.8201		0.0152
190	128.25	156.40		162.63	0.8200		0.0117
200	128.23	156.40		162.63	0.8199		0.0087
210	131.93	156.40		161.40	0.8435		0.0048
220	140.77	156.40		103.49	0.9001		...

Analysis of Load Case 16 : IP+VE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	20.36	113.10	-20.36	100.63	0.1800	0.2024	0.1689
20	21.29	113.10	-21.29	98.54	0.1882	0.2160	0.1798
30	20.98	113.10	-20.98	98.54	0.1855	0.2130	0.1769
40	20.73	86.02	-20.73	94.58	0.2410	0.2192	0.1813
50	130.00	156.40		73.10	0.8312		0.2539
60	121.26	156.40		165.53	0.7753		0.0267
70	122.44	156.40		164.86	0.7828		0.0250
80	121.57	156.40		164.86	0.7773		0.0233
90	122.97	156.40		164.18	0.7862		0.0214
100	122.95	156.40		164.18	0.7861		0.0194
110	122.94	156.40		164.18	0.7860		0.0170
120	125.07	156.40		163.47	0.7997		0.0139
130	125.07	156.40		163.47	0.7997		0.0125
150	124.78	156.40		163.80	0.7978		0.0207
160	124.77	156.40		163.80	0.7978		0.0183
170	128.27	156.40		162.63	0.8201		0.0163
180	128.25	156.40		162.63	0.8200		0.0137
190	128.24	156.40		162.63	0.8200		0.0111
200	128.23	156.40		162.63	0.8199		0.0087
210	131.93	156.40		161.40	0.8435		0.0048
220	140.77	156.40		103.49	0.9001		...

Analysis of Load Case 17 : NP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	27.81	113.10	-27.81	100.63	0.2459	0.2763	0.2478
20	29.19	113.10	-29.19	98.54	0.2581	0.2962	0.2653
30	28.89	113.10	-28.89	98.54	0.2554	0.2932	0.2624
40	28.63	86.02	-28.63	94.58	0.3329	0.3027	0.2702
50	11.81	156.40	-11.81	73.10	0.0755	0.1615	0.3620
60	5.78	156.40	-5.78	165.53	0.0369	0.0349	0.0299
70	5.49	156.40	-5.49	164.86	0.0351	0.0333	0.0283
80	5.21	156.40	-5.21	164.86	0.0333	0.0316	0.0266
90	4.72	156.40	-4.72	164.18	0.0302	0.0288	0.0248
100	4.36	156.40	-4.36	164.18	0.0279	0.0265	0.0224
110	3.88	156.40	-3.88	164.18	0.0248	0.0237	0.0194
120	3.35	156.40	-3.35	163.47	0.0214	0.0205	0.0159
130	3.05	156.40	-3.05	163.47	0.0195	0.0187	0.0140
150	4.53	156.40	-4.53	163.80	0.0289	0.0276	0.0242
160	4.09	156.40	-4.09	163.80	0.0261	0.0250	0.0209
170	3.68	156.40	-3.68	162.63	0.0236	0.0227	0.0181
180	3.27	156.40	-3.27	162.63	0.0209	0.0201	0.0152
190	2.76	156.40	-2.76	162.63	0.0176	0.0170	0.0117
200	2.35	156.40	-2.35	162.63	0.0150	0.0144	0.0087
210	1.36	156.40	-1.36	161.40	0.0087	0.0085	0.0048
220	0.96	156.40	-0.96	71.15	0.0062	0.0136	0.0141



Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	29.52	113.10	-29.52	100.63	0.2610	0.2933	0.2408
20	30.92	113.10	-30.92	98.54	0.2734	0.3138	0.2578
30	30.50	113.10	-30.50	98.54	0.2696	0.3095	0.2550
40	30.11	86.02	-30.11	94.58	0.3501	0.3184	0.2626
50	130.23	156.40		73.10	0.8327		0.3823
60	121.32	156.40		165.53	0.7757		0.0289
70	122.49	156.40		164.86	0.7832		0.0274
80	121.62	156.40		164.86	0.7776		0.0258
90	123.01	156.40		164.18	0.7865		0.0242
100	122.99	156.40		164.18	0.7864		0.0218
110	122.97	156.40		164.18	0.7862		0.0189
120	125.10	156.40		163.47	0.7999		0.0156
130	125.09	156.40		163.47	0.7998		0.0137
150	124.86	156.40		163.80	0.7983		0.0238
160	124.83	156.40		163.80	0.7981		0.0207
170	128.31	156.40		162.63	0.8204		0.0180
180	128.29	156.40		162.63	0.8203		0.0152
190	128.26	156.40		162.63	0.8201		0.0118
200	128.25	156.40		162.63	0.8200		0.0088
210	131.93	156.40		161.40	0.8436		0.0051
220	140.93	156.40		103.49	0.9011		...

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	29.52	113.10	-29.52	100.81	0.2610	0.2928	0.2401
20	30.92	113.10	-30.92	98.73	0.2734	0.3132	0.2570
30	30.50	113.10	-30.50	98.73	0.2696	0.3089	0.2542
40	30.11	86.02	-30.11	94.88	0.3501	0.3174	0.2613
50	12.57	156.40	-12.57	73.69	0.0804	0.1706	0.3793
60	6.19	156.40	-6.19	166.73	0.0396	0.0371	0.0286
70	5.90	156.40	-5.90	166.08	0.0377	0.0355	0.0272
80	5.60	156.40	-5.60	166.08	0.0358	0.0337	0.0256
90	5.11	156.40	-5.11	165.41	0.0327	0.0309	0.0239
100	4.72	156.40	-4.72	165.41	0.0302	0.0285	0.0216
110	4.22	156.40	-4.22	165.41	0.0270	0.0255	0.0187
120	3.67	156.40	-3.67	164.72	0.0235	0.0223	0.0154
130	3.36	156.40	-3.36	164.72	0.0215	0.0204	0.0135
150	5.62	156.40	-5.62	165.04	0.0359	0.0341	0.0235
160	5.05	156.40	-5.05	165.04	0.0323	0.0306	0.0204
170	4.54	156.40	-4.54	163.90	0.0290	0.0277	0.0178
180	3.99	156.40	-3.99	163.90	0.0255	0.0243	0.0150
190	3.34	156.40	-3.34	163.90	0.0213	0.0204	0.0116
200	2.79	156.40	-2.79	163.90	0.0178	0.0170	0.0087
210	1.68	156.40	-1.68	162.70	0.0107	0.0103	0.0050
220	1.46	156.40	-1.46	71.78	0.0093	0.0203	0.0184

Analysis of Load Case 20 : BL+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	27.81	113.10	-27.81	100.63	0.2459	0.2763	0.2478
20	29.19	113.10	-29.19	98.54	0.2581	0.2962	0.2653
30	28.89	113.10	-28.89	98.54	0.2554	0.2932	0.2624
40	28.63	86.02	-28.63	94.58	0.3329	0.3027	0.2702
50	129.58	156.40		73.10	0.8285		0.3620
60	121.29	156.40		165.53	0.7755		0.0299
70	122.47	156.40		164.86	0.7830		0.0283
80	121.60	156.40		164.86	0.7775		0.0266
90	122.99	156.40		164.18	0.7864		0.0248
100	122.97	156.40		164.18	0.7863		0.0224
110	122.95	156.40		164.18	0.7861		0.0194
120	125.08	156.40		163.47	0.7998		0.0159
130	125.07	156.40		163.47	0.7997		0.0140



150	124.80	156.40	163.80	0.7980	0.0242
160	124.78	156.40	163.80	0.7978	0.0209
170	128.28	156.40	162.63	0.8202	0.0181
180	128.26	156.40	162.63	0.8201	0.0152
190	128.25	156.40	162.63	0.8200	0.0117
200	128.23	156.40	162.63	0.8199	0.0087
210	131.93	156.40	161.40	0.8435	0.0048
220	140.77	156.40	103.49	0.9001	...

Absolute Maximum of the all of the Stress Ratio's 0.9315

Governing Element: Btm skirt-2
 Governing Load Case 4 : NP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Btm skirt-1	...	-22.7	4.4	63.7	45.4	+ Tens
Btm skirt-1	...	-22.7	-4.4	-63.7	-90.9	- Comp
Btm skirt-2	...	-23.8	4.7	63.3	44.1	+ Tens
Btm skirt-2	...	-23.8	-4.7	-63.3	-91.8	- Comp
Btm skirt-3	...	-23.6	4.6	56.8	37.8	+ Tens
Btm skirt-3	...	-23.6	-4.6	-56.8	-85.0	- Comp
Top skirt	...	-23.3	4.6	50.3	31.6	+ Tens
Top skirt	...	-23.3	-4.6	-50.3	-78.3	- Comp
BTM dish	...	-9.0	2.5	25.6	19.0	+ Tens
BTM dish	...	-9.0	-2.5	-25.6	-37.0	- Comp
shell-1	...	-4.4	1.3	13.2	10.1	+ Tens
shell-1	...	-4.4	-1.3	-13.2	-18.8	- Comp
shell-2	...	-4.1	1.2	11.7	8.8	+ Tens
shell-2	...	-4.1	-1.2	-11.7	-17.0	- Comp
shell-3	...	-3.9	1.2	10.1	7.4	+ Tens
shell-3	...	-3.9	-1.2	-10.1	-15.2	- Comp
shell-4	...	-3.6	1.0	8.8	6.2	+ Tens
shell-4	...	-3.6	-1.0	-8.8	-13.4	- Comp
shell-5	...	-3.3	1.0	7.4	5.1	+ Tens
shell-5	...	-3.3	-1.0	-7.4	-11.7	- Comp
shell-6	...	-2.8	1.0	6.1	4.2	+ Tens
shell-6	...	-2.8	-1.0	-6.1	-9.9	- Comp
shell-7	...	-2.3	1.0	5.0	3.7	+ Tens
shell-7	...	-2.3	-1.0	-5.0	-8.3	- Comp
shell-8	...	-2.0	1.0	4.3	3.2	+ Tens
shell-8	...	-2.0	-1.0	-4.3	-7.3	- Comp
cone	...	-1.4	0.7	2.8	2.2	+ Tens
cone	...	-1.4	-0.7	-2.8	-4.9	- Comp
shell-9	...	-3.2	1.3	11.8	9.9	+ Tens
shell-9	...	-3.2	-1.3	-11.8	-16.3	- Comp
shell-10	...	-2.8	1.3	9.1	7.6	+ Tens
shell-10	...	-2.8	-1.3	-9.1	-13.2	- Comp
shell-11	...	-2.4	1.3	6.9	5.8	+ Tens
shell-11	...	-2.4	-1.3	-6.9	-10.6	- Comp
shell-12	...	-2.0	1.2	4.8	4.0	+ Tens

shell-12	...	-2.0	-1.2	-4.8	-8.1	- Comp
shell-13	...	-1.6	1.2	3.0	2.6	+ Tens
shell-13	...	-1.6	-1.2	-3.0	-5.8	- Comp
shell-14	...	-1.2	1.2	1.6	1.6	+ Tens
shell-14	...	-1.2	-1.2	-1.6	-3.9	- Comp
shell-15	...	-0.7	0.7	0.6	0.6	+ Tens
shell-15	...	-0.7	-0.7	-0.6	-2.0	- Comp
TOP dish	...	-0.7	0.4	0.6	0.3	+ Tens
TOP dish	...	-0.7	-0.4	-0.6	-1.7	- Comp

Additional Results per Section 4.4, see 4.4.15

Nomenclature :

- L - Section Length mm
- Fxa - Allowable comp mem stress due to compression with lamc ≤ 0.15 , N./mm²
- Fba - Allowable all comp mem stress due to bending moment, N./mm²
- Fva - Allowable shear stress when only shear is present, N./mm²
- Fha - Allowable hoop comp memb stress under ext pressure alone, N./mm²
- Fxha - Allowable axial comp w/hoop compression and lamc ≤ 0.15 , N./mm²
- Fhxa - Allowable hoop comp w/axial compression and lamc ≤ 0.15 , N./mm²
- Faha - Allowable axial comp w/hoop compression and lamc > 0.15 , N./mm²
- Fbha - Allowable axial comp from bending w/hoop compression, N./mm²
- Fhba - Allowable hoop comp w/long compression due to moment, N./mm²

Allowable Stress Results for Case 1 : NP+EW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	100	107	72		6895	100	6895	6895	6895
20	0.408	9000	98	106	70		6895	98	6895	6895	6895
30	0.408	9000	98	106	70		6895	98	6895	6895	6895
40	0.403	9000	94	102	67		6895	94	6895	6895	6895
50						73	6895		6895	6895	6895
60	0.509	22850	165	165	97	55	6895	165	6895	6895	6895
70	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
80	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
90	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
100	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
110	0.507	22850	164	164	96	54	6895	164	6895	6895	6895
120	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
130	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
150	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
160	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
170	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
180	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
190	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
200	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
210	0.788	19916	161	161	90	40	6895	161	6895	6895	6895
220						71	6895		6895	6895	6895

Allowable Stress Results for Case 2 : NP+EW+EE+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	100	107	72		6895	100	6895	6895	6895
20	0.408	9000	98	106	70		6895	98	6895	6895	6895
30	0.408	9000	98	106	70		6895	98	6895	6895	6895
40	0.403	9000	94	102	67		6895	94	6895	6895	6895
50						138	6895		6895	6895	6895
60	0.509	22850	165	165	97	55	6895	165	6895	6895	6895
70	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
80	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
90	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
100	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
110	0.507	22850	164	164	96	54	6895	164	6895	6895	6895
120	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
130	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
150	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
160	0.791	19916	163	163	92	43	6895	163	6895	6895	6895



170	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
180	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
190	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
200	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
210	0.788	19916	161	161	90	40	6895	161	6895	6895	6895
220						136	6895		6895	6895	6895

Allowable Stress Results for Case 3 : NP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	100	107	72		6895	100	6895	6895	6895
20	0.408	9000	98	106	70		6895	98	6895	6895	6895
30	0.408	9000	98	106	70		6895	98	6895	6895	6895
40	0.403	9000	94	102	67		6895	94	6895	6895	6895
50						73	6895		6895	6895	6895
60	0.509	22850	165	165	97	55	6895	165	6895	6895	6895
70	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
80	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
90	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
100	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
110	0.507	22850	164	164	96	54	6895	164	6895	6895	6895
120	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
130	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
150	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
160	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
170	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
180	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
190	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
200	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
210	0.788	19916	161	161	90	40	6895	161	6895	6895	6895
220						71	6895		6895	6895	6895

Allowable Stress Results for Case 4 : NP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	100	107	72		6895	100	6895	6895	6895
20	0.408	9000	98	106	70		6895	98	6895	6895	6895
30	0.408	9000	98	106	70		6895	98	6895	6895	6895
40	0.403	9000	94	102	67		6895	94	6895	6895	6895
50						139	6895		6895	6895	6895
60	0.509	22850	165	165	97	55	6895	165	6895	6895	6895
70	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
80	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
90	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
100	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
110	0.507	22850	164	164	96	54	6895	164	6895	6895	6895
120	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
130	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
150	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
160	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
170	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
180	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
190	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
200	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
210	0.788	19916	161	161	90	40	6895	161	6895	6895	6895
220						136	6895		6895	6895	6895

Allowable Stress Results for Case 5 : NP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	102	109	74		6895	102	6895	6895	6895
20	0.408	9000	100	107	72		6895	100	6895	6895	6895
30	0.408	9000	100	107	72		6895	100	6895	6895	6895
40	0.400	9000	96	103	69		6895	96	6895	6895	6895
50							6895		6895	6895	6895
60	0.505	22850	168	168	99	58	6895	168	6895	6895	6895
70	0.505	22850	167	167	98	58	6895	167	6895	6895	6895
80	0.505	22850	167	167	98	58	6895	167	6895	6895	6895
90	0.504	22849	167	167	98	57	6895	167	6895	6895	6895
100	0.504	22849	167	167	98	57	6895	167	6895	6895	6895
110	0.504	22850	167	167	98	57	6895	167	6895	6895	6895
120	0.504	22850	166	166	97	56	6895	166	6895	6895	6895
130	0.504	22850	166	166	97	56	6895	166	6895	6895	6895



150	0.786	19916	166	166	93	46	6895	166	6895	6895	6895
160	0.786	19916	166	166	93	46	6895	166	6895	6895	6895
170	0.784	19916	165	165	92	44	6895	165	6895	6895	6895
180	0.784	19916	165	165	92	44	6895	165	6895	6895	6895
190	0.784	19916	165	165	92	44	6895	165	6895	6895	6895
200	0.784	19916	165	165	92	44	6895	165	6895	6895	6895
210	0.783	19916	164	164	91	43	6895	164	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 6 : NP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	102	109	74		6895	102	6895	6895	6895
20	0.408	9000	100	107	72		6895	100	6895	6895	6895
30	0.408	9000	100	107	72		6895	100	6895	6895	6895
40	0.400	9000	96	103	69		6895	96	6895	6895	6895
50							6895		6895	6895	6895
60	0.505	22850	168	168	99	58	6895	168	6895	6895	6895
70	0.505	22850	167	167	98	58	6895	167	6895	6895	6895
80	0.505	22850	167	167	98	58	6895	167	6895	6895	6895
90	0.504	22849	167	167	98	57	6895	167	6895	6895	6895
100	0.504	22849	167	167	98	57	6895	167	6895	6895	6895
110	0.504	22850	167	167	98	57	6895	167	6895	6895	6895
120	0.504	22850	166	166	97	56	6895	166	6895	6895	6895
130	0.504	22850	166	166	97	56	6895	166	6895	6895	6895
150	0.786	19916	166	166	93	46	6895	166	6895	6895	6895
160	0.786	19916	166	166	93	46	6895	166	6895	6895	6895
170	0.784	19916	165	165	92	44	6895	165	6895	6895	6895
180	0.784	19916	165	165	92	44	6895	165	6895	6895	6895
190	0.784	19916	165	165	92	44	6895	165	6895	6895	6895
200	0.784	19916	165	165	92	44	6895	165	6895	6895	6895
210	0.783	19916	164	164	91	43	6895	164	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 7 : IP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	100	107	72		6895	100	6895	6895	6895
20	0.408	9000	98	106	70		6895	98	6895	6895	6895
30	0.408	9000	98	106	70		6895	98	6895	6895	6895
40	0.403	9000	94	102	67		6895	94	6895	6895	6895
50						73	6895		6895	6895	6895
60	0.509	22850	165	165	97	55	6895	165	6895	6895	6895
70	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
80	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
90	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
100	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
110	0.507	22850	164	164	96	54	6895	164	6895	6895	6895
120	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
130	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
150	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
160	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
170	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
180	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
190	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
200	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
210	0.788	19916	161	161	90	40	6895	161	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 8 : IP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	100	107	72		6895	100	6895	6895	6895
20	0.408	9000	98	106	70		6895	98	6895	6895	6895
30	0.408	9000	98	106	70		6895	98	6895	6895	6895
40	0.403	9000	94	102	67		6895	94	6895	6895	6895
50						139	6895		6895	6895	6895
60	0.509	22850	165	165	97	55	6895	165	6895	6895	6895
70	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
80	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
90	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
100	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
110	0.507	22850	164	164	96	54	6895	164	6895	6895	6895



120	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
130	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
150	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
160	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
170	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
180	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
190	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
200	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
210	0.788	19916	161	161	90	40	6895	161	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 9 : EP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.407	9000	100	108	73		6895	100	6895	6895	6895
20	0.403	9000	98	106	70		6895	98	6895	6895	6895
30	0.403	9000	98	106	70		6895	98	6895	6895	6895
40	0.396	9000	94	102	67		6895	94	6895	6895	6895
50						73	6895		6895	6895	6895
60	0.501	22850	166	166	97	56	6895	166	6895	6895	6895
70	0.500	22850	166	166	97	56	6895	166	6895	6895	6895
80	0.500	22850	166	166	97	56	6895	166	6895	6895	6895
90	0.499	22849	165	165	96	55	6895	165	6895	6895	6895
100	0.499	22849	165	165	96	55	6895	165	6895	6895	6895
110	0.499	22850	165	165	96	55	6895	165	6895	6895	6895
120	0.499	22850	164	164	96	54	6895	164	6895	6895	6895
130	0.499	22850	164	164	96	54	6895	164	6895	6895	6895
150	0.779	19916	165	165	92	44	6895	165	6895	6895	6895
160	0.779	19916	165	165	92	44	6895	165	6895	6895	6895
170	0.777	19916	163	163	91	42	6895	163	6895	6895	6895
180	0.777	19916	163	163	91	42	6895	163	6895	6895	6895
190	0.777	19916	163	163	91	42	6895	163	6895	6895	6895
200	0.777	19916	163	163	91	42	6895	163	6895	6895	6895
210	0.776	19916	162	162	90	41	6895	162	6895	6895	6895
220						71	6895		6895	6895	6895

Allowable Stress Results for Case 10 : EP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.407	9000	100	108	73		6895	100	6895	6895	6895
20	0.403	9000	98	106	70		6895	98	6895	6895	6895
30	0.403	9000	98	106	70		6895	98	6895	6895	6895
40	0.396	9000	94	102	67		6895	94	6895	6895	6895
50						140	6895		6895	6895	6895
60	0.501	22850	166	166	97	56	6895	166	6895	6895	6895
70	0.500	22850	166	166	97	56	6895	166	6895	6895	6895
80	0.500	22850	166	166	97	56	6895	166	6895	6895	6895
90	0.499	22849	165	165	96	55	6895	165	6895	6895	6895
100	0.499	22849	165	165	96	55	6895	165	6895	6895	6895
110	0.499	22850	165	165	96	55	6895	165	6895	6895	6895
120	0.499	22850	164	164	96	54	6895	164	6895	6895	6895
130	0.499	22850	164	164	96	54	6895	164	6895	6895	6895
150	0.779	19916	165	165	92	44	6895	165	6895	6895	6895
160	0.779	19916	165	165	92	44	6895	165	6895	6895	6895
170	0.777	19916	163	163	91	42	6895	163	6895	6895	6895
180	0.777	19916	163	163	91	42	6895	163	6895	6895	6895
190	0.777	19916	163	163	91	42	6895	163	6895	6895	6895
200	0.777	19916	163	163	91	42	6895	163	6895	6895	6895
210	0.776	19916	162	162	90	41	6895	162	6895	6895	6895
220						137	6895		6895	6895	6895

Allowable Stress Results for Case 11 : HP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	102	109	74		6895	102	6895	6895	6895
20	0.408	9000	100	107	72		6895	100	6895	6895	6895
30	0.408	9000	100	107	72		6895	100	6895	6895	6895
40	0.400	9000	96	103	69		6895	96	6895	6895	6895
50							6895		6895	6895	6895
60	0.505	22850	168	168	99	58	6895	168	6895	6895	6895
70	0.505	22850	167	167	98	58	6895	167	6895	6895	6895
80	0.505	22850	167	167	98	58	6895	167	6895	6895	6895
90	0.504	22849	167	167	98	57	6895	167	6895	6895	6895



100	0.504	22849	167	167	98	57	6895	167	6895	6895	6895
110	0.504	22850	167	167	98	57	6895	167	6895	6895	6895
120	0.504	22850	166	166	97	56	6895	166	6895	6895	6895
130	0.504	22850	166	166	97	56	6895	166	6895	6895	6895
150	0.786	19916	166	166	93	46	6895	166	6895	6895	6895
160	0.786	19916	166	166	93	46	6895	166	6895	6895	6895
170	0.784	19916	165	165	92	44	6895	165	6895	6895	6895
180	0.784	19916	165	165	92	44	6895	165	6895	6895	6895
190	0.784	19916	165	165	92	44	6895	165	6895	6895	6895
200	0.784	19916	165	165	92	44	6895	165	6895	6895	6895
210	0.783	19916	164	164	91	43	6895	164	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 12 : HP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	102	109	74		6895	102	6895	6895	6895
20	0.408	9000	100	107	72		6895	100	6895	6895	6895
30	0.408	9000	100	107	72		6895	100	6895	6895	6895
40	0.400	9000	96	103	69		6895	96	6895	6895	6895
50							6895		6895	6895	6895
60	0.505	22850	168	168	99	58	6895	168	6895	6895	6895
70	0.505	22850	167	167	98	58	6895	167	6895	6895	6895
80	0.505	22850	167	167	98	58	6895	167	6895	6895	6895
90	0.504	22849	167	167	98	57	6895	167	6895	6895	6895
100	0.504	22849	167	167	98	57	6895	167	6895	6895	6895
110	0.504	22850	167	167	98	57	6895	167	6895	6895	6895
120	0.504	22850	166	166	97	56	6895	166	6895	6895	6895
130	0.504	22850	166	166	97	56	6895	166	6895	6895	6895
150	0.786	19916	166	166	93	46	6895	166	6895	6895	6895
160	0.786	19916	166	166	93	46	6895	166	6895	6895	6895
170	0.784	19916	165	165	92	44	6895	165	6895	6895	6895
180	0.784	19916	165	165	92	44	6895	165	6895	6895	6895
190	0.784	19916	165	165	92	44	6895	165	6895	6895	6895
200	0.784	19916	165	165	92	44	6895	165	6895	6895	6895
210	0.783	19916	164	164	91	43	6895	164	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 13 : IP+WE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	100	107	72		6895	100	6895	6895	6895
20	0.408	9000	98	106	70		6895	98	6895	6895	6895
30	0.408	9000	98	106	70		6895	98	6895	6895	6895
40	0.403	9000	94	102	67		6895	94	6895	6895	6895
50						73	6895		6895	6895	6895
60	0.509	22850	165	165	97	55	6895	165	6895	6895	6895
70	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
80	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
90	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
100	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
110	0.507	22850	164	164	96	54	6895	164	6895	6895	6895
120	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
130	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
150	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
160	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
170	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
180	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
190	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
200	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
210	0.788	19916	161	161	90	40	6895	161	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 14 : IP+WF+CW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	100	107	72		6895	100	6895	6895	6895
20	0.408	9000	98	106	70		6895	98	6895	6895	6895
30	0.408	9000	98	106	70		6895	98	6895	6895	6895
40	0.403	9000	94	102	67		6895	94	6895	6895	6895
50						73	6895		6895	6895	6895
60	0.509	22850	165	165	97	55	6895	165	6895	6895	6895
70	0.508	22850	164	164	97	55	6895	164	6895	6895	6895



80	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
90	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
100	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
110	0.507	22850	164	164	96	54	6895	164	6895	6895	6895
120	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
130	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
150	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
160	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
170	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
180	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
190	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
200	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
210	0.788	19916	161	161	90	40	6895	161	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 15 : IP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	100	107	72		6895	100	6895	6895	6895
20	0.408	9000	98	106	70		6895	98	6895	6895	6895
30	0.408	9000	98	106	70		6895	98	6895	6895	6895
40	0.403	9000	94	102	67		6895	94	6895	6895	6895
50						73	6895		6895	6895	6895
60	0.509	22850	165	165	97	55	6895	165	6895	6895	6895
70	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
80	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
90	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
100	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
110	0.507	22850	164	164	96	54	6895	164	6895	6895	6895
120	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
130	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
150	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
160	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
170	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
180	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
190	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
200	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
210	0.788	19916	161	161	90	40	6895	161	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 16 : IP+VE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	100	107	72		6895	100	6895	6895	6895
20	0.408	9000	98	106	70		6895	98	6895	6895	6895
30	0.408	9000	98	106	70		6895	98	6895	6895	6895
40	0.403	9000	94	102	67		6895	94	6895	6895	6895
50						73	6895		6895	6895	6895
60	0.509	22850	165	165	97	55	6895	165	6895	6895	6895
70	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
80	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
90	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
100	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
110	0.507	22850	164	164	96	54	6895	164	6895	6895	6895
120	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
130	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
150	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
160	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
170	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
180	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
190	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
200	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
210	0.788	19916	161	161	90	40	6895	161	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 17 : NP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	100	107	72		6895	100	6895	6895	6895
20	0.408	9000	98	106	70		6895	98	6895	6895	6895
30	0.408	9000	98	106	70		6895	98	6895	6895	6895
40	0.403	9000	94	102	67		6895	94	6895	6895	6895
50						73	6895		6895	6895	6895



60	0.509	22850	165	165	97	55	6895	165	6895	6895	6895
70	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
80	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
90	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
100	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
110	0.507	22850	164	164	96	54	6895	164	6895	6895	6895
120	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
130	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
150	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
160	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
170	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
180	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
190	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
200	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
210	0.788	19916	161	161	90	40	6895	161	6895	6895	6895
220						71	6895		6895	6895	6895

Allowable Stress Results for Case 18 : FS+BS+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	100	107	72		6895	100	6895	6895	6895
20	0.408	9000	98	106	70		6895	98	6895	6895	6895
30	0.408	9000	98	106	70		6895	98	6895	6895	6895
40	0.403	9000	94	102	67		6895	94	6895	6895	6895
50						73	6895		6895	6895	6895
60	0.509	22850	165	165	97	55	6895	165	6895	6895	6895
70	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
80	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
90	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
100	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
110	0.507	22850	164	164	96	54	6895	164	6895	6895	6895
120	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
130	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
150	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
160	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
170	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
180	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
190	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
200	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
210	0.788	19916	161	161	90	40	6895	161	6895	6895	6895
220							6895		6895	6895	6895

Allowable Stress Results for Case 19 : FS+BS+EP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.407	9000	100	108	73		6895	100	6895	6895	6895
20	0.403	9000	98	106	70		6895	98	6895	6895	6895
30	0.403	9000	98	106	70		6895	98	6895	6895	6895
40	0.396	9000	94	102	67		6895	94	6895	6895	6895
50						73	6895		6895	6895	6895
60	0.501	22850	166	166	97	56	6895	166	6895	6895	6895
70	0.500	22850	166	166	97	56	6895	166	6895	6895	6895
80	0.500	22850	166	166	97	56	6895	166	6895	6895	6895
90	0.499	22849	165	165	96	55	6895	165	6895	6895	6895
100	0.499	22849	165	165	96	55	6895	165	6895	6895	6895
110	0.499	22850	165	165	96	55	6895	165	6895	6895	6895
120	0.499	22850	164	164	96	54	6895	164	6895	6895	6895
130	0.499	22850	164	164	96	54	6895	164	6895	6895	6895
150	0.779	19916	165	165	92	44	6895	165	6895	6895	6895
160	0.779	19916	165	165	92	44	6895	165	6895	6895	6895
170	0.777	19916	163	163	91	42	6895	163	6895	6895	6895
180	0.777	19916	163	163	91	42	6895	163	6895	6895	6895
190	0.777	19916	163	163	91	42	6895	163	6895	6895	6895
200	0.777	19916	163	163	91	42	6895	163	6895	6895	6895
210	0.776	19916	162	162	90	41	6895	162	6895	6895	6895
220						71	6895		6895	6895	6895

Allowable Stress Results for Case 20 : BL+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10	0.411	9000	100	107	72		6895	100	6895	6895	6895
20	0.408	9000	98	106	70		6895	98	6895	6895	6895
30	0.408	9000	98	106	70		6895	98	6895	6895	6895



40	0.403	9000	94	102	67		6895	94	6895	6895	6895
50						73	6895		6895	6895	6895
60	0.509	22850	165	165	97	55	6895	165	6895	6895	6895
70	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
80	0.508	22850	164	164	97	55	6895	164	6895	6895	6895
90	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
100	0.507	22849	164	164	96	54	6895	164	6895	6895	6895
110	0.507	22850	164	164	96	54	6895	164	6895	6895	6895
120	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
130	0.507	22850	163	163	95	53	6895	163	6895	6895	6895
150	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
160	0.791	19916	163	163	92	43	6895	163	6895	6895	6895
170	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
180	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
190	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
200	0.790	19916	162	162	91	41	6895	162	6895	6895	6895
210	0.788	19916	161	161	90	40	6895	161	6895	6895	6895
220							6895		6895	6895	6895

Cone Junction Stresses and related Calculations are shown in the Conical Sections report.

Note:

Load cases involving Wind and Seismic are assumed to cause a net section bending moment. These load case components are: EQ, EE, WI, WE, VO, VE, VF, HE, HI, WF. Static bending moments due to the weights of nozzles etc. are considered to be local in nature and will not cause a net section bending moment. This is an important consideration when computing the unity check per section 4.4.12 equations 4.4.86, 4.4.87, 4.4.90, 4.4.91 and 4.4.92.

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

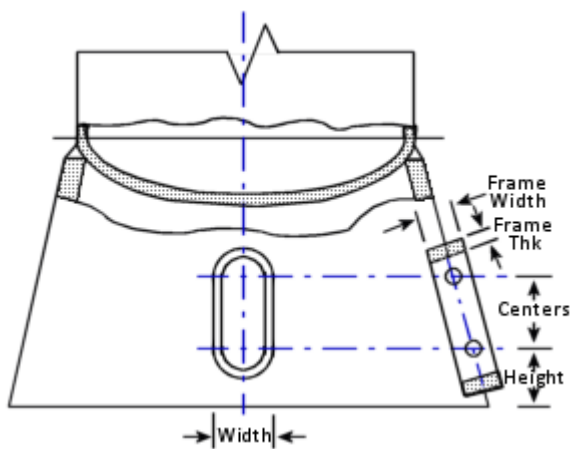
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Top skirt

Total vessel (empty) weight	Fg	1362634.375 Kg
Head Weight below section 2-2	DeltaFg	85803.141 Kg
Weight of vessel contents (liquid)	Ff	463746.188 Kg
Vessel Gross Weight	Fg + Ff	1826380.500 Kg
Skirt Mean Diameter	Dz	7333.600 mm
Skirt Analysis Thickness (Corroded)	ez	32.400 mm
Skirt Height	H	1000.000 mm
Additional Axial Force at section 1-1	F1	-6054.400 Kg
Moment at section 1-1	M1	6691570.000 Kg-m.
Applied Moment - Bottom of Skirt	Mb	6991045.500 Kg-m.
Weight at Bottom of Skirt	W	1827998.250 Kg

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Spherical
Head Mean Diameter at Skirt Top	Db	7274.299 mm
Head Corroded Thickness	eb	62.300 mm



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	501829568.000 cm ⁴
Moment of Inertia	Iyy =	501829568.000 cm ⁴
Section Modulus	W =	1368577536.000 mm ³
Section Metal Area	A =	7464.697 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{\frac{\min(Ixx, Iyy)}{Area}}$$

$$= \sqrt{\frac{\min(0.50183E+09, 0.50183E+09)}{7464.697}}$$

$$= 2592.819 \text{ mm}$$

Skirt Slenderness Ratio [rg/L]:

$$= rg / L$$

$$= 2592.819 / 1000.0$$

$$= 2.593$$

Correct the Moment [M4]:
 top of skirt is the critical end
 Controlling Moment [M4] = 6691570 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:
 $= -M / Z - W / A$
 $= -6691570 / 54743E+10 - 1827998 / 7464.69678$
 $= -36.003 \text{ N./mm}^2$

Upwind Side:
 $= +(M) / Z - W / A$
 $= +6691570 / 54743E+10 - 1827998 / 7464.69678$
 $= -12.028 \text{ N./mm}^2$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:
 $= F1 - Fg - Ff + 4 \cdot M1 / Dz$
 $= -6054.4 - 1362634 - 463746.188 + 4 \cdot 6691570 / 7333.6$
 $= 1817307 \text{ Kgf}$

Force at the joint, 16.12-2 [Fzq]:
 $= F1 - Fg - Ff - 4 \cdot M1 / Dz$
 $= -6054.4 - 1362634 - 463746.188 - 4 \cdot 6691570 / 7333.6$
 $= -5482177.5 \text{ Kgf}$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:
 $= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (1817307 + 85803.141 + 463746.188) / (\pi \cdot 7274.299 \cdot 62.3) + 4550.0 \cdot 7274.299 / (4 \cdot 62.3)$
 $= 149.13 \text{ N./mm}^2$

Stress [σ_{1qm}]:
 $= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (-0.5E+07 + 85803.141 + 463746.188) / (\pi \cdot 7274.299 \cdot 62.3) + 4550.0 \cdot 7274.299 / (4 \cdot 62.3)$
 $= 98.85 \text{ N./mm}^2$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:
 $= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$
 $= (463746.188 + 85803.141) / (\pi \cdot 7274.299 \cdot 62.3) + 4550.0 \cdot 7274.299 / (4 \cdot 62.3)$
 $= 136.61 \text{ N./mm}^2$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:
 $= Fzp / (\pi \cdot Dz \cdot ez)$
 $= 1817307 / (\pi \cdot 7333.6 \cdot 32.4)$
 $= 23.88 \text{ N./mm}^2$

Stress [σ_{3qm}]:
 $= Fzq / (\pi \cdot Dz \cdot ez)$
 $= -5482177.500 / (\pi \cdot 7333.6 \cdot 32.4)$
 $= -72.02 \text{ N./mm}^2$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:
 $= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$
 $= 1 - 0.5(7274.3 + 62.3 - 7333.6 + 32.4) / (3606.0 + 62.3)$
 $= 0.9952$

$\gamma = \arccos(\cos(\gamma)) = \arccos(0.995) = 5.631 \text{ deg}$

Dimension [a]:
 $= 0.5 \cdot \sqrt{ (eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma)) }$
 $= 0.5 \cdot \sqrt{ (62.3^2 + 32.4^2 + 2 \cdot 62.3 \cdot 32.4 \cdot 0.995) }$
 $= 47.299 \text{ mm}$

Bending moment at verification points p and q [Mp & Mq]:

$$\begin{aligned} M_p &= a \cdot F_{zp} = 47.29 \cdot 1817307.2 = 85957.70 \text{ Kg-m.} \\ M_q &= a \cdot F_{zq} = 47.29 \cdot -5482177.5 = -259304.16 \text{ Kg-m.} \end{aligned}$$

Correction factor [C]:

$$\begin{aligned} &= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2 \\ &= 0.63 - 0.057 \left(\frac{62.3}{32.4} \right)^2 \\ &= 0.419 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2) \\ &= 0.419 \cdot 6 \cdot 85957.695 / (\pi \cdot 7274.299 \cdot 62.3^2) \\ &= 23.91 \text{ N./mm}^2 \end{aligned}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2) \\ &= 0.419 \cdot 6 \cdot -259304.156 / (\pi \cdot 7274.299 \cdot 62.3^2) \\ &= -72.12 \text{ N./mm}^2 \end{aligned}$$

Stress [$\sigma_{3pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2) \\ &= 0.419 \cdot 6 \cdot 85957.695 / (\pi \cdot 7333.6 \cdot 32.4^2) \\ &= 87.67 \text{ N./mm}^2 \end{aligned}$$

Stress [$\sigma_{3qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2) \\ &= 0.419 \cdot 6 \cdot -259304.156 / (\pi \cdot 7333.6 \cdot 32.4^2) \\ &= -264.48 \text{ N./mm}^2 \end{aligned}$$

Geometry variables, 16.12-32 to 36:

$$\begin{aligned} eb/D_b &= 2.453/286.39 = 0.0086 \\ y &= 125 \cdot eb/D_b = 125 \cdot 62.3/7274.299 = 1.0705 \\ \alpha &= 0 \end{aligned}$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= 0.0 \text{ N./mm}^2 \text{ (16.12-31)}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$\begin{aligned} &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 149.128 - 23.906 + 0.0 \\ &= 125.22 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1potot}]:

$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 149.128 + 23.906 - 0.0 \\ &= 173.03 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1qitot}]:

$$\begin{aligned} &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\ &= 98.849 - 72.117 + 0.0 \\ &= 170.97 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{1qotot}]:

$$\begin{aligned} &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= 98.849 + 72.117 - 0.0 \\ &= 26.73 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{2pitot}]:

$$\begin{aligned} &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 136.61 + 23.906 + 0.0 \\ &= 160.52 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{2potot}]:

$$\begin{aligned}
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 136.61 - 23.906 - 0.0 \\
 &= 112.70 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{2qitot}]: \\
 &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\
 &= 136.61 + -72.117 + 0.0 \\
 &= 64.49 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{2qotot}]: \\
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 136.61 - -72.117 - 0.0 \\
 &= 208.73 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3pitot}]: \\
 &= \sigma_{3pm} - \sigma_{3pb} \\
 &= 23.875 - 87.675 \\
 &= -63.80 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3potot}]: \\
 &= \sigma_{3pm} + \sigma_{3pb} \\
 &= 23.875 + 87.675 \\
 &= 111.55 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qitot}]: \\
 &= \sigma_{3qm} - \sigma_{3qb} \\
 &= -72.023 - -264.484 \\
 &= 192.46 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qotot}]: \\
 &= \sigma_{3qm} + \sigma_{3qb} \\
 &= -72.023 + -264.484 \\
 &= -336.51 \text{ N./mm}^2
 \end{aligned}$$

Allowable stresses:

$$\begin{aligned}
 \text{For } \sigma_{1pitot} \text{ and } \sigma_{1potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\
 &= 156.4(3 - (1/1.5)) \cdot (149.128/156.4)^2 \\
 &= 374.40 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{1qitot} \text{ and } \sigma_{1qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\
 &= 156.4(3 - (1/1.5)) \cdot (98.849/156.4)^2 \\
 &= 427.55 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2pitot} \text{ and } \sigma_{2potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\
 &= 156.4(3 - (1/1.5)) \cdot (136.61/156.4)^2 \\
 &= 389.65 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2qitot} \text{ and } \sigma_{2qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2 \\
 &= 156.4(3 - (1/1.5)) \cdot (136.61/156.4)^2 \\
 &= 389.65 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3pitot} \text{ \& } \sigma_{3potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2 \\
 &= 156.4(3 - (1/1.5)) \cdot (23.875/156.4)^2 \\
 &= 466.77 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3qitot} \text{ \& } \sigma_{3qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2 \\
 &= 156.4(3 - (1/1.5)) \cdot (-72.023/156.4)^2 \\
 &= 447.09 \text{ N./mm}^2
 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm²)



Stress	Actual	Allowed	Result	Equation
Stot1pi	125.22	374.40	Pass	46
Stot1po	173.03	374.40	Pass	47
Stot1qi	170.97	427.55	Pass	48
Stot1qo	26.73	427.55	Pass	49
Stot2pi	160.52	389.65	Pass	50
Stot2po	112.70	389.65	Pass	51
Stot2qi	64.49	389.65	Pass	52
Stot2qo	208.73	389.65	Pass	53
Stot3pi	-63.80	466.77	Pass	54
Stot3po	111.55	466.77	Pass	55
Stot3qi	192.46	447.09	Pass	56
Stot3qo	-336.51	447.09	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
 Packing is installed in the Shop.
 Trays are installed in the Shop.
 Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	30383.094 mm
Center of Gravity of the Packing	25515.621 mm
Center of Gravity of the Liquid	11841.040 mm
Center of Gravity of the Insulation	27342.502 mm
Center of Gravity of the Nozzles	18883.906 mm
Center of Gravity of the Trays	50349.973 mm
Center of Gravity of the Added Weights (Operating)	31400.443 mm
Center of Gravity of the Added Weights (Empty)	31400.443 mm
Center of Gravity of Bare Shell New and Cold	23747.055 mm
Center of Gravity of Bare Shell Corroded	23793.545 mm
Vessel CG in the Operating Condition	21994.742 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	25676.422 mm
Vessel CG in the Test Condition	25525.189 mm

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	52299.99 mm
Total vessel weight (No Liquid) Twt	1287459.50 Kgf
Impact weight multiplication factor Imp	1.20
Design lifting weight, DWT = Imp * Twt	1544951.50 Kgf
Elevation of the Tailing Lug (bottom)	0.00 mm
Elevation of the Lifting Lug (top)	40000.00 mm
Design Reaction force at the tailing lug	553230.88 Kgf
Design Reaction force at the lifting lug	991720.62 Kgf
CG distance during field lifting operation	25676.42 mm
CG distance from tailing lug	25676.42 mm
CG distance from the nearer lifting lug/trunnion	14323.58 mm
Compressive Allowable Increase Factor	1.20

Critical Values: (N./mm², mm)

	Max Stress	Elevation	Allowables
Bending	30.02	8800.00	107.50
Shear	6.71	39750.02	122.50 (0.7*5)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kgf	Shear Stress N./mm ²
0.00	2.9	0.0	446150.3	5.3
19200.00	6316836.5	12.3	5659.6	0.0
29990.01	4185971.8	8.3	455636.2	1.6
41550.03	1667663.5	13.1	210091.1	1.9
52300.04	43500.7	0.6	6571.6	0.1

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.2792 at elevation 8800.0 mm - Must be <=1



Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	7366.0000	mm
Skirt Thickness	STHK	36.0000	mm
Skirt Internal Corrosion Allowance	SCA	1.6000	mm
Skirt External Corrosion Allowance		0.0000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	52.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	175.00	N./mm ²
Basing Yield Stress		262.00	N./mm ²
Inside Diameter of Basing	DI, D4	7013.9995	mm
Outside Diameter of Basing	DOU, D5	7774.0000	mm
Nominal Diameter of Bolts	BND, d(b0)	76.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Root Area of a Single Bolt	Area	31.3100	cm ²
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	138.00	N./mm ²
Number of Bolts	RN, nb	52	
Diameter of Bolt Circle	DC, DBC	7570.0000	mm
Bolt Allowable Shear Stress		82.800	N./mm ²
Ultimate Comp. Strength of Concrete	FPC	41.4	N./mm ²
Allowable Comp. Strength of Concrete	FC	22.5	N./mm ²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	25.0000	mm
Width of Gussets at Top Plate	TWDT	194.0000	mm
Width of Gussets at Base Plate	BWDT	194.0000	mm
Gusset Plate Elastic Modulus	E	201734000.0	KPa.
Gusset Plate Yield Stress	SY	262.0	N./mm ²
Height of Gussets	HG, h1	334.0000	mm
Distance between Gussets	RG, b6	140.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	70.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	64.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	204.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	102.0000	mm
External Corrosion Allowance	CA	1.6000	mm
Dead Weight of Vessel	DW	1287459.4	Kgf
Operating Weight of Vessel	ROW	1895043.6	Kgf
Earthquake Moment on Basing	EQMOM	9641689.0	Kg-m.
Wind Moment on Basing	WIMOM	1181000.2	Kg-m.
User Moment at the Base [Earthquake Case]		349105.3	Kg-m. (UEQMOM)
User Moment at the Base [WindLoad Case]		349105.3	Kg-m. (UWIMOM)
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 9990794/7570.0) - 1835805)/52$$

$$= 66216.2 \text{ Kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 1530106/7570.0) - 1287459)/52$$

$$= -9210.9 \text{ Kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/7570.0) - 1895044)/52$$

$$= -36443.1 \text{ Kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abss : 31.3100 cm²
 Area Available in all the Bolts Abss * RN : 1628.1200 cm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.300	0.451	1.511	2.441	0.438	0.781	661175.9	2496980.8	6743.6
4	0.376	0.413	1.705	2.277	0.421	0.784	697604.9	2533410.0	7626.5
6	0.394	0.404	1.751	2.236	0.417	0.784	707012.7	2542817.5	7871.7
8	0.408	0.406	1.785	2.205	0.414	0.785	714151.3	2549956.2	8063.0
10	0.405	0.405	1.777	2.213	0.414	0.784	712359.9	2548164.8	8014.5
12	0.405	0.405	1.777	2.212	0.414	0.784	712509.0	2548314.0	8018.6
13	0.405	0.405	1.777	2.213	0.414	0.784	712434.4	2548239.5	8016.5

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 712434 / (6.846 \cdot 7570.0 \cdot 2.213)$$

$$= 121.856 \text{ N./mm}^2, \text{ Should be } \leq 138.0 \text{ N./mm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot Bolt \text{ Area} / (\pi \cdot Dc)$$

$$= 52 \cdot 31.31 / (\pi \cdot 7570.0)$$

$$= 6.846 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 8.436[(2 \cdot 0.405 \cdot 7570.0 + 380.0) / (2 \cdot 0.405 \cdot 7570.0)]$$

$$= 8.958 \text{ N./mm}^2, \text{ Should be } \leq 22.5 \text{ N./mm}^2$$

Values for table 10.3, l = 204.0, b = 228.672, l/b = 0.892108

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(4002.817, 5852.662) = 5852.662 \text{ Kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 5852.662 / 172.9)} + 1.6$$

$$= 46.227 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 204.0, b = 140.0, b/l = 0.686275
 As b/l (0.686) is less than 1, inverting b/l = 1.457.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l) / (\pi \cdot g))) + 1)] - [(0.7 - g2)P / (4\pi)]$$

$$= 44058.88 / (4 \cdot \pi) [1.3(\ln((2 \cdot 204.0 \cdot \sin(\pi \cdot 102.0/204.0)) / (\pi \cdot 58.737))) + 1] - [(0.7 - 0.076) \cdot 44058.88 / (4 \cdot \pi)]$$

$$= 4933.5591 \text{ Kgf}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(4933.56) / 174.66) + 1.6}$$

$$= 42.3680 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 44058.88 / (124.11 \cdot 194.0) + 1.6$$

$$= 19.545 \text{ (not less than } 9.525 + 1.6 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 7570.0 / 52 = 457.343 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76(P \cdot a / (m \cdot (h + tba)) \cdot 1.5 \cdot \text{Sktope})^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76(44059 \cdot 102.0 / (457.34 \cdot 386.0 \cdot 242))^{2/3} \cdot 3683.01^{1/3} + \text{Ca}$$

$$= 27.723 + 1.6 = 29.323 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 300906.7 / (31.31 \cdot 52)$$

$$= 18.1 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	46.2270	mm
Actual Basing Thickness as entered by user	52.0000	mm
Required Thickness of Chair Cap	42.3680	mm
Actual Top Ring Thickness as entered by user	64.0000	mm
Required Gusset thickness, + CA	19.5454	mm
Actual Gusset Thickness as entered by user	25.0000	mm
Required Thickness of Skirt for Local Stress	29.3226	mm
Given Thickness of Skirt	36.0000	mm
Required Gusset Height to meet local stress	227.2550	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 90.885 \cdot (36.0 - 1.6)$$

$$= 318.8 \text{ Kg/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 319 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707]$$

$$= 22.785 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 44058.9 / (122.0 + 2 \cdot (334.0 + 64.0))$$

$$= 47.994 \text{ Kg/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 44058.9 \cdot 102.0 / (122.0 \cdot (398.0) + 0.6667 \cdot (398.0)^2)$$

$$= 29.152 \text{ Kg/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{47.99^2 + 29.15^2}$$

$$= 56.154 \text{ Kg/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:



$$\begin{aligned} &= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 56.15 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707] \\ &= 4.013 \text{ mm} \end{aligned}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$\begin{aligned} &= \text{Bolt Load} / (2 \cdot \text{TopWth}) \\ &= 44058.9 / (2 \cdot 194.0) \\ &= 113.554 \text{ Kg/mm} \end{aligned}$$

Weld Load [Wh]:

$$\begin{aligned} &= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth}) \\ &= 44058.9 \cdot 102.0 / (2 \cdot 398.0 \cdot 194.0) \\ &= 29.102 \text{ Kg/mm} \end{aligned}$$

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{W_v^2 + W_h^2} \\ &= \sqrt{113.55^2 + 29.1^2} \\ &= 117.224 \text{ Kg/mm} \end{aligned}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$\begin{aligned} &= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 117.22 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707] \\ &= 8.378 \text{ mm} \end{aligned}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	22.7852	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	8.3782	mm
Required Gusset to Top Plate Double Fillet Weld Size	8.3782	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Conical Calculations per ASME VIII-2, 4.3.11

Input Values:

cone

Design Internal Pressure, Small End	4563.000	KPa.
Design Internal Pressure, Large End	4563.000	KPa.
Temperature for Internal Pressure	130.0	°C
Design External Pressure	0.000	KPa.
Temperature for External Pressure	0.0	°C

CONE Properties:

Cone Material		SA-516 70 +IT
Cone Allowable Stress at Temperature	Ss	156.40 N./mm ²
Cone Allowable Stress At Ambient		175.00 N./mm ²
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00
Circumferential Joint Efficiency of Cone		1.00
Actual Thickness of Cone	Tc	139.5000 mm
Corrosion Allowance for Cone	Cac	6.0000 mm
Diameter Basis for Cone and Cylinders		ID
Diameter of Small End of Cone	Ds	4600.000 mm
Diameter of Large End of Cone	Dl	7200.000 mm
Half Apex Angle for Cone		27.47 degrees
Axial Length of Cone	Lc	2500.000 mm

SMALL-END Properties:

Attached Small End Material Specification		SA-516 70 +IT
Operating Allowable Stress	Ss	156.40 N./mm ²
Ambient Allowable Stress		175.00 N./mm ²
Longitudinal Joint Efficiency	Es,E1	1.00
Actual Thickness	Ts	76.0000 mm
Corrosion Allowance	Cas	3.0000 mm
Axial Length of Attached Element	Ls	19916.668 mm

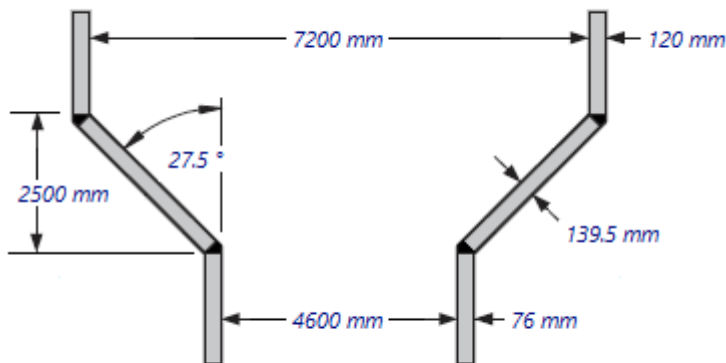
LARGE-END Properties:

Attached Large End Material Specification		SA-516 70 +IT
Operating Allowable Stress	Sl	156.40 N./mm ²
Ambient Allowable Stress		175.00 N./mm ²
Longitudinal Joint Efficiency	El,E1	1.00
Actual Thickness	Tl	120.0000 mm
Corrosion Allowance	Cal	6.0000 mm
Axial Length of Attached Element	Ll	22850.000 mm

Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: None

Warning: rtk -6.000mm, must be > Tl, 114.000mm, see 4.3.12



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 0 °C

Elastic Modulus of Cone Material	0.203E+09 KPa. at 0 °C
Elastic Modulus of Small Cylinder Material	0.204E+09 KPa. at 0 °C
Elastic Modulus of Large Cylinder Material	0.203E+09 KPa. at 0 °C
Elastic Modulus of Large End Reinforcement	0.203E+09 KPa. at 0 °C
Elastic Modulus of Small End Reinforcement	0.204E+09 KPa. at 0 °C

Elastic Modulus Data from ASME Section II Part D at 130 °C

Elastic Modulus of Cone Material	0.195E+09 KPa. at 130 °C
Elastic Modulus of Small Cylinder Material	0.196E+09 KPa. at 130 °C
Elastic Modulus of Large Cylinder Material	0.195E+09 KPa. at 130 °C
Elastic Modulus of Large End Reinforcement	0.195E+09 KPa. at 130 °C
Elastic Modulus of Small End Reinforcement	0.196E+09 KPa. at 130 °C

Forces and Moments on the ends of the Cone:

Axial Force on Small End of Cone	-300795.72 Kg-f
Axial Force on Large End of Cone	-433104.66 Kg-f
Moment on Small End of Cone	1649576.88 Kg-m.
Moment on Large End of Cone	2284475.50 Kg-m.

The moments above are computed from the Wind & Seismic loads as well as any user defined moments and include effects due to eccentric masses. The vertical forces include the weight of the elements and details above and below the cone as well as any user defined vertical forces. Vertical forces also include loads cause from vertical accelerations due to earthquake loading.

Maximum Centroid Reinforcement Distance Large End	25.4000 mm
Maximum Centroid Reinforcement Distance Small End	25.4000 mm

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

Junction Calculations per 4.3.11:

Calculations for the Large End Cone/Shell Junction - Without Knuckle

Fhe per 4.1.3 or 4.3.11.2 Step 6 [Fhe]:
 $= 0.4 \cdot E(tc/D)$
 $= 0.4 \cdot 0.20271E+09(133.5/7440.0)$
 $= 1455.0 \text{ N./mm}^2$

Factor of Safety [FS]:
 Since $Fic [234.4] \geq Fy[234.4]$, FS = 1.667

Allowable Stress per 4.3.11 Step 6 [Fha]:

$$= F_{ic}/FS$$

$$= 234.4/1.667$$

$$= 140.612 \text{ N./mm}^2$$

Large End under Internal Pressure:

Cone Length along Cone surface per 4.3.46 [ConeMinLen]:

$$= 2 \cdot \sqrt{(RL \cdot Tc/\cos(\alpha)) + 1.4 \cdot \sqrt{(Rs \cdot tc/\cos(\alpha))}}$$

$$= 2 \cdot \sqrt{(3606.763 \cdot 133.5/\cos(27.47)) + 1.4 \cdot \sqrt{(2303.381 \cdot 133.5/\cos(27.47))}}$$

$$= 2297.519 \text{ mm}$$

Results for the Large End Cylinder Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	75.351	234.600	Pass
Sigma sm-	65.734	234.600	Pass
Sigma thetam+	17.383	234.600	Pass
Sigma thetam-	32.812	234.600	Pass
Sigma sm+ + Sigma	-166.180	469.200	Pass
Sigma sm+ - Sigma	316.882	469.200	Pass
Sigma sm- + Sigma	-143.558	469.200	Pass
Sigma sm- - Sigma	275.027	469.200	Pass
Sigma thetam+ +	-55.076	469.200	Pass
Sigma thetam+ -	89.842	469.200	Pass
Sigma thetam- +	-29.976	469.200	Pass
Sigma thetam- -	95.600	469.200	Pass

Results for the Large End Cone Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	64.172	234.600	Pass
Sigma sm-	55.982	234.600	Pass
Sigma thetam+	8.087	234.600	Pass
Sigma thetam-	24.225	234.600	Pass
Sigma sm+ + Sigma	-111.953	469.200	Pass
Sigma sm+ - Sigma	240.297	469.200	Pass
Sigma sm- + Sigma	-96.635	469.200	Pass
Sigma sm- - Sigma	208.598	469.200	Pass
Sigma thetam+ +	-44.750	469.200	Pass
Sigma thetam+ -	60.925	469.200	Pass
Sigma thetam- +	-21.560	469.200	Pass
Sigma thetam- -	70.010	469.200	Pass

Cone Large end without a Knuckle Cylinder Section:

Net Axial Force ML +ve [XL+]:

$$= FL / (2 \cdot \pi \cdot RL) + ML / (\pi \cdot RL^2)$$

$$= -433104.656 / (2 \cdot 3.141 \cdot 3606.763) + 2284476 / (3.142 \cdot 3606.763^2)$$

$$= 36.786 \text{ Kg/mm}$$

Net Axial Force ML -ve [XL-]:

$$= FL / (2 \cdot \pi \cdot RL) - ML / (\pi \cdot RL^2)$$

$$= -433104.656 / (2 \cdot 3.142 \cdot 3606.763) - 2284476 / (3.141 \cdot 3606.763^2)$$

$$= -75.0091 \text{ Kg/mm}$$

Thickness Ratio [n]:

$$= tc/tL = 133.5/114.0 = 1.1711$$

Curve Fitting Geometry [H and B]:

$$H = \text{Sqrt}(0.5 \cdot D/tL) = \text{Sqrt}(0.5 \cdot 7213.525/114.0) = 5.6248$$

$$B = \tan(27.474) = 0.52$$

This analysis is preformed per Table 4.3.1 for the Large End:

Using Data from Table 4.3.3 and 4.3.4 knowing n
 Table 4.3.3 - n: $1.0 < (n=1.1711) < 1.25$

Use n to construct a table: C1, C2, C3..... by interpolation

Table 4.3.3 [MsN1]:

C1 = -3.098354 C2 = 3.687418 C3 = 0.759848 C4 = -0.233480
 C5 = -0.077657 C6 = 0.067976 C7 = 0.026909 C8 = -0.015581
 C9 = 0.001343 C10 = -0.010853 C11 = 0.000000

Compute using Table 4.3.3 [MsN1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) + C11 \cdot \ln(H) \cdot \ln(B)^2$$

$$= C1 + C2 \cdot \ln(5.625) + C3 \cdot \ln(0.52) + C4 \cdot (\ln(5.625))^2 + C5 \cdot (\ln(0.52))^2 + C6 \cdot \ln(5.625) \cdot \ln(0.52) + C7 \cdot (\ln(5.625))^3 + C8 \cdot (\ln(0.52))^3 + C9 \cdot \ln(5.625) \cdot (\ln(0.52))^2 + C10 \cdot (\ln(5.625))^2 \cdot \ln(0.52) + C11 \cdot \ln(5.625) \cdot (\ln(0.52))^2$$

$$X = 2.1323, \text{ MsN1} = -\exp(X) = -8.4341$$

Table 4.3.3 [Qn1]:

C1 = -1.934263 C2 = 2.329532 C3 = 0.524897 C4 = -0.091792
 C5 = -0.087663 C6 = 0.188851 C7 = 0.008801 C8 = -0.020148
 C9 = 0.000836 C10 = -0.028395 C11 = 0.000000

Compute using Table 4.3.3 [Qn1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) + C11 \cdot \ln(H) \cdot \ln(B)^2$$

$$= C1 + C2 \cdot \ln(5.625) + C3 \cdot \ln(0.52) + C4 \cdot (\ln(5.625))^2 + C5 \cdot (\ln(0.52))^2 + C6 \cdot \ln(5.625) \cdot \ln(0.52) + C7 \cdot (\ln(5.625))^3 + C8 \cdot (\ln(0.52))^3 + C9 \cdot \ln(5.625) \cdot (\ln(0.52))^2 + C10 \cdot (\ln(5.625))^2 \cdot \ln(0.52) + C11 \cdot \ln(5.625) \cdot (\ln(0.52))^2$$

$$X = 1.3284, \text{ Qn} = -\exp(X) = -3.775$$

Table 4.3.4 MsN2:

C1 = -5.717905 C2 = 0.005839 C3 = 0.473249 C4 = -0.213052
 C5 = 2.250697 C6 = 0.000017 C7 = 0.003089 C8 = -0.001948
 C9 = -0.215461 C10 = -0.000030 C11 = -0.106248

Compute using Table 4.3.4 [MsN2]:

$$X1 = C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A) + C10 \cdot \ln(H^2) \cdot \ln(A)$$

$$= C1 + C3 \cdot \ln(5.625^2) + C5 \cdot \ln(0.48) + C7 \cdot (\ln(5.625^2))^2 + C9 \cdot (\ln(0.48))^2 + C11 \cdot \ln(5.625^2) \cdot \ln(0.48) + C10 \cdot \ln(5.625^2) \cdot \ln(0.48)$$

$$X2 = 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR) + C11 \cdot \ln(H^2) \cdot \ln(AR)$$

$$= 1 + C2 \cdot \ln(5.625) + C4 \cdot \ln(0.48) + C6 \cdot (\ln(5.625))^2 + C8 \cdot (\ln(0.48))^2 + C10 \cdot \ln(5.625) \cdot \ln(0.48) + C11 \cdot \ln(5.625) \cdot \ln(0.48)$$

$$\text{MsN2} = \exp(X1/X2) = \exp(-0.17/0.293) = -0.5587$$

Table 4.3.4 QN2:

C1 = -5.014468 C2 = 0.015113 C3 = -0.039161 C4 = -0.196933
 C5 = 2.101579 C6 = -0.000286 C7 = 0.000055 C8 = -0.004771
 C9 = -0.217121 C10 = -0.001688 C11 = 0.009119

Compute using Table 4.3.4 [QN2]:

$$X3 = C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A) + C10 \cdot \ln(H^2) \cdot \ln(A)$$

$$= C1 + C3 \cdot \ln(5.625^2) + C5 \cdot \ln(0.48) + C7 \cdot (\ln(5.625^2))^2 + C9 \cdot (\ln(0.48))^2 + C11 \cdot \ln(5.625^2) \cdot \ln(0.48) + C10 \cdot \ln(5.625^2) \cdot \ln(0.48)$$

$$X4 = 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR) + C11 \cdot \ln(H^2) \cdot \ln(AR)$$

$$= 1 + C2 \cdot \ln(5.625) + C4 \cdot \ln(0.48) + C6 \cdot (\ln(5.625))^2 + C8 \cdot (\ln(0.48))^2 + C10 \cdot \ln(5.625) \cdot \ln(0.48) + C11 \cdot \ln(5.625) \cdot \ln(0.48)$$

$$\text{Qn2} = \exp(X3/X4) = \exp(-0.465/0.325) = -0.2386$$

Stresses for the Large End Cylinder per Table 4.3.1

Stress Resultant Value [Msp]:

$$= P \cdot t \cdot L^2 \cdot \text{MsN1} = 4563.0 \cdot 114.0^2 \cdot -8.434 = -51003.1562 \text{ Kgf}$$

Compute with XL+ [Msx+]:

$$= XL \cdot t \cdot L \cdot \text{MsN2} = 36.786 \cdot 114.0 \cdot -0.559 = -2342.9429 \text{ Kgf}$$

Compute with XL- [Msx-]:

$$= XL \cdot t \cdot L \cdot \text{MsN2} = -75.009 \cdot 114.0 \cdot -0.559 = 4777.4111 \text{ Kgf}$$

Compute with XL+ [Ms+]:

$$= \text{MsP} + \text{Msx} = -51003.156 + -2342.943 = -53346.0977 \text{ Kgf}$$

Compute with XL- [Ms-]:

$$= \text{MsP} + \text{Msx} = -51003.156 + 4777.411 = -46225.7422 \text{ Kgf}$$

Compute [Qp]:
 $= P \cdot tL \cdot Qn1 = 4563.0 \cdot 114.0 \cdot -3.775 = -200.2473 \text{ Kgf/mm}$

Compute with XL+ [Qx+]:
 $= XL \cdot QN2 = 36.786 \cdot -0.004 = -8.7779 \text{ Kgf/mm}$

Compute with XL- [Qx-]:
 $= XL \cdot QN2 = -75.009 \cdot -0.004 = 17.8987 \text{ Kgf/mm}$

Compute with XL+ [Q+]:
 $= Qp + Qx = -200.247 + -8.778 = -209.0252 \text{ Kgf/mm}$

Compute with XL- [Q-]:
 $= Qp + Qx = -200.247 + 17.899 = -182.3486 \text{ Kgf/mm}$

Compute [β_{cy}]:
 $= 3 \cdot (1 - v^2) / (RL^2 \cdot tL^2) \cdot 0.25$
 $= (3 \cdot (1 - 0.3^2) / (3606.763^2 \cdot 114.0^2)) \cdot 0.25 = 0.002 \text{ 1/mm}$

Compute with XL+ [Ns+]:
 $= P \cdot RL/2 + XL = 4563.0 \cdot 3606.763/2 + 36.786 = 875.9264 \text{ Kgf/mm}$

Compute with XL- [Ns-]:
 $= P \cdot RL/2 + XL = 4563.0 \cdot 3606.763/2 + -75.009 = 764.1312 \text{ Kgf/mm}$

Compute with XL+ [Ntheta+]:
 $= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q)$
 $= 4563.0 \cdot 3606.763 + 2 \cdot 0.002 \cdot 3606.763 \cdot (-53346.098 \cdot 0.002 + -209.025) = 202.0702 \text{ Kgf/mm}$

Compute with XL- [Ntheta-]:
 $= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q)$
 $= 4563.0 \cdot 3606.763 + 2 \cdot 0.002 \cdot 3606.763 \cdot (-46225.742 \cdot 0.002 + -182.349) = 381.4225 \text{ Kgf/mm}$

Note: Kpc has been set to 1.0

σ_{Sm} with XL+ [σ_{sm+}]:
 $= Ns / tL$
 $= 875.926/114.0$
 $= 75.351 \text{ N./mm}^2$

σ_{Sm} with XL- [σ_{sm-}]:
 $= Ns / tL$
 $= 764.131/114.0$
 $= 65.734 \text{ N./mm}^2$

σ_{Sb} with XL+ [σ_{sb+}]:
 $= 6 \cdot Ms / (tL^2 \cdot Kpc)$
 $= 6 \cdot -53346.098 / (114.0^2 \cdot 1.0)$
 $= -241.531 \text{ N./mm}^2$

σ_{Sb} with XL- [σ_{sb-}]:
 $= 6 \cdot Ms / (tL^2 \cdot Kpc)$
 $= 6 \cdot -46225.742 / (114.0^2 \cdot 1.0)$
 $= -209.293 \text{ N./mm}^2$

σ_{Thetam} with XL+ [$\sigma_{thetam+}$]:
 $= Ntheta / tL$
 $= 202.07/114.0 \text{ N./mm}^2$
 $= 17.383 \text{ N./mm}^2$

σ_{Thetam} with XL- [$\sigma_{thetam-}$]:
 $= Ntheta / tL$
 $= 381.422/114.0$
 $= 32.812 \text{ N./mm}^2$

σ_{Thetab} with XL+ [$\sigma_{thetab+}$]:
 $= 6 \cdot v \cdot Ms / (tL^2 \cdot Kpc)$

$$= 6 \cdot 0.3 \cdot -53346.098 / (114.0^2 \cdot 1.0)$$

$$= -72.459 \text{ N./mm}^2$$

σ_{Thetab} with XL- [σ_{thetab}]:

$$= 6 \cdot v \cdot Ms / (tL^2 \cdot Kpc)$$

$$= 6 \cdot 0.3 \cdot -46225.742 / (114.0^2 \cdot 1.0)$$

$$= -62.788 \text{ N./mm}^2$$

For the Large End Cone Junction per Table 4.3.1

Compute [McsP]:

$$= Msp = Msp = -51003.1562 \text{ Kgf}$$

McsX with XL+ [McsX+]:

$$= MsX = -2342.9429 \text{ Kgf}$$

McsX with XL- [McsX-]:

$$= MsX = 4777.4111 \text{ Kgf}$$

Mcs with XL+ [Mcs+]:

$$= McsP + MsX = -51003.156 + -2342.943$$

$$= -53346.098 \text{ Kgf}$$

Mcs with XL- [Mcs-]:

$$= McsP + MsX = -51003.156 + 4777.411$$

$$= -46225.742 \text{ Kgf}$$

Compute with XL+ [McsX+]:

$$= MsX = -2342.9429 \text{ Kgf}$$

Compute with XL- [McsX-]:

$$= MsX = 4777.4111 \text{ Kgf}$$

Compute with XL+ [Qc+]:

$$= Q \cdot \cos(a) + Ns \cdot \sin(a)$$

$$= -209.025 \cdot \cos(27.474) + 875.926 \cdot \sin(27.474)$$

$$= 218.660 \text{ Kgf/mm}$$

Compute with XL- [Qc-]:

$$= Q \cdot \cos(a) - Ns \cdot \sin(a)$$

$$= -182.349 \cdot \cos(27.474) + 764.131 \cdot \sin(27.474)$$

$$= 190.751 \text{ Kgf/mm}$$

Compute [Rc]:

$$= RL/\cos(a) = 3606.763/\cos(27.474) = 4065.2554 \text{ mm}$$

Compute [β -co]:

$$= (3 \cdot (1 - v^2) / (Rc^2 \cdot tC^2))^{1/4}$$

$$= (3 \cdot (1 - 0.3^2) / (4065.255^2 \cdot 133.5^2))^{1/4}$$

$$= 0.002 \text{ 1/mm}$$

Compute with XL +ve [Ncs+]:

$$= Ns \cdot \cos(a) - Q \cdot \sin(a)$$

$$= 875.926 \cdot \cos(27.474) - -209.025 \cdot \sin(27.474)$$

$$= 873.571 \text{ Kgf/mm}$$

Compute with XL -ve [Ncs-]:

$$= Ns \cdot \cos(a) + Q \cdot \sin(a)$$

$$= 764.131 \cdot \cos(27.474) - -182.349 \cdot \sin(27.474)$$

$$= 762.077 \text{ Kgf/mm}$$

Compute with XL +ve [NcTheta+]:

$$= P \cdot RL/\cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc)$$

$$= 4563.0 \cdot 3606.763/\cos(27.474) + 2 \cdot 0.002 \cdot 4065.255 \cdot$$

$$(-53346.098 \cdot 0.002 - 218.66)$$

$$= 110.092 \text{ Kgf/mm}$$

Compute with XL -ve [NcTheta-]:

$$= P \cdot RL/\cos(A) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc)$$

$$= 4563.0 \cdot 3606.763/\cos(27.474) + 2 \cdot 0.002 \cdot 4065.255 \cdot$$

$$(-46225.742 \cdot 0.002 - 190.751)$$

$$= 329.772 \text{ Kgf/mm}$$

Note: $K_{cpc} = 1.0$

Compute with XL +ve $[\sigma_{sm+}]$:

$$= N_{cs}/t_c = 873.571/133.5 = 64.172 \text{ N./mm}^2$$

Compute with XL -ve $[\sigma_{sm-}]$:

$$= N_{cs}/t_c = 762.077/133.5 = 55978.4805 \text{ N./mm}^2$$

Compute with XL +ve $[\sigma_{sb+}]$

$$= 6 \cdot M_{cs}/(t_c^2 \cdot K_{cpc})$$

$$= 6 \cdot -53346.098/(133.5^2 \cdot 1.0)$$

$$= -176.125 \text{ N./mm}^2$$

Compute with XL -ve $[\sigma_{sb-}]$

$$= 6 \cdot M_{cs}/(t_c^2 \cdot K_{cpc})$$

$$= 6 \cdot -46225.742/(133.5^2 \cdot 1.0)$$

$$= -152.616 \text{ N./mm}^2$$

Compute with XL +ve $[\sigma_{\theta_{tam+}}]$:

$$= N_{C\theta}/t_c = 110.092/133.5 = 8.09 \text{ N./mm}^2$$

Compute with XL -ve $[\sigma_{\theta_{tam-}}]$:

$$= N_{C\theta}/t_c = 329.772/133.5 = 24.22 \text{ N./mm}^2$$

Compute with XL +ve $[\sigma_{\theta_{tab+}}]$:

$$= 6 \cdot v \cdot M_{cs}/(t_c^2 \cdot K_{cpc})$$

$$= 6 \cdot 0.3 \cdot -53346.098/(133.5^2 \cdot 1.0)$$

$$= -52.837 \text{ N./mm}^2$$

Compute with XL -ve $[\sigma_{\theta_{tab-}}]$:

$$= 6 \cdot v \cdot M_{cs} / (t_c^2 \cdot K_{cpc})$$

$$= 6 \cdot 0.3 \cdot -46225.742/(133.5^2 \cdot 1.0)$$

$$= -45.785 \text{ N./mm}^2$$

Calculations for the Small End Cone/Shell Junction

Fhe per 4.1.3 or 4.3.11.2 Step 6 [Fhe]:

$$= 0.4 \cdot E(t_c/D)$$

$$= 0.4 \cdot 0.20409E+09(133.5/4752.0)$$

$$= 2293.6 \text{ N./mm}^2$$

Factor of Safety [FS]:

Since $F_{ic} [234.4] \geq F_y [234.4]$, FS = 1.667

Allowable Stress per 4.3.11 Step 6 [Fha]:

$$= F_{ic}/FS$$

$$= 234.4/1.667$$

$$= 140.612 \text{ N./mm}^2$$

Small End under Internal Pressure:

Results for Cylinder Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	82.500	234.600	Pass
Sigma sm-	55.901	234.600	Pass
Sigma thetam+	222.456	234.600	Pass
Sigma thetam-	187.499	234.600	Pass
Sigma sm+ + Sigma	309.872	469.200	Pass
Sigma sm- + Sigma	197.485	469.200	Pass
Sigma sm+ - Sigma	-144.873	469.200	Pass
Sigma sm- - Sigma	-85.684	469.200	Pass
Sigma thetam+ +	290.668	469.200	Pass
Sigma thetam- +	229.975	469.200	Pass
Sigma thetam+ -	154.244	469.200	Pass

Sigma thetam- - 145.024 469.200 Pass

Results for the Cone Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	43.583	234.600	Pass
Sigma sm-	29.242	234.600	Pass
Sigma thetam+	209.668	234.600	Pass
Sigma thetam-	179.032	234.600	Pass
Sigma sm+ + Sigma	111.569	469.200	Pass
Sigma sm- + Sigma	71.577	469.200	Pass
Sigma sm+ - Sigma	-24.404	469.200	Pass
Sigma sm- - Sigma	-13.093	469.200	Pass
Sigma thetam+ +	230.064	469.200	Pass
Sigma thetam- +	191.732	469.200	Pass
Sigma thetam+ -	189.272	469.200	Pass
Sigma thetam- -	166.331	469.200	Pass

Small End Cylinder Without a Flare

Compute the ratio equation 4.3.60 [n]:
 $= tc/ts = 133.5/73.0 = 1.8288$

Compute the ratio equation 4.3.61 [H]:
 $= \sqrt{(Rs/ts)} = \sqrt{(2303.381/73.0)} = 5.6172$

Compute the ratio equation 4.3.62 [B]:
 $= \tan(a) = \tan(27.474) = 0.52$

Compute Ms+ from equation 4.3.59 [Xs+]:
 $= Fs / (2 \cdot \pi \cdot Rs) + Ms / (\pi \cdot Rs^2)$
 $= -300795.719 / (2 \cdot 3.142 \cdot 2303.381) + 0.16495E+10 / (3.141 \cdot 2303.381^2)$
 $= 78.2107 \text{ Kgf/mm}$

Compute Ms- from equation 4.3.59 [Xs-]:
 $= Fs / (2 \cdot \pi \cdot Rs) - Ms / (\pi \cdot Rs^2) + XS'$
 $= -300795.719 / (2 \cdot 3.142 \cdot 2303.381) - 0.16495E+10 / (3.141 \cdot 2303.381^2) + 0.0$
 $= -119.7852 \text{ Kgf/mm}$

From Table 4.3.5 MsN:

C1 = -15.432117 C2 = 3.108220 C3 = 6.587597 C4 = -0.162176
 C5 = -1.482143 C6 = -0.390489 C7 = 0.007941 C8 = 0.144160
 C9 = 0.042587 C10 = 0.008039 C11 = 0.000000

Compute from Table 4.3.5 [MsN]:
 $= \exp[C1 + C2 \cdot \ln(H^2) + C3 \cdot \ln(a) + C4 \cdot (\ln(H^2))^2 + C5 \cdot \ln(a)^2 + C6 \cdot \ln(H^2) \cdot \ln(a) +$
 $C7 \cdot \ln(H^2)^3 + C8 \cdot \ln(a)^3 + C9 \cdot \ln(H^2) \cdot \ln(a)^2 + C10 \cdot \ln(H^2)^2 \cdot \ln(a)]$
 $= \exp[-15.432 + 3.108 \cdot \ln(5.617^2) + 6.588 \cdot \ln(27.474) +$
 $-0.162 \cdot (\ln(5.617^2))^2 + -1.482 \cdot \ln(27.474)^2 +$
 $-0.39 \cdot \ln(5.617^2) \cdot \ln(27.474) + 0.008 \cdot \ln(5.617^2)^3$
 $0.144 \cdot \ln(27.474)^3 + 0.043 \cdot \ln(5.617^2) \cdot \ln(27.474)^2 +$
 $0.008 \cdot \ln(5.617^2)^2 \cdot \ln(27.474)] = 7.0668$

Compute from Table 4.3.2 [Msp]:
 $= p \cdot tL^2 \cdot Msn = 4563.0 \cdot 73.0^2 \cdot 7.06678 = 17523.2363 \text{ Kgf}$

From table 4.3.6 MsN:

C1 = 0.007058 C2 = 0.000304 C3 = -0.000922 C4 = 0.121542
 C5 = 0.010611 C6 = -0.000011 C7 = -0.000008 C8 = 0.005878
 C9 = 0.001229 C10 = 0.000190 C11 = 0.193748

Compute from Table 4.3.6 [MsN]:
 $= (C1 + C3 \cdot H + C5 \cdot B + C7 \cdot H^2 + C9 \cdot B^2 + C11 \cdot H \cdot B) /$
 $(1 + C2 \cdot H + C4 \cdot B + C6 \cdot H^2 + C8 \cdot B^2 + C10 \cdot H \cdot B)$
 $= (0.00706 + -0.00092 \cdot 5.617 + 0.01061 \cdot 27.474 +$
 $-0.00001 \cdot 5.617^2 + 0.00123 \cdot 27.474^2 +$
 $0.19375 \cdot 5.617 \cdot 27.474) / (1 + 0.0003 \cdot 5.617 +$
 $0.12154 \cdot 27.474 + -0.00001 \cdot 5.617^2 + 0.00588 \cdot 27.474^2 +$
 $0.00019 \cdot 5.617 \cdot 27.474)$

$$= 0.5375$$

Compute from Table 4.3.2 [Msx+]:

$$= X_s \cdot t_s \cdot M_s N = 78.211 \cdot 73.0 \cdot 0.538 = 3069.0486 \text{ Kgf}$$

Compute from Table 4.3.2 [Msx-]:

$$= X_s \cdot t_s \cdot M_s N = -119.785 \cdot 73.0 \cdot 0.538 = -4700.4678 \text{ Kgf}$$

Compute from Table 4.3.2 [Ms+]:

$$= M_s + M_{sx+} = 17523.236 + 3069.049 = 20592.2852 \text{ Kgf}$$

Compute from Table 4.3.2 [Ms-]:

$$= M_s + M_{sx-} = 17523.236 + -4700.468 = 12822.7686 \text{ Kgf}$$

From table 4.3.5 Qn:

$$\begin{aligned} C1 &= 0.579654 & C2 &= -0.000028 & C3 &= 0.008556 & C4 &= 0.002685 \\ C5 &= -0.002941 & C6 &= 0.000000 & C7 &= -0.000005 & C8 &= -0.000117 \\ C9 &= -0.000087 & C10 &= 0.000001 & C11 &= -0.003760 \end{aligned}$$

Compute from Table 4.3.5 [Qn]:

$$\begin{aligned} &= (C1 + C3 \cdot H^2 + C5 \cdot \alpha + C7 \cdot H^4 + C9 \cdot \alpha^2 + C11 \cdot H^2 \cdot \alpha) / \\ &\quad (1 + C2 \cdot H^2 + C4 \cdot \alpha + C6 \cdot H^4 + C8 \cdot \alpha^2 + C10 \cdot H^2 \cdot \alpha) \\ &= (0.57965 + 0.00856 \cdot 5.617^2 + -0.00294 \cdot 27.474 \\ &\quad -0.00001 \cdot 5.617^4 + -0.00009 \cdot 27.474^2 + -0.00376 \cdot 5.617^2 \\ &\quad \cdot 27.474) / (1 + -0.00003 \cdot 5.617^2 + 0.00268 \cdot 27.474 + \\ &\quad 0 \cdot 5.617^4 + -0.00012 \cdot 27.474^2 + 0.000001 \cdot 5.617^2 \cdot 27.474) \\ &= -2.598 \end{aligned}$$

Compute from Table 4.3.2 [Qp]:

$$= P \cdot t_s \cdot Q_n = 4563.0 \cdot 73.0 \cdot -2.598 = -88.2615 \text{ Kgf/mm}$$

From table 4.3.6 Qn:

$$\begin{aligned} C1 &= -0.407003 & C2 &= 0.021640 & C3 &= -0.324089 & C4 &= -0.004099 \\ C5 &= -0.110758 & C6 &= 0.002213 & C7 &= 0.000265 & C8 &= -0.014361 \\ C9 &= 0.000823 & C10 &= 0.000107 & C11 &= 0.000000 \end{aligned}$$

Compute from Table 4.3.6 [Qn]:

$$\begin{aligned} &= C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot \ln(H)^2 + C5 \cdot \ln(B)^2 + C6 \cdot \ln(H) \cdot \ln(B) + \\ &\quad C7 \cdot \ln(H)^3 + C8 \cdot \ln(B)^3 + C9 \cdot \ln(H) \cdot \ln(B)^2 + C10 \cdot \ln(H)^2 \cdot \ln(B) \\ &= -0.407 + 0.02164 \cdot \ln(5.617) + -0.32409 \cdot \ln(0.52) + \\ &\quad -0.0041 \cdot \ln(5.617)^2 + -0.11076 \cdot (\ln(0.52))^2 + \\ &\quad 0.00221 \cdot \ln(5.617) \cdot \ln(0.52) + 0.00027 \cdot \ln(5.617)^3 + \\ &\quad -0.01436 \cdot \ln(0.52)^3 + 0.00082 \cdot \ln(5.617) \cdot \ln(0.52)^2 + \\ &\quad 0.00011 \cdot \ln(5.617)^2 \cdot \ln(0.52) \\ &= -0.2140 \end{aligned}$$

Compute with Xs+ [Qx+]:

$$= X_{s+} \cdot Q_n = 78.211 \cdot -0.214 = -16.7383 \text{ Kgf/mm}$$

Compute with Xs- [Qx-]:

$$= X_{s-} \cdot Q_n = -119.785 \cdot -0.214 = 25.6359 \text{ Kgf/mm}$$

Compute with Xs+ [Q+]:

$$= Q_p + Q_{x+} = -88.262 + -16.738 = -105.0 \text{ Kgf/mm}$$

Compute with Xs- [Q-]:

$$= Q_p + Q_{x-} = -88.262 + 25.636 = -62.6256 \text{ Kgf/mm}$$

Compute angle [βcy]:

$$\begin{aligned} &= (3 \cdot (1 - v^2) / (R_s^2 \cdot t_s^2))^{0.25} = (3 \cdot (1 - 0.3^2) / (2303.381^2 \cdot 73.0^2))^{0.25} \\ &= 0.0031 \text{ 1/mm} \end{aligned}$$

Compute with Xs+ [Ns+]:

$$= P \cdot R_s / 2 + X_{s+} = 4563.0 \cdot 2303.381 / 2 + 78.211 = 614.1096 \text{ Kgf/mm}$$

Compute with Xs- [Ns-]:

$$= P \cdot R_s / 2 + X_{s-} = 4563.0 \cdot 2303.381 / 2 + -119.785 = 416.1137 \text{ Kgf/mm}$$

Compute with Xs+ [NTheta+]:

$$= P \cdot R_s + 2 \cdot \beta_{cy} \cdot R_s \cdot (-M_s \cdot \beta_{cy} - Q)$$

$$= 4563.0 \cdot 2303.381 + 2 \cdot 0.003 \cdot 2303.381 \cdot (-20592.285 \cdot 0.003 - -209.025) = 1655.9178 \text{ Kgf/mm}$$

Compute with Xs- [NTheta-]:
 $= P \cdot Rs + 2 \cdot \beta_{cy} \cdot Rs \cdot (-Ms \cdot \beta_{cy} - Q)$
 $= 4563.0 \cdot 2303.381 + 2 \cdot 0.003 \cdot 2303.381 \cdot (-12822.769 \cdot 0.003 - -182.349) = 1395.7062 \text{ Kgf/mm}$

Compute Stress with Xs+ [σ_{sm}]:
 $= Ns(+)/ts = 614.11/73.0 = 82.4996 \text{ N./mm}^2$

Compute Stress with Xs- [σ_{sm}]:
 $= Ns(-)/ts = 416.114/73.0 = 55.9008 \text{ N./mm}^2$

Compute stress with Xs+ [σ_{sb}]:
 $= 6 \cdot Ms(+)/(ts^2 \cdot Kpc) = 6 \cdot 20592.285/(73.0^2 \cdot 1.0) = 227.3729 \text{ N./mm}^2$

Compute stress with Xs- [σ_{sb}]:
 $= 6 \cdot Ms(-)/(ts^2 \cdot Kpc) = 6 \cdot 12822.769/(73.0^2 \cdot 1.0) = 141.5846 \text{ N./mm}^2$

Compute stress with Xs+ [$\sigma_{\theta_{sm}}$]:
 $= N_{\theta}(+)/ts = 1655.918/73.0 = 222.4562 \text{ N./mm}^2$

Compute stress with Xs- [$\sigma_{\theta_{sm}}$]:
 $= N_{\theta}(-)/ts = 1395.706/73.0 = 187.4994 \text{ N./mm}^2$

Compute stress with Xs+ [$\sigma_{\theta_{sb}}$]:
 $= 6 \cdot v \cdot Ms(+)/(ts^2 \cdot Kpc) = 6 \cdot 0.3 \cdot 20592.285/(73.0^2 \cdot 1.0) = 68.2119 \text{ N./mm}^2$

Compute stress with Xs- [$\sigma_{\theta_{sb}}$]:
 $= 6 \cdot v \cdot Ms(-)/(ts^2 \cdot Kpc) = 6 \cdot 0.3 \cdot 12822.769/(73.0^2 \cdot 1.0) = 42.4754 \text{ N./mm}^2$

Small End Cone Without a Flare

Compute with Xs+ [Qc+]:
 $= Q \cdot \cos(a) + Ns \cdot \sin(a) = -105.0 \cdot \cos(27.47) + 614.1 \cdot \sin(27.47)$
 $= 190.1635 \text{ Kgf/mm}$

Compute with Xs- [Qc-]:
 $= Q \cdot \cos(a) + Ns \cdot \sin(a) = -62.6 \cdot \cos(27.47) + 416.1 \cdot \sin(27.47)$
 $= 136.4127 \text{ Kgf/mm}$

Compute effective radius [Rc]:
 $= Rs/\cos(a) = 2303.381/\cos(27.474) = 2596.1877 \text{ mm}$

Compute the angle [β_{co}]:
 $= (3 \cdot (1 - v^2)/(Rc^2 \cdot tc^2))^{0.25} = (3 \cdot (1 - 0.3^2)/(2596.188^2 \cdot 133.5^2))^{0.25}$
 $= 0.0022 \text{ 1/mm}$

Compute with Ms+ [Ncs+]:
 $= Ns \cdot \cos(a) - Q \cdot \sin(a)$
 $= 614.11 \cdot \cos(27.47) - 105.0 \cdot \sin(27.47) = 593.2903 \text{ Kgf/mm}$

Compute with Ms- [Ncs-]:
 $= Ns \cdot \cos(a) - Q \cdot \sin(a)$
 $= 416.114 \cdot \cos(27.47) - 62.626 \cdot \sin(27.47) = 398.0756 \text{ Kgf/mm}$

Compute with Ms+ [Nctheta+]:
 $= P \cdot Rs/\cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} + Qc)$
 $= 4563.0 \cdot 2303.381/\cos(27.47) + 2 \cdot 0.0022 \cdot 2596.188 \cdot (-20592.285 \cdot 0.0022 + 190.163) = 2854.2048 \text{ Kgf/mm}$

Compute with Ms- [Nctheta-]:
 $= P \cdot Rs/\cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} + Qc)$
 $= 4563.0 \cdot 2303.381/\cos(27.47) + 2 \cdot 0.0022 \cdot (-12822.769 \cdot 0.0022 + 136.413) = 2437.1521 \text{ Kgf/mm}$

Compute the stress with Ms+ [σ_{sm}]:
 $= Ncs/tc = 593.29/133.5 = 43.5828 \text{ N./mm}^2$



Compute the stress with Ms- [σ_{sm}]:
= $N_{cs}/t_c = 398.076/133.5 = 29.2424 \text{ N./mm}^2$

Compute the stress with Ms+ [σ_{sb}]:
= $6 \cdot M_{cs}/(t_c^2 \cdot K_{cpc}) = 6 \cdot 20592.285/(133.5^2 \cdot 1.0) = 67.9864 \text{ N./mm}^2$

Compute the stress with Ms- [σ_{sb}]:
= $6 \cdot M_{cs}/(t_c^2 \cdot K_{cpc}) = 6 \cdot 12822.769/(133.5^2 \cdot 1.0) = 42.335 \text{ N./mm}^2$

Compute stress with Ms+ [$\sigma_{\theta am}$]:
= $N_{\theta}/t_c = 2854.205/133.5 = 209.668 \text{ N./mm}^2$

Compute stress with Ms- [$\sigma_{\theta am}$]:
= $N_{\theta}/t_c = 2437.152/133.5 = 179.032 \text{ N./mm}^2$

Compute stress with Ms+ [$\sigma_{\theta ab}$]:
= $6 \cdot v \cdot M_{cs}/(t_c^2 \cdot K_{cpc}) = 6 \cdot 0.3 \cdot 20592.285/(133.5^2 \cdot 1.0) = 20.3959 \text{ N./mm}^2$

Compute stress with Ms- [$\sigma_{\theta ab}$]:
= $6 \cdot v \cdot M_{cs}/(t_c^2 \cdot K_{cpc}) = 6 \cdot 0.3 \cdot 12822.769/(133.5^2 \cdot 1.0) = 12.7005 \text{ N./mm}^2$

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Input, Nozzle Desc: C

From: 50

Pressure for Reinforcement Calculations	P	4740.966	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Hemispherical Head	D	7200.00	mm
Head Finished (Minimum) Thickness	t	68.3000	mm
Head Internal Corrosion Allowance	c	6.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

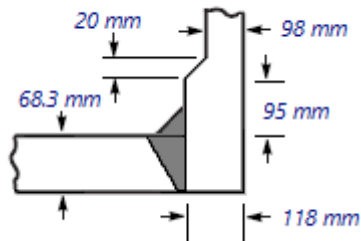
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	98.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	95.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		118.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	
Flange Series		B	
Piping Exit Angle [North Clockwise reference]		135.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		4500.0	mm
Centerline Distance to Tangent		5500.0	mm

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: C

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	36.000	in.
Actual Thickness Used in Calculation	3.858	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 926.4 \cdot (\exp(4740.966/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 15.1206 + 6.0 + 0.0 = 21.1206 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 92.0 + 926.4) / 926.4) - 0.0$
 $= 26739.061 - 0.0 = 26739.0605 \text{ KPa.}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 14.340 mm
 Min. required shell thickness, internal p + c : 61.075 mm

Nozzle neck minimum required thickness : 14.340 mm

Minimum nozzle neck thickness 97.993 mm , is greater than required thickness 14.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 14.4	Allowable	: 147.6 N./mm ²	Passed
Expansion	: 0.0	Allowable	: 376.3 N./mm ²	Passed
Occasional	: 10.9	Allowable	: 196.3 N./mm ²	Passed
Shear	: 2.1	Allowable	: 103.3 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[147.6/156.4, 1]$
 $= 0.944$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 118.0 - 6.0$
 $= 112.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 98.0 - 6.0$
 $= 92.000 \text{ mm}$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 7200.0/2 + 6.763$



$$= 3606.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{R_{eff} \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3606.0 \cdot 62.3}, 2 \cdot 463.2)$$

$$= 473.977 \text{ mm}$$

Intermediate value [Cn]:

$$= \min((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min((62.3 + 0.0)/112.0)^{0.35}, 1)$$

$$= \min(0.814, 1)$$

$$= 0.814$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{R_n \cdot t_n}, L_{pr1} + t)$$

$$= \min(62.3 + 0.0 + 0.814 \cdot \sqrt{463.2 \cdot 112.0}, 500.0 + 62.3)$$

$$= 247.797 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot f_{rp}) / LR)$$

$$= 62.3 + ((0.0 \cdot 1.0) / 473.977)$$

$$= 62.300 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 62.3 \cdot 473.977$$

$$= 295.287 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2C}$$

$$= 176.176 + 85.258$$

$$= 261.434 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + f_{rn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 295.287 + 0.944(261.434 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 542.511 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 112.0 / \ln[1 + 112.0/463.2]$$

$$= 517.180 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 62.3 / \ln[1 + 62.3/3606.0]$$

$$= 3637.060 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N/mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 4740.966 \cdot 517.18 \cdot 247.797$$

$$= 61958.7 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 4740.966 \cdot 3637.06 \cdot (473.977 + 112.0) / 2$$

$$= 515186.1 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot Rxs \cdot Rnc) / 2$$

$$= (4740.966 \cdot 3637.06 \cdot 463.2) / 2$$

$$= 407241.9 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (515186 + 407241.9 + 61958.7) / 4740.966$$

$$= 20361.2 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / (2 \cdot teff))$$

$$= 234.6 / (2 \cdot 20361.152 / 542.511 - 3637.06 / (2 \cdot 62.3))$$

$$= 5113.9 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t / Rxs]$$

$$= 2 \cdot 156.4 [62.3 / 3637.06]$$

$$= 5357.7 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(5113.855, 5357.708)$$

$$= 5113.855 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (61958.676 + 515186 + 407241.875) / 542.511$$

$$= 177.945 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 4740.966 \cdot 3637.06 / (2 \cdot 62.3)$$

$$= 138.4 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 177.945 - 138.396, 138.396)$$

$$= 217.5 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	217.49	N./mm ²
Maximum Allowable Working Pressure	Pmax	5113.86	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (463.2 + 112.0) / 463.2$$

$$= 1.242 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (463.2 + 112.0)$$

$$= 903.522 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 112.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(407242 \cdot 1.24, 1.5 \cdot 147.6(300.323 + 0.0), \pi / 4 \cdot 4741.0 \cdot 463.2^2 \cdot 1.24^2)$$

$$= 125629.352 \text{ Kgf}$$

Weld Stress Value [τ]:
 $= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 125629.352 / (903.522(0.49 \cdot 7.071 + 0.6 \cdot 37.0 + 0.49 \cdot 0.0))$
 $= 53.130 < \text{or} = \text{to } 156.400 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240 , the MDMT is : -104 °C
 The computed stress ratio is : 0.147

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (13.503 \cdot 1.0) / (98.0 - 6.0 - 0.0)$
 $= 0.147$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 46.2491 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 614.5492 mm

Input Echo, WRC107/537 Item	1,	Description: C	:
Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	6.0000	mm
Vessel Diameter	Dv	7200.000	mm
Vessel Thickness	Tv	68.300	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	156.40	N./mm ²
Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	6.0000	mm
Nozzle Diameter	Dn	914.400	mm
Nozzle Thickness	Tn	118.000	mm
Nozzle Material		SA-266 2	
Nozzle UNS Number		K03506	
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	147.60	N./mm ²
Design Internal Pressure	Dp	4740.966	KPa.
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

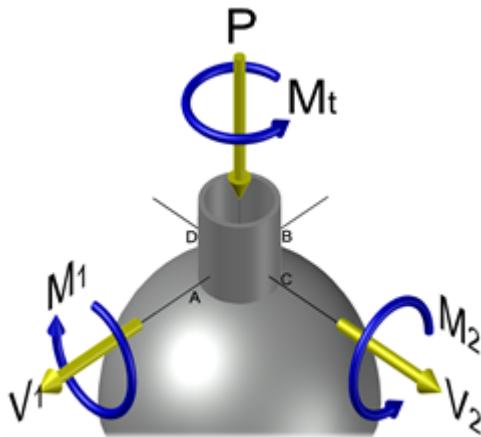
Radial Load (SUS)	P	8257.1	Kgf
Longitudinal Shear (SUS)	(V1) V1	5963.4	Kgf
Circumferential Shear (SUS)	(Vc) V2	5963.4	Kgf
Circumferential Moment (SUS)	(Mc) M1	16514.9	Kg-m.

Longitudinal Moment (SUS) (Ml) M2 16514.9 Kg-m.
 Torsional Moment (SUS) Mt 18992.2 Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))}$$

$$= 1150.4 + 2 \cdot 1.65 \cdot \sqrt{(3637.15 (68.3 - 6.0))}$$

$$= 2721.264 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	8257.1	Kgf
Circumferential Shear	(VC)	V2	5963.4	Kgf
Longitudinal Shear	(VL)	V1	5963.4	Kgf
Circumferential Moment	(MC)	M1	16514.9	Kg-m.
Longitudinal Moment	(ML)	M2	16514.9	Kg-m.
Torsional Moment		MT	18992.2	Kg-m.

Dimensionless Param: U = 1.21 TAU = 5.00 (4.64) RHO = 0.56

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03360	(A,B,C,D)
$M(x) / P$	SP 1	0.03280	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / MC$	SM 1	0.03720	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / MC$	SM 1	0.05211	(A,B,C,D)
$N(x) * T * \text{SQRT}(Rm * T) / ML$	SM 1	0.03720	(A,B,C,D)
$M(x) * \text{SQRT}(Rm * T) / ML$	SM 1	0.05211	(A,B,C,D)
$N(y) * T / P$	SP 1	0.05331	(A,B,C,D)
$M(y) / P$	SP 1	0.00957	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / MC$	SM 1	0.05251	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / MC$	SM 1	0.01718	(A,B,C,D)
$N(y) * T * \text{SQRT}(Rm * T) / ML$	SM 1	0.05251	(A,B,C,D)
$M(y) * \text{SQRT}(Rm * T) / ML$	SM 1	0.01718	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7	-0.7
Rad. Bend. P		-4.1	4.1	-4.1	4.1	-4.1	4.1	-4.1	4.1
Rad. Memb. MC		0.0	0.0	0.0	0.0	-3.3	-3.3	3.3	3.3
Rad. Bend. MC		0.0	0.0	0.0	0.0	-27.4	27.4	27.4	-27.4
Rad. Memb. ML		-3.3	-3.3	3.3	3.3	0.0	0.0	0.0	0.0
Rad. Bend. ML		-27.4	27.4	27.4	-27.4	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-35.5	27.6	25.9	-20.7	-35.5	27.6	25.9	-20.7
Tang. Memb. P		-1.1	-1.1	-1.1	-1.1	-1.1	-1.1	-1.1	-1.1
Tang. Bend. P		-1.2	1.2	-1.2	1.2	-1.2	1.2	-1.2	1.2
Tang. Memb. MC		0.0	0.0	0.0	0.0	-4.6	-4.6	4.6	4.6
Tang. Bend. MC		0.0	0.0	0.0	0.0	-9.0	9.0	9.0	-9.0
Tang. Memb. ML		-4.6	-4.6	4.6	4.6	0.0	0.0	0.0	0.0
Tang. Bend. ML		-9.0	9.0	9.0	-9.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-16.0	4.5	11.3	-4.3	-16.0	4.5	11.3	-4.3
Shear VC		0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT		1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4
Tot. Shear		2.0	2.0	0.9	0.9	0.9	0.9	2.0	2.0
Str. Int.		35.7	27.7	25.9	20.8	35.5	27.6	26.1	21.0

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		138.4	138.4	138.4	138.4	138.4	138.4	138.4	138.4
Rad. Pl (SUS)		-4.0	-4.0	2.6	2.6	-4.0	-4.0	2.6	2.6
Rad. Q (SUS)		-31.5	31.5	23.3	-23.3	-31.5	31.5	23.3	-23.3
Long. Pm (SUS)		138.4	138.4	138.4	138.4	138.4	138.4	138.4	138.4
Long. Pl (SUS)		-5.7	-5.7	3.5	3.5	-5.7	-5.7	3.5	3.5
Long. Q (SUS)		-10.2	10.2	7.8	-7.8	-10.2	10.2	7.8	-7.8
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear Q (SUS)		1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4
Pm (SUS)		138.4	138.4	138.4	138.4	138.4	138.4	138.4	138.4
Pm+Pl (SUS)		134.6	134.6	142.1	142.1	134.6	134.6	142.1	142.1
Pm+Pl+Q (Total)		122.6	166.1	164.3	134.1	122.5	166.0	164.5	134.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	138.40	156.40	Passed
Pm+Pl (SUS)	142.12	234.60	Passed
Pm+Pl+Q (TOTAL)	166.11	469.20	Passed



The $Pm+PI+Q$ allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The $Pm+PI+Q$ allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: A

From: 80

Pressure for Reinforcement Calculations	P	4619.402	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	7200.00	mm
Shell Finished (Minimum) Thickness	t	124.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		8100.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

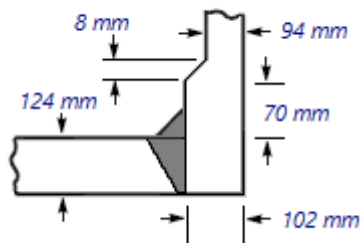
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	94.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	8.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		102.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: A

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	3.701	in.



Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 621.6 \cdot (\exp(4619.402/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 9.8814 + 6.0 + 0.0 = 15.8814 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 88.0 + 621.6) / 621.6) - 0.0$
 $= 36796.086 - 0.0 = 36796.0859 \text{ KPa.}$

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 441.87 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 1028.40 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 14.340 mm
 Min. required shell thickness, internal p + c : 114.109 mm

Nozzle neck minimum required thickness : 14.340 mm

Minimum nozzle neck thickness 94.005 mm , is greater than required thickness 14.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	10.7,	Allowable	:	147.6 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	380.1 N./mm ²	Passed
Occasional	:	7.1,	Allowable	:	196.3 N./mm ²	Passed
Shear	:	2.2,	Allowable	:	103.3 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[147.6/156.4, 1]$
 $= 0.944$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 102.0 - 6.0$
 $= 96.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 94.0 - 6.0$
 $= 88.000 \text{ mm}$



Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7212.0/118.0$$

$$= 61.119 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 7200.0/2 + 6.763$$

$$= 3606.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3606.0 \cdot 118.0}, 2 \cdot 310.8)$$

$$= 621.600 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 118.0, 0.0] + \sqrt{310.8 \cdot 96.0}$$

$$= 172.733 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 400.0$$

$$= 400.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(118.0 + 0.0)$$

$$= 944.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[172.733, 400.0, 944.0] + 118.0$$

$$= 290.733 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 118.0 + ((0.0 \cdot 1.0) / 621.6)$$

$$= 118.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 310.8 + 96.0) / (\sqrt{(7212.0 + 118.0) \cdot 118.0}))$$

$$= 0.772$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 118.0 \cdot 621.6 \cdot \max((0.772/5)^{0.85}, 1)$$

$$= 733.488 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2C$$

$$= 180.48 + 90.725$$

$$= 271.205 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^4$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 733.488 + 0.944(271.205 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 989.934 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 96.0 / \ln[1 + 96.0/310.8]$$

$$= 356.649 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 118.0 / \ln[1 + 118.0/3606.0]$$

$$= 3664.689 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 4619.402 \cdot 356.649 \cdot 290.733$$

$$= 48844.8 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 4619.402 \cdot 3664.689 (621.6 + 96.0)$$

$$= 1238801.8 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 4619.402 \cdot 3664.689 \cdot 310.8$$

$$= 536537.9 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (1238802 + 536538 + 48844.8) / 4619.402$$

$$= 38724.6 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 234.6 / (2 \cdot 38724.559 / 989.934 - 3664.689 / 118.0)$$

$$= 4972.2 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 156.4 [118.0 / 3664.689]$$

$$= 5035.7 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(4972.159, 5035.66)$$

$$= 4972.159 \text{ KPa.}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (48844.797 + 1238802 + 536538) / 989.934$$

$$= 180.714 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 4619.402 \cdot 3664.689 / 118.0$$

$$= 143.5 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 180.714 - 143.472, 143.472)$$

$$= 218.0 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	217.96	N./mm ²
Maximum Allowable Working Pressure	Pmax	4972.16	KPa.

Strength of Nozzle Attachment Welds per 4.5.14



Discontinuity Force Factor [ky]:

= (Rnc + tn) / Rnc
 = (310.8 + 96.0)/310.8
 = 1.309 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lt]:

= $\pi/2 \cdot (Rn + tn)$
 = $\pi/2 \cdot (310.8 + 96.0)$
 = 639.000 mm

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 0.000, 0.000, mm

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be >= min(0.7 * 96.000, 6.000) Passed

Weld Load Value [fwelds]:

= min(fy • ky, 1.5 • Sn(A2 + A3), $\pi/4 \cdot P \cdot Rn^2 \cdot ky^2$)
 = min(536538•1.31, 1.5•147.6(271.205+0.0), $\pi/4 \cdot 4619.4 \cdot 310.8^2 \cdot 1.31^2$)
 = 61225.734 Kgf

Weld Stress Value [τ]:

= fwelds/(Lt(0.49•L41T + 0.6•tw1 + 0.49•L43T))
 = 61225.734/(639.0(0.49•7.071 + 0.6•37.0 + 0.49•0.0))
 = 36.612 < or = to 156.400 Weld Stress Passed

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle away from the intersecting weld by 129.924 mm.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.100

The computed thickness ratio [sr]:

= (tr • E*)/(tn - ci - cext)
 = (8.825 • 1.0)/(94.0 - 6.0 - 0.0)
 = 0.100

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 23.0579 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 547.0580 mm

Input Echo, WRC297 Item 1, Description: A :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	6.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	7200.000	mm
Shell Thickness	124.0000	mm
Shell Stress Concentration Factor	1.000	



Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	156.40	N./mm ²
Diameter Basis for Nozzle		ID	
Nozzle Corrosion Allowance		6.0000	mm
Nozzle Diameter		609.600	mm
Nozzle Thickness		102.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	147.60	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	4619.40	KPa.
Radial Load	P	5504.71	Kgf
Circumferential Shear	Vc	3975.63	Kgf
Longitudinal Shear	Vl	3975.63	Kgf
Circumferential Moment	Mc	7340.00	Kg-m.
Longitudinal Moment	Ml	7340.00	Kg-m.
Torsional Moment	Mt	8441.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (7.976) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 813.6 + 2 \cdot 1.65 \cdot \sqrt{(3665.0(124.0 - 6.0))}$
 $= 2983.761 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	7330.000 mm
Nozzle Outside Diameter	(d) =	813.600 mm
Vessel Thickness used	(T) =	118.000 mm
Nozzle Thickness used	(t) =	96.000 mm
T / t	=	1.229
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.875$$

Nr/P	=	0.140
Mr/P	=	0.189
M0/P	=	0.071
N0/P	=	0.133
MrD/Mc	=	0.268
NrDL/Mc	=	0.157
M0d/Mc	=	0.121
N0DL/Mc	=	-0.058
MrD/Ml	=	0.203
NrDL/Ml	=	0.129
M0D/Ml	=	0.091
N0DL/Ml	=	0.108

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-1	1	-1	1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-3	3	3	-3
Normal Pressure Stress	138	143	138	143

Outplane Stress Summary	132	147	140	141
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-4	4	-4	4
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-7	7	7	-7
Inplane Pressure Stress	69	69	69	69

Inplane Stress Summary	55	80	73	66
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	132	147	140	141
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-1	1	-1	1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-4	4	4	-4
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	69	69	69	69

Outplane Stress Summary	63	75	71	65
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-4	4	-4	4
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-10	10	10	-10
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0



Inplane Pressure Stress	138	143	138	143
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Inplane Stress Summary	122	156	145	138
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	122	156	145	138

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	21	15	21
Hoop Stress Summary	14	19	15	21

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-4	4	-4	4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-7	7	7	-7
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	-6	17	13	5

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	21	20	15	21

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	21	15	21
Hoop Stress Summary	15	20	14	20

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-4	4	-4	4
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-10	10	10	-10
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	-9	20	16	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	25	21	16	20

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	138	143	138	143	138	143	138	143
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-4	4	2	-2	-14	14	6	-6
Long. Pm (SUS)	69	69	69	69	69	69	69	69
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-11	11	3	-3	-5	5	3	-3
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	138.0	143.0	138.0	143.0	138.0	143.0	138.0	143.0
Pm+Pl (SUS)	138.0	143.0	138.0	143.0	138.0	143.0	138.0	143.0
Pm+Pl+Q (Total)	134.0	147.0	140.0	141.0	124.0	157.0	144.0	137.0



Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	143.00	156.40	Passed
Pm+Pl (SUS)	143.00	234.60	Passed
Pm+Pl+Q (TOTAL)	157.00	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	21	15	21	15	21	15	21
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-11	11	3	-3	-14	14	6	-6
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	15.0	21.0	15.0	21.0	15.0	21.0	15.0	21.0
Pm+Pl (SUS)	15.0	21.0	15.0	21.0	15.0	21.0	15.0	21.0
Pm+Pl+Q (Total)	21.0	21.0	15.0	21.0	24.0	21.0	15.0	21.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	21.00	147.60	Passed
Pm+Pl (SUS)	21.00	221.40	Passed
Pm+Pl+Q (TOTAL)	24.00	442.80	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: MH-6
From: 90

Pressure for Reinforcement Calculations	P	4616.805	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	7200.00	mm
Shell Finished (Minimum) Thickness	t	122.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		9800.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

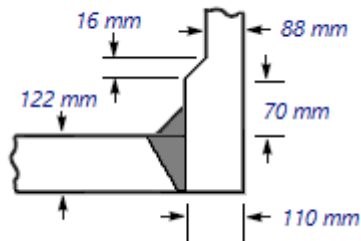
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		300.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	88.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	16.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-6

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	3.465	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 621.6 \cdot (\exp(4616.805/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 9.8758 + 6.0 + 0.0 = 15.8758 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 82.0 + 621.6) / 621.6) - 0.0$
 $= 34558.68 - 0.0 = 34558.6797 \text{ KPa.}$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[147.6/156.4, 1]$
 $= 0.944$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 110.0 - 6.0$
 $= 104.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 88.0 - 6.0$
 $= 82.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 7212.0/116.0$
 $= 62.172 \text{ must be less than 400.}$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 7200.0/2 + 6.763$
 $= 3606.000 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{3606.0 \cdot 116.0}, 2 \cdot 310.8)$
 $= 621.600 \text{ mm}$

Thickness Limit Candidate [LH1]:
 $= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$
 $= \min[1.5 \cdot 116.0, 0.0] + \sqrt{310.8 \cdot 104.0}$
 $= 179.787 \text{ mm}$

Thickness Limit Candidate [LH2]:
 $= L_{pr1}$

$$= 400.0$$

$$= 400.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(116.0 + 0.0)$$

$$= 928.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[179.787, 400.0, 928.0] + 116.0$$

$$= 295.787 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 116.0 + ((0.0 \cdot 1.0) / 621.6)$$

$$= 116.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + t_n) / (\sqrt{ (D_i + teff) \cdot teff }))$$

$$= \min(12, (2 \cdot 310.8 + 104.0) / (\sqrt{ (7212.0 + 116.0) \cdot 116.0 }))$$

$$= 0.787$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 116.0 \cdot 621.6 \cdot \max((0.787/5)^{0.85}, 1)$$

$$= 721.056 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2C$$

$$= 193.44 + 91.785$$

$$= 285.225 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 721.056 + 0.944(285.225 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 990.733 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/Rn]$$

$$= 104.0 / \ln[1 + 104.0/310.8]$$

$$= 360.302 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 116.0 / \ln[1 + 116.0/3606.0]$$

$$= 3663.692 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 4616.805 \cdot 360.302 \cdot 295.787$$

$$= 50174.5 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + t_n)$$

$$= 4616.805 \cdot 3663.692(621.6 + 104.0)$$

$$= 1251567.5 \text{ Kg}$$



Discontinuity Force from Internal Pressure [FY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 4616.805 \cdot 3663.692 \cdot 310.8$$

$$= 536090.4 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (1251568 + 536090 + 50174.5) / 4616.805$$

$$= 39036.2 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 234.6 / (2 \cdot 39036.234 / 990.733 - 3663.692 / 116.0)$$

$$= 4968.0 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / t / R_{xs}$$

$$= 156.4 [116.0 / 3663.692]$$

$$= 4951.7 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(4968.028, 4951.657)$$

$$= 4951.657 \text{ KPa.}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (50174.48 + 1251568 + 536090) / 990.733$$

$$= 181.919 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 4616.805 \cdot 3663.692 / 116.0$$

$$= 145.8 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 181.919 - 145.824, 145.824)$$

$$= 218.0 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	218.01	N./mm ²
Maximum Allowable Working Pressure	Pmax	4951.66	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (310.8 + 104.0) / 310.8$$

$$= 1.335 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (310.8 + 104.0)$$

$$= 651.566 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 104.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi / 4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(536090 \cdot 1.33, 1.5 \cdot 147.6(285.225 + 0.0), \pi / 4 \cdot 4616.8 \cdot 310.8^2 \cdot 1.33^2)$$

$$= 63621.707 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (L_t(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

= $63621.707 / (651.566(0.49 \cdot 7.071 + 0.6 \cdot 37.0 + 0.49 \cdot 0.0))$
= 37.311 < or = to 156.400 Weld Stress Passed

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle away from the intersecting weld by 237.922 mm.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -12 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.108

The computed thickness ratio [sr]:
= $(tr \cdot E^*) / (tnom - ci - cext)$
= $(8.82 \cdot 1.0) / (88.0 - 6.0 - 0.0)$
= 0.108

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

[Note: This Element/Detail was specified as being Impact Tested.](#)

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 23.9768 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 545.9767 mm

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Input, Nozzle Desc: MH-5

From: 110

Pressure for Reinforcement Calculations	P	4616.805	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	7200.00	mm
Shell Finished (Minimum) Thickness	t	122.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		16400.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

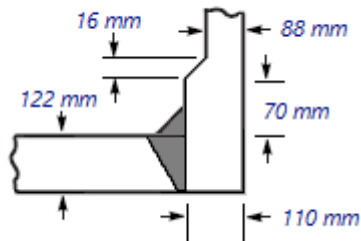
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		300.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	88.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	16.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-5

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	3.465	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 621.6 \cdot (\exp(4616.805/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 9.8758 + 6.0 + 0.0 = 15.8758 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 82.0 + 621.6) / 621.6) - 0.0$$

$$= 34558.68 - 0.0 = 34558.6797 \text{ KPa.}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 110.0 - 6.0$$

$$= 104.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 88.0 - 6.0$$

$$= 82.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 7212.0/116.0$$

$$= 62.172 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 7200.0/2 + 6.763$$

$$= 3606.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{3606.0 \cdot 116.0}, 2 \cdot 310.8)$$

$$= 621.600 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 116.0, 0.0] + \sqrt{310.8 \cdot 104.0}$$

$$= 179.787 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$



$$= 400.0$$

$$= 400.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(116.0 + 0.0)$$

$$= 928.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[179.787, 400.0, 928.0] + 116.0$$

$$= 295.787 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 116.0 + ((0.0 \cdot 1.0) / 621.6)$$

$$= 116.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + t_n) / (\sqrt{ (Di + teff) \cdot teff }))$$

$$= \min(12, (2 \cdot 310.8 + 104.0) / (\sqrt{ (7212.0 + 116.0) \cdot 116.0 }))$$

$$= 0.787$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 116.0 \cdot 621.6 \cdot \max((0.787/5)^{0.85}, 1)$$

$$= 721.056 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2C$$

$$= 193.44 + 91.785$$

$$= 285.225 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 721.056 + 0.944(285.225 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 990.733 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/Rn]$$

$$= 104.0 / \ln[1 + 104.0/310.8]$$

$$= 360.302 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 116.0 / \ln[1 + 116.0/3606.0]$$

$$= 3663.692 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 4616.805 \cdot 360.302 \cdot 295.787$$

$$= 50174.5 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + t_n)$$

$$= 4616.805 \cdot 3663.692(621.6 + 104.0)$$

$$= 1251567.5 \text{ Kg}$$



Discontinuity Force from Internal Pressure [FY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 4616.805 \cdot 3663.692 \cdot 310.8$$

$$= 536090.4 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (1251568 + 536090 + 50174.5) / 4616.805$$

$$= 39036.2 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 234.6 / (2 \cdot 39036.234 / 990.733 - 3663.692 / 116.0)$$

$$= 4968.0 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 156.4 [116.0 / 3663.692]$$

$$= 4951.7 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(4968.028, 4951.657)$$

$$= 4951.657 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (50174.48 + 1251568 + 536090) / 990.733$$

$$= 181.919 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 4616.805 \cdot 3663.692 / 116.0$$

$$= 145.8 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 181.919 - 145.824, 145.824)$$

$$= 218.0 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	218.01	N./mm ²
Maximum Allowable Working Pressure	Pmax	4951.66	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (310.8 + 104.0) / 310.8$$

$$= 1.335 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (310.8 + 104.0)$$

$$= 651.566 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 104.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(536090 \cdot 1.33, 1.5 \cdot 147.6(285.225 + 0.0), \pi / 4 \cdot 4616.8 \cdot 310.8^2 \cdot 1.33^2)$$

$$= 63621.707 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$= 63621.707 / (651.566(0.49 \cdot 7.071 + 0.6 \cdot 37.0 + 0.49 \cdot 0.0))$
 $= 37.311 < \text{or} = \text{to } 156.400 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -12 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240 , the MDMT is : -104 °C
The computed stress ratio is : 0.108

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (8.82 \cdot 1.0) / (88.0 - 6.0 - 0.0)$
 $= 0.108$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

[Note: This Element/Detail was specified as being Impact Tested.](#)

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 23.9768 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 545.9767 mm

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Input, Nozzle Desc: MH-4
From: 130

Pressure for Reinforcement Calculations	P	4616.805	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	7200.00	mm
Shell Finished (Minimum) Thickness	t	120.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		20600.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

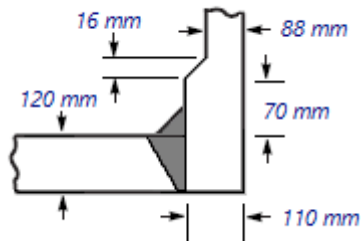
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		300.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	88.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	16.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-4

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	3.465	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 621.6 \cdot (\exp(4616.805/(147.6 \cdot 1.0)) - 1) + 6.0 + 0.0$
 $= 9.8758 + 6.0 + 0.0 = 15.8758 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 82.0 + 621.6) / 621.6) - 0.0$
 $= 34558.68 - 0.0 = 34558.6797 \text{ KPa.}$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 944.03 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 1036.40 mm

This nozzle or its diameter limit intersects weld seam # [8] on this element. The distance between the nozzle and seam is, 341.54 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 1036.40 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[147.6/156.4, 1]$
 $= 0.944$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 110.0 - 6.0$
 $= 104.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 88.0 - 6.0$
 $= 82.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 7212.0/114.0$
 $= 63.263 \text{ must be less than 400.}$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 7200.0/2 + 6.763$
 $= 3606.000 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{3606.0 \cdot 114.0}, 2 \cdot 310.8)$
 $= 621.600 \text{ mm}$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, te] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 114.0, 0.0] + \sqrt{(310.8 \cdot 104.0)}$$

$$= 179.787 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 400.0$$

$$= 400.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + te)$$

$$= 8(114.0 + 0.0)$$

$$= 912.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[179.787, 400.0, 912.0] + 114.0$$

$$= 293.787 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot frp) / LR)$$

$$= 114.0 + ((0.0 \cdot 1.0) / 621.6)$$

$$= 114.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 310.8 + 104.0) / (\sqrt{(7212.0 + 114.0) \cdot 114.0}))$$

$$= 0.794$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 114.0 \cdot 621.6 \cdot \max((0.794/5)^{0.85}, 1)$$

$$= 708.624 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2C}$$

$$= 191.36 + 91.785$$

$$= 283.145 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg_{41}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 708.624 + 0.944(283.145 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 976.338 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 104.0 / \ln[1 + 104.0/310.8]$$

$$= 360.302 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 114.0 / \ln[1 + 114.0/3606.0]$$

$$= 3662.708 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 4616.805 \cdot 360.302 \cdot 293.787$$

$$= 49835.2 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 4616.805 \cdot 3662.708(621.6 + 104.0)$$

$$= 1251231.2 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 4616.805 \cdot 3662.708 \cdot 310.8$$

$$= 535946.4 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (1251231 + 535946 + 49835.2)/4616.805$$

$$= 39018.8 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 234.6 / (2 \cdot 39018.824/976.338 - 3662.708/114.0)$$

$$= 4907.7 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 156.4[114.0/3662.708]$$

$$= 4867.6 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(4907.67, 4867.591)$$

$$= 4867.591 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (49835.219 + 1251231 + 535946)/976.338$$

$$= 184.519 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 4616.805 \cdot 3662.708/114.0$$

$$= 148.3 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 184.519 - 148.342, 148.342)$$

$$= 220.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	220.70	N./mm ²
Maximum Allowable Working Pressure	Pmax	4867.59	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (310.8 + 104.0)/310.8$$

$$= 1.335 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (310.8 + 104.0)$$

$$= 651.566 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 104.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:
= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(535946 \cdot 1.33, 1.5 \cdot 147.6(283.145 + 0.0), \pi/4 \cdot 4616.8 \cdot 310.8^2 \cdot 1.33^2)$
= 63621.707 Kgf

Weld Stress Value [τ]:
= $fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot tw_1 + 0.49 \cdot L_{43T}))$
= $63621.707 / (651.566(0.49 \cdot 7.071 + 0.6 \cdot 37.0 + 0.49 \cdot 0.0))$
= 37.311 < or = to 156.400 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -12 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.108

The computed thickness ratio [sr]:
= $(tr \cdot E^*) / (t_{nom} - c_i - c_{ext})$
= $(8.82 \cdot 1.0) / (88.0 - 6.0 - 0.0)$
= 0.108

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 23.9768 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 543.9769 mm

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Input, Nozzle Desc: E

From: 140

Pressure for Reinforcement Calculations	P	4616.805	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cone at Nozzle Location	D	6187.25	mm
Cone Half Apex Angle	Alpha	27.47	Degrees
Shell Finished (Minimum) Thickness	t	154.0000	mm
Shell Internal Corrosion Allowance	c	6.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		22623.80	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

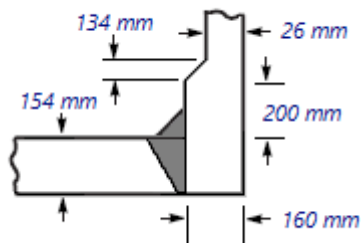
Material		SA-765 I	+IT
Material UNS Number		K03046	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	123.60	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.0000	mm
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	200.0000	mm
Height of Beveled Transition	L'	134.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm
Corrosion Allowance	can	6.0000	mm
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		62.5256	deg.
Flange Class		600	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: E

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	36.000	in.
Actual Thickness Used in Calculation	1.024	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 926.4 \cdot (\exp(4616.805/(123.6 \cdot 1.0)) - 1) + 6.0 + 0.0$$

$$= 17.63 + 6.0 + 0.0 = 23.63 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 123.6 \cdot 1.0 \cdot \ln((2 \cdot 20.0 + 926.4) / 926.4) - 0.0$$

$$= 5224.476 - 0.0 = 5224.4756 \text{ KPa.}$$

This nozzle or its diameter limit intersects weld seam # [5] on this element. The distance between the nozzle and seam is, 759.63 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 1294.59 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 14.340 mm
 Min. required shell thickness, internal p + c : 110.707 mm

Nozzle neck minimum required thickness : 14.340 mm

Minimum nozzle neck thickness 26.010 mm , is greater than required thickness 14.340 mm

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[123.6/156.4, 1]$$

$$= 0.790$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 160.0 - 6.0$$

$$= 154.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 26.0 - 6.0$$

$$= 20.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 6200.771/148.0$$

$$= 41.897 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 6187.245/2 + 6.763$$

$$= 3100.385 \text{ mm}$$

Effective Length of Vessel Wall [LR]:



$$= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot Rn)$$

$$= \min(\sqrt{(3100.385 \cdot 148.0)}, 2 \cdot 463.2)$$

$$= 677.390 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, te] + \sqrt{Rn \cdot tn}$$

$$= \min[1.5 \cdot 148.0, 0.0] + \sqrt{463.2 \cdot 154.0}$$

$$= 267.082 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= Lpr1$$

$$= 500.0$$

$$= 500.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + te)$$

$$= 8(148.0 + 0.0)$$

$$= 1184.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t / \cos(\alpha) \text{ for set-in nozzles}$$

$$= \min[267.082, 500.0, 1184.0] + 148.0 / \cos(27.474)$$

$$= 433.896 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 148.0 + ((0.0 \cdot 1.0) / 677.39)$$

$$= 148.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, ((2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff})))$$

$$= \min(12, (2 \cdot 463.2 + 154.0) / (\sqrt{(6200.77 + 148.0) \cdot 148.0}))$$

$$= 1.115$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 564.893 + 80.806$$

$$= 645.700 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 154.0 / \ln[1 + 154.0/463.2]$$

$$= 536.522 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 4616.805 \cdot 536.522 \cdot 433.896$$

$$= 109600.0 \text{ Kg}$$

Effective Radius for Nozzles in Cones (perp to axis) [Rnc]:

$$= Rn / \cos(\alpha)$$

$$= 463.2 / \cos(27.474)$$

$$= 522.082 \text{ mm}$$

Area Contributed by the Vessel Wall, Cone, perpendicular [A1]:

$$= t \cdot LR \cdot \max((\lambda / 5)^{0.85}, 1)$$

$$= 148.0 \cdot 677.39 \cdot \max((1.115/5)^{0.85}, 1)$$

$$= 1002.537 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:
 $= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$
 $= 1002.537 + 0.79(645.7 + 0.0) + 0.5 + 0.0 + 0.0 + 1.0(0.0)$
 $= 1513.321 \text{ cm}^2$

Shell Radius for Force Calculation [Rxs]:
 $= \text{teff} / \ln[1 + \text{teff} \cdot \cos[\alpha] / (\text{reff} + \text{Lc} \cdot \sin[\alpha])]$
 $= 148.0 / \ln[1 + 148.0 \cdot \cos[27.474] / (3100.385 + 1373.049 \cdot \sin[27.474])]$
 $= 4282.067 \text{ mm}$

Force acting on the Nozzle on the Cone [fS]:
 $= P / \cos(\alpha) [\text{Reff} + \text{Rnc} \cdot \sin[\alpha] + \text{Lt} \cdot \sin[\alpha] / 2] (\text{Lt})$
 $= 4616.805 / 0.89 [4282.07 + 522.08 \cdot 0.461 + 851.0 \cdot 0.46 / 2] (850.97)$
 $= 1597434.0 \text{ Kg}$

Discontinuity Force from Pressure [fY]:
 $= P / \cos(\alpha) (\text{Reff} + \text{Rnc} \cdot \sin[\alpha] / 2) \text{Rnc}$
 $= 4616.805 / 0.89 \cdot (4282.07 + 522.082 \cdot 0.46 / 2) \cdot 522.082$
 $= 892304.7 \text{ Kg}$

Cone Length Term [Lc]:
 $= \text{Lt} + \text{Rnc}$
 $= 850.966 + 522.082 + 0.0$
 $= 1373.049 \text{ mm}$

Area Resisting Internal Pressure [Ap]:
 $= (fN + fS + fY) / P$
 $= (109600.0 + 1597434 + 892305) / 4616.805$
 $= 55210.9 \text{ cm}^2$

Maximum Allowable Working Pressure Candidate [Pmax1]:
 $= \text{Sallow} / (2 \cdot \text{Ap} / \text{AT} - \text{Rxs} / \text{teff})$
 $= 234.6 / (2 \cdot 55210.91 / 1513.321 - 4282.067 / 148.0)$
 $= 5327.4 \text{ KPa}$

Maximum Allowable Working Pressure Candidate [Pmax2]:
 $= S[t / \text{Rxs}]$
 $= 156.4 [148.0 / 4282.067]$
 $= 5405.3 \text{ KPa}$

Maximum Allowable Working Pressure [Pmax]:
 $= \min(\text{Pmax1}, \text{Pmax2})$
 $= \min(5327.435, 5405.299)$
 $= 5327.435 \text{ KPa}$

Average Primary Membrane Stress [σ_{Avg}]:
 $= (fN + fS + fY) / \text{AT}$
 $= (109600.023 + 1597434 + 892305) / 1513.321$
 $= 168.446 \text{ N./mm}^2$

General Primary Membrane Stress [σ_{Circ}]:
 $= P \cdot \text{Rxs} / \text{teff}$
 $= 4616.805 \cdot 4282.067 / 148.0$
 $= 133.6 \text{ N./mm}^2$

Maximum Local Primary Membrane Stress [PL]:
 $= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$
 $= \max(2 \cdot 168.446 - 133.585, 133.585)$
 $= 203.3 \text{ N./mm}^2$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	203.31	N./mm ²
Maximum Allowable Working Pressure	Pmax	5327.43	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
 $= (\text{Rnc} + \text{tn}) / \text{Rnc}$



$$= (522.082 + 154.0) / 522.082$$
$$= 1.295 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [Lt]:

$$= \pi/2 \cdot \sqrt{((R_{nc} + T_n)^2 + (R_n + T_n)^2) / 2}$$
$$= \pi/2 \cdot \sqrt{((522.082 + 154.0)^2 + (463.2 + 154.0)^2) / 2}$$
$$= 1016.794 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 154.000, 6.000)$ Passed

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$
$$= \min(892305 \cdot 1.29, 1.5 \cdot 123.6(645.7 + 0.0), \pi/4 \cdot 4616.8 \cdot 463.2^2 \cdot 1.29^2)$$
$$= 133041.266 \text{ Kg}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$
$$= 133041.266 / (1016.794(0.49 \cdot 7.071 + 0.6 \cdot 10.0 + 0.49 \cdot 0.0))$$
$$= 135.572 < \text{or} = \text{to } 156.400 \text{ Weld Stress Passed}$$

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle farther away from the intersecting weld.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -57 °C

The Drop for this Nozzle is : 62.1931 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 716.1930 mm

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Input, Nozzle Desc: P
From: 150

Pressure for Reinforcement Calculations	P	4616.805	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4600.00	mm
Shell Finished (Minimum) Thickness	t	76.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		24300.01	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

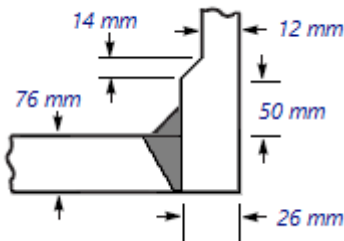
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	14.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		26.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: P

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	1.500	in.
Actual Thickness Used in Calculation	0.472	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 44.1 \cdot (\exp(4616.805/(147.6 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.7006 + 3.0 + 0.0 = 3.7006 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 9.0 + 44.1) / 44.1) - 0.0$$

$$= 50518.516 - 0.0 = 50518.5156 \text{ KPa.}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 7.520 mm
 Min. required shell thickness, internal p + c : 72.005 mm

Nozzle neck minimum required thickness : 7.520 mm

Minimum nozzle neck thickness 11.989 mm , is greater than required thickness 7.520 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	4.7,	Allowable :	147.6 N./mm ²	Passed
Expansion :	0.0,	Allowable :	386.1 N./mm ²	Passed
Occasional :	4.7,	Allowable :	196.3 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 26.0 - 3.0$$

$$= 23.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 3.0$$

$$= 9.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 4606.0/73.0$$

$$= 63.096 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 4600.0/2 + 3.381$$



$$= 2303.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{2303.0 \cdot 73.0}, 2 \cdot 22.05)$$

$$= 44.100 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 73.0, 0.0] + \sqrt{22.05 \cdot 23.0}$$

$$= 22.520 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 200.0$$

$$= 200.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(73.0 + 0.0)$$

$$= 584.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[22.52, 200.0, 584.0] + 73.0$$

$$= 95.520 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot \text{frp}) / LR)$$

$$= 73.0 + ((0.0 \cdot 1.0) / 44.1)$$

$$= 73.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + \text{teff}) \cdot \text{teff}}))$$

$$= \min(12, (2 \cdot 22.05 + 23.0) / (\sqrt{(4606.0 + 73.0) \cdot 73.0}))$$

$$= 0.115$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 73.0 \cdot 44.1 \cdot \max((0.115/5)^{0.85}, 1)$$

$$= 32.193 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 23.0 \cdot 95.52$$

$$= 21.970 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 32.193 + 0.944(21.97 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 53.426 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 23.0 / \ln[1 + 23.0/22.05]$$

$$= 32.192 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= \text{teff} / \ln[1 + \text{teff}/\text{Reff}]$$

$$= 73.0 / \ln[1 + 73.0/2303.0]$$

$$= 2339.313 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:



$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 4616.805 \cdot 32.192 \cdot 95.52$$

$$= 1447.7 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (LR + t_n)$$

$$= 4616.805 \cdot 2339.313 (44.1 + 23.0)$$

$$= 73900.7 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 4616.805 \cdot 2339.313 \cdot 22.05$$

$$= 24284.8 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (73900.7 + 24284.8 + 1447.7) / 4616.805$$

$$= 2116.2 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 234.6 / (2 \cdot 2116.247 / 53.426 - 2339.313 / 73.0)$$

$$= 4972.6 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / R_{xs}$$

$$= 156.4 [73.0 / 2339.313]$$

$$= 4880.3 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(4972.624, 4880.295)$$

$$= 4880.295 \text{ KPa.}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$

$$= (1447.711 + 73900.742 + 24284.82) / 53.426$$

$$= 182.884 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 4616.805 \cdot 2339.313 / 73.0$$

$$= 148.0 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 182.884 - 147.956, 147.956)$$

$$= 217.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	217.81	N./mm ²
Maximum Allowable Working Pressure	Pmax	4880.30	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (22.05 + 23.0) / 22.05$$

$$= 2.043 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (22.05 + 23.0)$$

$$= 70.764 \text{ mm}$$



Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
 = 7.071, 0.000, 0.000, mm

Required Weld Throat Dimensions :
 tc, L41T = 7.071, must be $\geq \min(0.7 * 23.000, 6.000)$ Passed

Weld Load Value [fwelds]:
 = $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 = $\min(24285 \cdot 2.04, 1.5 \cdot 147.6(21.97 + 0.0), \pi/4 \cdot 4616.8 \cdot 22.05^2 \cdot 2.04^2)$
 = 750.443 Kgf

Weld Stress Value [τ]:
 = $fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 = $750.443 / (70.764(0.49 \cdot 7.071 + 0.6 \cdot 10.0 + 0.49 \cdot 0.0))$
 = 10.988 < or = to 156.400 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.070

The computed thickness ratio [sr]:
 = $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 = $(0.626 \cdot 1.0) / (12.0 - 3.0 - 0.0)$
 = 0.070

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 0.4413 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 276.4413 mm

Input Echo, WRC297 Item 1, Description: P :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	4600.000	mm	
Shell Thickness	76.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	156.40	N./mm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	38.100	mm	
Nozzle Thickness	26.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-266 2		
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	147.60	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	4616.80	KPa.
Radial Load	P	0.00	Kgf
Circumferential Shear	Vc	0.00	Kgf
Longitudinal Shear	Vl	0.00	Kgf
Circumferential Moment	Mc	0.00	Kg-m.
Longitudinal Moment	Ml	0.00	Kg-m.
Torsional Moment	Mt	0.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (1.186) is < 5
 check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (3.465) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 90.1 + 2 \cdot 1.65 \cdot \sqrt{(2339.5(76.0 - 3.0))}$
 $= 1453.857 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4679.000 mm
Nozzle Outside Diameter	(d) =	90.100 mm
Vessel Thickness used	(T) =	73.000 mm
Nozzle Thickness used	(t) =	23.000 mm
T / t	=	3.174
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.154$$

Nr/P	=	0.286
Mr/P	=	0.255
M0/P	=	0.276
N0/P	=	0.181
MrD/Mc	=	0.185
NrDL/Mc	=	0.184
M0d/Mc	=	0.232
N0DL/Mc	=	-0.135
MrD/Ml	=	0.186
NrDL/Ml	=	0.189
M0D/Ml	=	0.230
N0DL/Ml	=	-0.158

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	0	0	0	0
Normal Pressure Stress	143	148	143	148
Outplane Stress Summary	143	148	143	148

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	71	71	71	71
Inplane Stress Summary	71	71	71	71

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	143	148	143	148

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	0	0	0	0
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	71	71	71	71
Outplane Stress Summary	71	71	71	71

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	0	0	0	0
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	143	148	143	148
Inplane Stress Summary	143	148	143	148

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0



Shear Stress Summary 0 0 0 0

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	143	148	143	148

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	9	3	9

Hoop Stress Summary 3 9 3 9

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary 1 1 1 1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary 0 0 0 0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	3	9	3	9

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	9	3	9

Hoop Stress Summary 3 9 3 9



Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	1	1	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	3	9	3	9

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	143	148	143	148	143	148	143	148
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	71	71	71	71	71	71	71	71
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	0	0	0	0	0	0
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	143.0	148.0	143.0	148.0	143.0	148.0	143.0	148.0
Pm+Pl (SUS)	143.0	148.0	143.0	148.0	143.0	148.0	143.0	148.0
Pm+Pl+Q (Total)	143.0	148.0	143.0	148.0	143.0	148.0	143.0	148.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	148.00	156.40	Passed
Pm+Pl (SUS)	148.00	234.60	Passed
Pm+Pl+Q (TOTAL)	148.00	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:



Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	9	3	9	3	9	3	9
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	0	0	0	0	0	0
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	3.0	9.0	3.0	9.0	3.0	9.0	3.0	9.0
Pm+Pl (SUS)	3.0	9.0	3.0	9.0	3.0	9.0	3.0	9.0
Pm+Pl+Q (Total)	3.0	9.0	3.0	9.0	3.0	9.0	3.0	9.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	147.60	Passed
Pm+Pl (SUS)	9.00	221.40	Passed
Pm+Pl+Q (TOTAL)	9.00	442.80	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: MH-3

From: 170

Pressure for Reinforcement Calculations	P	4616.805	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4600.00	mm
Shell Finished (Minimum) Thickness	t	74.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		31650.01	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

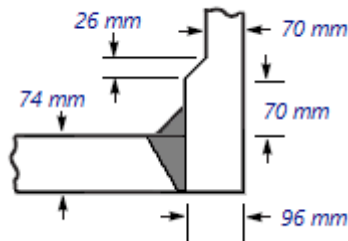
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		225.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	70.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	26.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		96.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-3

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	2.756	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 615.6 \cdot (\exp(4616.805/(147.6 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 9.7805 + 3.0 + 0.0 = 12.7805 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 67.0 + 615.6) / 615.6) - 0.0$
 $= 29066.996 - 0.0 = 29066.9961 \text{ KPa.}$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 565.56 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 805.17 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[147.6/156.4, 1]$
 $= 0.944$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 96.0 - 3.0$
 $= 93.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 70.0 - 3.0$
 $= 67.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 4606.0/71.0$
 $= 64.873 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 4600.0/2 + 3.381$
 $= 2303.000 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{2303.0 \cdot 71.0}, 2 \cdot 307.8)$
 $= 404.367 \text{ mm}$

Thickness Limit Candidate [LH1]:
 $= \min[1.5t_e, \sqrt{R_n \cdot t_n}]$
 $= \min[1.5 \cdot 71.0, 0.0] + \sqrt{307.8 \cdot 93.0}$

= 169.190 mm

Thickness Limit Candidate [LH2]:

= Lpr1
 = 400.0
 = 400.000 mm

Thickness Limit Candidate [LH3]:

= 8(t + te)
 = 8(71.0 + 0.0)
 = 568.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:

= min[LH1, LH2, LH3] + t for set-in nozzles
 = min[169.19, 400.0, 568.0] + 71.0
 = 240.190 mm

Effective Vessel Thickness [teff]:

= t + ((A5 • frp) / LR)
 = 71.0+((0.0•1.0)/404.367)
 = 71.000 mm

Parameter [λ]:

= min(12, (2•Rn + tn)/(√((Di + teff) • teff)))
 = min(12, (2•307.8 + 93.0)/(√((4606.0 + 71.0) • 71.0)))
 = 1.230

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

= t • LR • max((λ/5)^{0.85}, 1)
 = 71.0 • 404.367 • max((1.23/5)^{0.85}, 1)
 = 287.101 cm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

= A2A + A2C
 = 131.13 + 69.838
 = 200.968 cm²

Area Contributed by the Outside Fillet Weld [A41]:

= 0.5 • Leg⁴¹²
 = 0.5 • 10.0²
 = 0.500 cm²

The total area contributed by A1 through A43 [AT]:

= A1 + frn(A2 + A3) + A41 + A42 + A43
 = 287.101+0.944(200.968+0.0)+0.5+0.0+0.0
 = 477.261 cm²

Nozzle Radius for Force Calculation [Rxn]:

= tn / ln[1 + tn/Rn]
 = 93.0/ln[1 + 93.0/307.8]
 = 352.256 mm

Shell Radius for Force Calculation [Rxs]:

= teff / ln[1 + teff/Reff]
 = 71.0/ln[1 + 71.0/2303.0]
 = 2338.323 mm

Allowable Local Primary Membrane Stress [Sallow]:

= 1.5 • S • E
 = 1.5 • 156.4 • 1.0
 = 234.6 N./mm²

Force acting on the Nozzle [fN]:

= P • Rxn • LH
 = 4616.805 • 352.256 • 240.19
 = 39833.9 Kgf

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot Rxs(LR + tn) \\
 &= 4616.805 \cdot 2338.323(404.367 + 93.0) \\
 &= 547545.2 \text{ Kgf}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot Rxs \cdot Rnc \\
 &= 4616.805 \cdot 2338.323 \cdot 307.8 \\
 &= 338852.9 \text{ Kgf}
 \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= (fS + fY + fN)/P \\
 &= (547545 + 338852.9 + 39833.9)/4616.805 \\
 &= 19673.5 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\
 &= 234.6 / (2 \cdot 19673.506/477.261 - 2338.323/71.0) \\
 &= 4738.2 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S[t/Rxs] \\
 &= 156.4[71.0/2338.323] \\
 &= 4748.6 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(Pmax1, Pmax2) \\
 &= \min(4738.231, 4748.597) \\
 &= 4738.231 \text{ KPa.}
 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned}
 &= (fN + fS + fY) / AT \\
 &= (39833.859 + 547545 + 338852.938)/477.261 \\
 &= 190.324 \text{ N./mm}^2
 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned}
 &= P \cdot Rxs / teff \\
 &= 4616.805 \cdot 2338.323/71.0 \\
 &= 152.1 \text{ N./mm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\
 &= \max(2 \cdot 190.324 - 152.059, 152.059) \\
 &= 228.6 \text{ N./mm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	228.59	N./mm ²
Maximum Allowable Working Pressure	Pmax	4738.23	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned}
 &= (Rnc + tn) / Rnc \\
 &= (307.8 + 93.0)/307.8 \\
 &= 1.302 \text{ For set-in Nozzles}
 \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned}
 &= \pi/2 \cdot (Rn + tn) \\
 &= \pi/2 \cdot (307.8 + 93.0) \\
 &= 629.575 \text{ mm}
 \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 93.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned}
 &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\
 &= \min(338853 \cdot 1.3, 1.5 \cdot 147.6(200.968 + 0.0), \pi/4 \cdot 4616.8 \cdot 307.8^2 \cdot 1.3^2)
 \end{aligned}$$



= 59399.566 Kgf

Weld Stress Value [τ]:

= $f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot tw_1 + 0.49 \cdot L_{43T}))$
= $59399.566 / (629.575(0.49 \cdot 7.071 + 0.6 \cdot 37.0 + 0.49 \cdot 0.0))$
= 36.052 < or = to 156.400 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -12 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.130

The computed thickness ratio [sr]:

= $(tr \cdot E^*) / (t_{nom} - ci - c_{ext})$
= $(8.735 \cdot 1.0) / (70.0 - 3.0 - 0.0)$
= 0.130

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 35.1911 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 509.1910 mm

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Input, Nozzle Desc: MH-2

From: 200

Pressure for Reinforcement Calculations	P	4616.805	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4600.00	mm
Shell Finished (Minimum) Thickness	t	74.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		40150.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

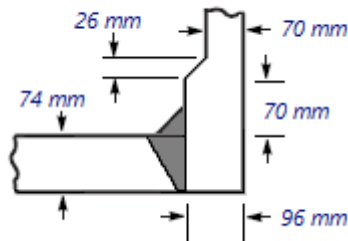
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		300.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	70.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	26.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		96.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-2

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	2.756	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 615.6 \cdot (\exp(4616.805/(147.6 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 9.7805 + 3.0 + 0.0 = 12.7805 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 67.0 + 615.6) / 615.6) - 0.0$
 $= 29066.996 - 0.0 = 29066.9961 \text{ KPa.}$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[147.6/156.4, 1]$
 $= 0.944$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 96.0 - 3.0$
 $= 93.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 70.0 - 3.0$
 $= 67.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 4606.0/71.0$
 $= 64.873 \text{ must be less than 400.}$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 4600.0/2 + 3.381$
 $= 2303.000 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{2303.0 \cdot 71.0}, 2 \cdot 307.8)$
 $= 404.367 \text{ mm}$

Thickness Limit Candidate [LH1]:
 $= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$
 $= \min[1.5 \cdot 71.0, 0.0] + \sqrt{307.8 \cdot 93.0}$
 $= 169.190 \text{ mm}$

Thickness Limit Candidate [LH2]:
 $= L_{pr1}$

$$= 400.0$$

$$= 400.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(71.0 + 0.0)$$

$$= 568.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[169.19, 400.0, 568.0] + 71.0$$

$$= 240.190 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 71.0 + ((0.0 \cdot 1.0) / 404.367)$$

$$= 71.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + t_n) / (\sqrt{ (D_i + teff) \cdot teff }))$$

$$= \min(12, (2 \cdot 307.8 + 93.0) / (\sqrt{ (4606.0 + 71.0) \cdot 71.0 }))$$

$$= 1.230$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 71.0 \cdot 404.367 \cdot \max((1.23/5)^{0.85}, 1)$$

$$= 287.101 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2C$$

$$= 131.13 + 69.838$$

$$= 200.968 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 287.101 + 0.944(200.968 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 477.261 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n / R_n]$$

$$= 93.0 / \ln[1 + 93.0 / 307.8]$$

$$= 352.256 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff / R_{eff}]$$

$$= 71.0 / \ln[1 + 71.0 / 2303.0]$$

$$= 2338.323 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 4616.805 \cdot 352.256 \cdot 240.19$$

$$= 39833.9 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + t_n)$$

$$= 4616.805 \cdot 2338.323(404.367 + 93.0)$$

$$= 547545.2 \text{ Kgf}$$



Discontinuity Force from Internal Pressure [FY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 4616.805 \cdot 2338.323 \cdot 307.8$$

$$= 338852.9 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (547545 + 338852.9 + 39833.9) / 4616.805$$

$$= 19673.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 234.6 / (2 \cdot 19673.506 / 477.261 - 2338.323 / 71.0)$$

$$= 4738.2 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 156.4 [71.0 / 2338.323]$$

$$= 4748.6 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(4738.231, 4748.597)$$

$$= 4738.231 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (39833.859 + 547545 + 338852.938) / 477.261$$

$$= 190.324 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 4616.805 \cdot 2338.323 / 71.0$$

$$= 152.1 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 190.324 - 152.059, 152.059)$$

$$= 228.6 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	228.59	N./mm ²
Maximum Allowable Working Pressure	Pmax	4738.23	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (307.8 + 93.0) / 307.8$$

$$= 1.302 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (307.8 + 93.0)$$

$$= 629.575 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 93.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(338853 \cdot 1.3, 1.5 \cdot 147.6(200.968 + 0.0), \pi / 4 \cdot 4616.8 \cdot 307.8^2 \cdot 1.3^2)$$

$$= 59399.566 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

= $59399.566 / (629.575(0.49 \cdot 7.071 + 0.6 \cdot 37.0 + 0.49 \cdot 0.0))$
= 36.052 < or = to 156.400 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -12 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.130

The computed thickness ratio [sr]:
= $(tr \cdot E^*) / (tnom - ci - cext)$
= $(8.735 \cdot 1.0) / (70.0 - 3.0 - 0.0)$
= 0.130

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

[Note: This Element/Detail was specified as being Impact Tested.](#)

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 35.1911 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 509.1910 mm

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Input, Nozzle Desc: D

From: 200

Pressure for Reinforcement Calculations	P	4616.805	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4600.00	mm
Shell Finished (Minimum) Thickness	t	74.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		39650.00	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

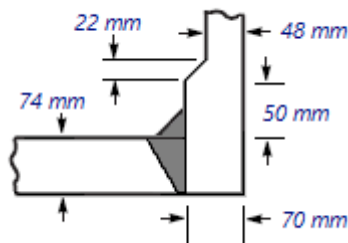
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	48.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	22.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		70.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: D

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	16.000	in.
Actual Thickness Used in Calculation	1.890	in.



Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 412.4 \cdot (\exp(4616.805/(147.6 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 6.5521 + 3.0 + 0.0 = 9.5521 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 45.0 + 412.4) / 412.4) - 0.0$
 $= 29134.98 - 0.0 = 29134.9805 \text{ KPa.}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
 Min. required shell thickness, internal p + c : 72.005 mm

Nozzle neck minimum required thickness : 11.340 mm

Minimum nozzle neck thickness 48.006 mm , is greater than required thickness 11.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 16.8,	Allowable	: 147.6 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 374.0 N./mm ²	Passed
Occasional	: 9.5,	Allowable	: 196.3 N./mm ²	Passed
Shear	: 4.3,	Allowable	: 103.3 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[147.6/156.4, 1]$
 $= 0.944$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 70.0 - 3.0$
 $= 67.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 48.0 - 3.0$
 $= 45.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 4606.0/71.0$



= 64.873 must be less than 400.

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 4600.0/2 + 3.381$$

$$= 2303.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{R_{eff} \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{2303.0 \cdot 71.0}, 2 \cdot 206.2)$$

$$= 404.367 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 71.0, 0.0] + \sqrt{206.2 \cdot 67.0}$$

$$= 117.539 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 400.0$$

$$= 400.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(71.0 + 0.0)$$

$$= 568.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[117.539, 400.0, 568.0] + 71.0$$

$$= 188.539 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 71.0 + ((0.0 \cdot 1.0) / 404.367)$$

$$= 71.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 206.2 + 67.0) / (\sqrt{(4606.0 + 71.0) \cdot 71.0}))$$

$$= 0.832$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 71.0 \cdot 404.367 \cdot \max((0.832/5)^{0.85}, 1)$$

$$= 287.101 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2C$$

$$= 81.07 + 32.813$$

$$= 113.883 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 20.0^2$$

$$= 2.000 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 287.101 + 0.944(113.883 + 0.0) + 2.0 + 0.0 + 0.0$$

$$= 396.576 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 67.0 / \ln[1 + 67.0/206.2]$$

$$= 238.131 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:



$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 71.0 / \ln[1 + 71.0/2303.0]$$

$$= 2338.323 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 4616.805 \cdot 238.131 \cdot 188.539$$

$$= 21137.6 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 4616.805 \cdot 2338.323 (404.367 + 67.0)$$

$$= 518922.1 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 4616.805 \cdot 2338.323 \cdot 206.2$$

$$= 227002.9 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (518922 + 227002.9 + 21137.6) / 4616.805$$

$$= 16292.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 234.6 / (2 \cdot 16292.69 / 396.576 - 2338.323 / 71.0)$$

$$= 4764.8 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / t / R_{xs}$$

$$= 156.4 [71.0 / 2338.323]$$

$$= 4748.6 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(4764.846, 4748.597)$$

$$= 4748.597 \text{ KPa.}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (21137.578 + 518922 + 227002.859) / 396.576$$

$$= 189.685 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 4616.805 \cdot 2338.323 / 71.0$$

$$= 152.1 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 189.685 - 152.059, 152.059)$$

$$= 227.3 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	227.31	N./mm ²
Maximum Allowable Working Pressure	Pmax	4748.60	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (206.2 + 67.0) / 206.2$$

$$= 1.325 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (206.2 + 67.0)$$

$$= 429.142 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 14.142, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 14.142, \text{ must be } \geq \min(0.7 \cdot 67.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(227003 \cdot 1.32, 1.5 \cdot 147.6(113.883 + 0.0), \pi/4 \cdot 4616.8 \cdot 206.2^2 \cdot 1.32^2)$$

$$= 27598.744 \text{ Kg}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$

$$= 27598.744 / (429.142(0.49 \cdot 14.142 + 0.6 \cdot 20.0 + 0.49 \cdot 0.0))$$

$$= 33.318 < \text{or } = \text{ to } 156.400 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.130

The computed thickness ratio [sr]:

$$= (t_r \cdot E) / (t_{nom} - c_i - c_{ext})$$

$$= (5.851 \cdot 1.0) / (48.0 - 3.0 - 0.0)$$

$$= 0.130$$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 16.2833 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 490.2832 mm

Input Echo, WRC297 Item 1, Description: D :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	4600.000	mm
Shell Thickness	74.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 156.40	N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	406.400	mm
Nozzle Thickness	70.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	



Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	147.60	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4616.80	KPa.
Radial Load	P	3669.81	Kgf
Circumferential Shear	Vc	2650.42	Kgf
Longitudinal Shear	Vl	2650.42	Kgf
Circumferential Moment	Mc	3262.23	Kg-m.
Longitudinal Moment	Ml	3262.23	Kg-m.
Torsional Moment	Mt	3751.55	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (7.806) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 546.4 + 2 \cdot 1.65 \cdot \sqrt{(2338.5(74.0 - 3.0))}$
 $= 1891.058 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	4677.000 mm
Nozzle Outside Diameter	(d) =	546.400 mm
Vessel Thickness used	(T) =	71.000 mm
Nozzle Thickness used	(t) =	67.000 mm
T / t	=	1.060
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.948$$

Nr/P	=	0.132
Mr/P	=	0.192
M0/P	=	0.067
N0/P	=	0.125
MrD/Mc	=	0.277
NrDL/Mc	=	0.149
M0d/Mc	=	0.111
N0DL/Mc	=	-0.049
MrD/Ml	=	0.202
NrDL/Ml	=	0.118
M0D/Ml	=	0.080
N0DL/Ml	=	0.111

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-2	2	-2	2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-1	-1	1	1
Outplane Bending (Ml)	-5	5	5	-5
Normal Pressure Stress	147	152	147	152
Outplane Stress Summary	136	158	150	149



Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-8	8	-8	8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-14	14	14	-14
Inplane Pressure Stress	73	73	73	73
Inplane Stress Summary	49	93	80	68

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	136	158	150	149

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	-2	2	-2	2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-7	7	7	-7
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	73	73	73	73
Outplane Stress Summary	62	84	77	67

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	-8	8	-8	8
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-19	19	19	-19
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	147	152	147	152
Inplane Stress Summary	117	176	159	141

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0



Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	1	1
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	117	176	159	141

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	20	14	20
Hoop Stress Summary	12	18	15	20

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-6	6	-6	6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-11	11	11	-11
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	-13	21	15	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	25	21	15	20

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	20	14	20

Hoop Stress Summary 14 19 13 18

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-6	6	-6	6
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-15	15	15	-15
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7

Axial Stress Summary -17 25 19 0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Shear stress)

	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary 0 0 1 1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside

Two * Max Shear Stress 32 26 20 18

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	147	152	147	152	147	152	147	152
Circ. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Circ. Q (SUS)	-7	7	3	-3	-27	27	11	-11
Long. Pm (SUS)	73	73	73	73	73	73	73	73
Long. Pl (SUS)	-1	-1	1	1	0	0	0	0
Long. Q (SUS)	-22	22	6	-6	-9	9	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	147.0	152.0	147.0	152.0	147.0	152.0	147.0	152.0
Pm+Pl (SUS)	146.0	151.0	148.0	153.0	146.0	151.0	148.0	153.0
Pm+Pl+Q (Total)	139.0	158.0	151.0	150.0	119.0	178.0	159.0	142.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	152.00	156.40	Passed
Pm+Pl (SUS)	153.00	234.60	Passed
Pm+Pl+Q (TOTAL)	178.01	469.20	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	14	20	14	20	14	20	14	20
Circ. Pl (SUS)	-1	-1	1	1	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-17	17	5	-5	-21	21	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	14.0	20.0	14.0	20.0	14.0	20.0	14.0	20.0
Pm+Pl (SUS)	13.0	19.0	15.0	21.0	14.0	20.0	14.0	20.0
Pm+Pl+Q (Total)	25.1	22.3	15.6	21.1	30.1	26.2	18.2	20.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	147.60	Passed
Pm+Pl (SUS)	21.00	221.40	Passed
Pm+Pl+Q (TOTAL)	30.07	442.80	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: MH-1

From: 210

Pressure for Reinforcement Calculations	P	4616.805	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	4600.00	mm
Shell Finished (Minimum) Thickness	t	72.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		42449.99	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

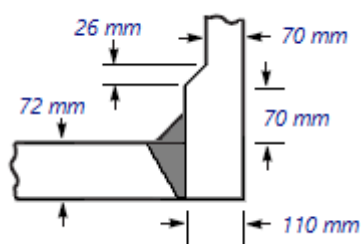
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		300.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	70.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	26.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: MH-1

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	2.756	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 615.6 \cdot (\exp(4616.805/(147.6 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 9.7805 + 3.0 + 0.0 = 12.7805 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 67.0 + 615.6) / 615.6) - 0.0$$

$$= 29066.996 - 0.0 = 29066.9961 \text{ KPa.}$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 602.91 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 813.43 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[147.6/156.4, 1]$$

$$= 0.944$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 110.0 - 3.0$$

$$= 107.000 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 70.0 - 3.0$$

$$= 67.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 4606.0/69.0$$

$$= 66.754 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 4600.0/2 + 3.381$$

$$= 2303.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{2303.0 \cdot 69.0}, 2 \cdot 307.8)$$

$$= 398.631 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t_e, \sqrt{R_n \cdot t_n}]$$

$$= \min[1.5 \cdot 69.0, 0.0] + \sqrt{307.8 \cdot 107.0}$$



= 181.479 mm

Thickness Limit Candidate [LH2]:

= Lpr1
 = 400.0
 = 400.000 mm

Thickness Limit Candidate [LH3]:

= 8(t + te)
 = 8(69.0 + 0.0)
 = 552.000 mm

Effective Nozzle Wall Length Outside the Vessel [LH]:

= min[LH1, LH2, LH3] + t for set-in nozzles
 = min[181.479, 400.0, 552.0] + 69.0
 = 250.479 mm

Effective Vessel Thickness [teff]:

= t + ((A5 • frp) / LR)
 = 69.0 + ((0.0 • 1.0) / 398.631)
 = 69.000 mm

Parameter [λ]:

= min(12, (2 • Rn + tn) / (√((Di + teff) • teff)))
 = min(12, (2 • 307.8 + 107.0) / (√((4606.0 + 69.0) • 69.0)))
 = 1.272

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

= t • LR • max((λ/5)^{0.85}, 1)
 = 69.0 • 398.631 • max((1.272/5)^{0.85}, 1)
 = 275.056 cm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

= A2A + A2C
 = 148.73 + 75.048
 = 223.778 cm²

Area Contributed by the Outside Fillet Weld [A41]:

= 0.5 • Leg⁴¹²
 = 0.5 • 10.0²
 = 0.500 cm²

The total area contributed by A1 through A43 [AT]:

= A1 + frn(A2 + A3) + A41 + A42 + A43
 = 275.056 + 0.944(223.778 + 0.0) + 0.5 + 0.0 + 0.0
 = 486.743 cm²

Nozzle Radius for Force Calculation [Rxn]:

= tn / ln[1 + tn/Rn]
 = 107.0 / ln[1 + 107.0/307.8]
 = 358.644 mm

Shell Radius for Force Calculation [Rxs]:

= teff / ln[1 + teff/Reff]
 = 69.0 / ln[1 + 69.0/2303.0]
 = 2337.334 mm

Allowable Local Primary Membrane Stress [Sallow]:

= 1.5 • S • E
 = 1.5 • 156.4 • 1.0
 = 234.6 N./mm²

Force acting on the Nozzle [fN]:

= P • Rxn • LH
 = 4616.805 • 358.644 • 250.479
 = 42293.4 Kg

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot Rxs(LR + tn) \\
 &= 4616.805 \cdot 2337.334(398.631 + 107.0) \\
 &= 556407.4 \text{ Kgf}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot Rxs \cdot Rnc \\
 &= 4616.805 \cdot 2337.334 \cdot 307.8 \\
 &= 338709.6 \text{ Kgf}
 \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= (fS + fY + fN)/P \\
 &= (556407 + 338709.6 + 42293.4)/4616.805 \\
 &= 19910.9 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\
 &= 234.6 / (2 \cdot 19910.939/486.743 - 2337.334/69.0) \\
 &= 4893.5 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S[t/Rxs] \\
 &= 156.4[69.0/2337.334] \\
 &= 4616.8 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(Pmax1, Pmax2) \\
 &= \min(4893.481, 4616.787) \\
 &= 4616.787 \text{ KPa.}
 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned}
 &= (fN + fS + fY) / AT \\
 &= (42293.379 + 556407 + 338709.562)/486.743 \\
 &= 188.868 \text{ N./mm}^2
 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned}
 &= P \cdot Rxs / teff \\
 &= 4616.805 \cdot 2337.334/69.0 \\
 &= 156.4 \text{ N./mm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\
 &= \max(2 \cdot 188.868 - 156.401, 156.401) \\
 &= 221.3 \text{ N./mm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	221.34	N./mm ²
Maximum Allowable Working Pressure	Pmax	4616.79	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned}
 &= (Rnc + tn) / Rnc \\
 &= (307.8 + 107.0)/307.8 \\
 &= 1.348 \text{ For set-in Nozzles}
 \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned}
 &= \pi/2 \cdot (Rn + tn) \\
 &= \pi/2 \cdot (307.8 + 107.0) \\
 &= 651.566 \text{ mm}
 \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 107.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned}
 &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\
 &= \min(338710 \cdot 1.35, 1.5 \cdot 147.6(223.778+0.0), \pi/4 \cdot 4616.8 \cdot 307.8^2 \cdot 1.35^2)
 \end{aligned}$$



= 63621.723 Kgf

Weld Stress Value [τ]:

= $\text{fwelds}/(\text{Lt}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T}))$
= $63621.723/(651.566(0.49 \cdot 7.071 + 0.6 \cdot 37.0 + 0.49 \cdot 0.0))$
= 37.311 < or = to 156.400 Weld Stress Passed

Nozzle/Structural Discontinuity Proximity Warning:

This nozzle's diameter limit intersects a major structural discontinuity. Reduce the diameter limit in the nozzle dialog so that there is no interference or move the nozzle away from the intersecting weld by 514.955 mm.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -12 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.130

The computed thickness ratio [sr]:

= $(\text{tr} \cdot E^*)/(\text{tnom} - \text{ci} - \text{cext})$
= $(8.735 \cdot 1.0)/(70.0 - 3.0 - 0.0)$
= 0.130

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 37.7132 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 509.7133 mm

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Input, Nozzle Desc: B

From: 220

Pressure for Reinforcement Calculations	P	4616.805	KPa.
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	156.40	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Hemispherical Head	D	4600.00	mm
Head Finished (Minimum) Thickness	t	40.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

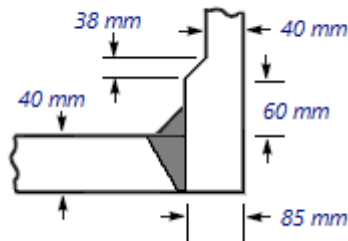
Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.60	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	40.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	38.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		85.0000	mm
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	37.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: B

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	1.575	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 615.6 \cdot (\exp(4616.805/(147.6 \cdot 1.0)) - 1) + 3.0 + 0.0$
 $= 9.7805 + 3.0 + 0.0 = 12.7805 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 147.6 \cdot 1.0 \cdot \ln((2 \cdot 37.0 + 615.6) / 615.6) - 0.0$
 $= 16753.738 - 0.0 = 16753.7383 \text{ KPa.}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	11.340 mm
Min. required shell thickness, internal p + c :	37.246 mm
Nozzle neck minimum required thickness :	11.340 mm

Minimum nozzle neck thickness 40.005 mm , is greater than required thickness 11.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	27.5,	Allowable :	147.6 N./mm ²	Passed
Expansion :	0.0,	Allowable :	363.3 N./mm ²	Passed
Occasional :	18.1,	Allowable :	196.3 N./mm ²	Passed
Shear :	5.3,	Allowable :	103.3 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[147.6/156.4, 1]$
 $= 0.944$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 85.0 - 3.0$
 $= 82.000 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 40.0 - 3.0$
 $= 37.000 \text{ mm}$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 4600.0/2 + 3.381$

$$= 2303.000 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{(2303.0 \cdot 37.0)}, 2 \cdot 307.8)$$

$$= 291.909 \text{ mm}$$

Intermediate value [Cn]:

$$= \min((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min((37.0 + 0.0)/82.0)^{0.35}, 1)$$

$$= \min(0.757, 1)$$

$$= 0.757$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{R_n \cdot t_n}, L_{pr1} + t)$$

$$= \min(37.0 + 0.0 + 0.757 \cdot \sqrt{(307.8 \cdot 82.0)}, 400.0 + 37.0)$$

$$= 157.248 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot \text{frp}) / LR)$$

$$= 37.0 + ((0.0 \cdot 1.0) / 291.909)$$

$$= 37.000 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 37.0 \cdot 291.909$$

$$= 108.006 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2C}$$

$$= 79.54 + 30.799$$

$$= 110.339 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + \text{frn}(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 108.006 + 0.944(110.339 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 212.637 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 82.0 / \ln[1 + 82.0/307.8]$$

$$= 347.188 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= \text{teff} / \ln[1 + \text{teff}/\text{Reff}]$$

$$= 37.0 / \ln[1 + 37.0/2303.0]$$

$$= 2321.458 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 156.4 \cdot 1.0$$

$$= 234.6 \text{ N/mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 4616.805 \cdot 347.188 \cdot 157.248$$

$$= 25703.2 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} \cdot (LR + t_n)/2$$

$$= 4616.805 \cdot 2321.458 \cdot (291.909 + 82.0) / 2$$

$$= 204331.3 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot Rxs \cdot Rnc) / 2$$

$$= (4616.805 \cdot 2321.458 \cdot 307.8) / 2$$

$$= 168204.4 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (204331.3 + 168204.4 + 25703.2) / 4616.805$$

$$= 8458.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / (2 \cdot teff))$$

$$= 234.6 / (2 \cdot 8458.74 / 212.637 - 2321.458 / (2 \cdot 37.0))$$

$$= 4868.0 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t / Rxs]$$

$$= 2 \cdot 156.4[37.0 / 2321.458]$$

$$= 4985.2 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(4868.005, 4985.199)$$

$$= 4868.005 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (25703.17 + 204331.328 + 168204.422) / 212.637$$

$$= 183.668 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 4616.805 \cdot 2321.458 / (2 \cdot 37.0)$$

$$= 144.8 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 183.668 - 144.842, 144.842)$$

$$= 222.5 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	234.60	N./mm ²
Local Primary Membrane Stress	PL	222.49	N./mm ²
Maximum Allowable Working Pressure	Pmax	4868.01	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (307.8 + 82.0) / 307.8$$

$$= 1.266 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (307.8 + 82.0)$$

$$= 612.296 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 82.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(168204 \cdot 1.27, 1.5 \cdot 147.6(110.339 + 0.0), \pi / 4 \cdot 4616.8 \cdot 307.8^2 \cdot 1.27^2)$$

$$= 56183.863 \text{ Kgf}$$

Weld Stress Value [τ]:
 $= \text{fwelds} / (L_T (0.49 \cdot L_{41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot L_{43T}))$
 $= 56183.863 / (612.296 (0.49 \cdot 7.071 + 0.6 \cdot 37.0 + 0.49 \cdot 0.0))$
 $= 35.062 < \text{or} = \text{to } 156.400 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : -12 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
 The computed stress ratio is : 0.236

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (8.735 \cdot 1.0) / (40.0 - 3.0 - 0.0)$
 $= 0.236$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 33.2720 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 473.2721 mm

Input Echo, WRC107/537 Item	1,	Description: B	:
Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	4600.000	mm
Vessel Thickness	Tv	40.000	mm
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	175.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	156.40	N./mm ²
Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	609.600	mm
Nozzle Thickness	Tn	85.000	mm
Nozzle Material		SA-266 2	
Nozzle UNS Number		K03506	
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	147.60	N./mm ²
Design Internal Pressure	Dp	4616.805	KPa.
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

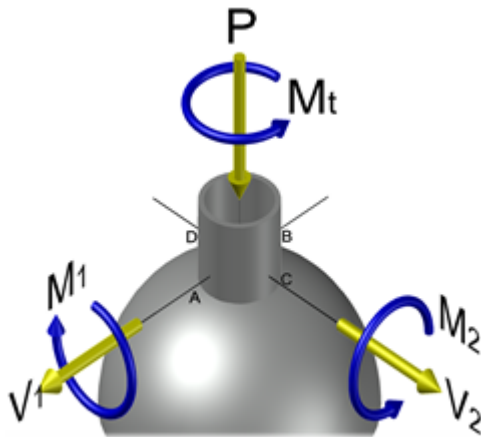
Radial Load (SUS)	P	5504.7	Kgf
Longitudinal Shear (SUS)	(V1) V1	3975.6	Kgf
Circumferential Shear (SUS)	(Vc) V2	3975.6	Kgf
Circumferential Moment (SUS)	(Mc) M1	7340.0	Kg-m.

Longitudinal Moment (SUS) (Ml) M2 7340.0 Kg-m.
 Torsional Moment (SUS) Mt 8441.0 Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 779.6 + 2 \cdot 1.65 \cdot \sqrt{(2321.5 (40.0 - 3.0))}$$

$$= 1746.762 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	5504.7	Kgf
Circumferential Shear	(VC) V2	3975.6	Kgf
Longitudinal Shear	(VL) V1	3975.6	Kgf
Circumferential Moment	(MC) M1	7340.0	Kg-m.
Longitudinal Moment	(ML) M2	7340.0	Kg-m.
Torsional Moment	MT	8441.0	Kg-m.

Dimensionless Param: U = 1.33 TAU = 5.00 (4.25) RHO = 0.45

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03015	(A,B,C,D)
$M(x) / P$	SP 1	0.02922	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.03151	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.04299	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.03151	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.04299	(A,B,C,D)
$N(y) * T / P$	SP 1	0.04549	(A,B,C,D)
$M(y) / P$	SP 1	0.00827	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.04325	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01368	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.04325	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01368	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		-1.2	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2	-1.2
Rad. Bend. P		-6.9	6.9	-6.9	6.9	-6.9	6.9	-6.9	6.9
Rad. Memb. MC		0.0	0.0	0.0	0.0	-5.7	-5.7	5.7	5.7
Rad. Bend. MC		0.0	0.0	0.0	0.0	-46.3	46.3	46.3	-46.3
Rad. Memb. ML		-5.7	-5.7	5.7	5.7	0.0	0.0	0.0	0.0
Rad. Bend. ML		-46.3	46.3	46.3	-46.3	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-60.0	46.4	43.8	-34.9	-60.0	46.4	43.8	-34.9
Tang. Memb. P		-1.8	-1.8	-1.8	-1.8	-1.8	-1.8	-1.8	-1.8
Tang. Bend. P		-2.0	2.0	-2.0	2.0	-2.0	2.0	-2.0	2.0
Tang. Memb. MC		0.0	0.0	0.0	0.0	-7.8	-7.8	7.8	7.8
Tang. Bend. MC		0.0	0.0	0.0	0.0	-14.7	14.7	14.7	-14.7
Tang. Memb. ML		-7.8	-7.8	7.8	7.8	0.0	0.0	0.0	0.0
Tang. Bend. ML		-14.7	14.7	14.7	-14.7	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-26.2	7.1	18.7	-6.8	-26.2	7.1	18.7	-6.8
Shear VC		0.9	0.9	-0.9	-0.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Shear MT		2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3
Tot. Shear		3.2	3.2	1.5	1.5	1.5	1.5	3.2	3.2
Str. Int.		60.3	46.6	43.9	35.0	60.1	46.4	44.2	35.3

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		144.8	144.8	144.8	144.8	144.8	144.8	144.8	144.8
Rad. Pl (SUS)		-6.8	-6.8	4.5	4.5	-6.8	-6.8	4.5	4.5
Rad. Q (SUS)		-53.2	53.2	39.4	-39.4	-53.2	53.2	39.4	-39.4
Long. Pm (SUS)		144.8	144.8	144.8	144.8	144.8	144.8	144.8	144.8
Long. Pl (SUS)		-9.6	-9.6	6.0	6.0	-9.6	-9.6	6.0	6.0
Long. Q (SUS)		-16.7	16.7	12.8	-12.8	-16.7	16.7	12.8	-12.8
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.9	0.9	-0.9	-0.9	-0.9	-0.9	0.9	0.9
Shear Q (SUS)		2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3
Pm (SUS)		144.8	144.8	144.8	144.8	144.8	144.8	144.8	144.8
Pm+Pl (SUS)		138.3	138.3	151.2	151.2	138.3	138.3	151.2	151.2
Pm+Pl+Q (Total)		118.9	191.5	188.8	138.1	118.7	191.2	189.1	138.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	144.84	156.40	Passed
Pm+Pl (SUS)	151.20	234.60	Passed
Pm+Pl+Q (TOTAL)	191.45	469.20	Passed



The $Pm+PI+Q$ allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The $Pm+PI+Q$ allowable was based on a temperature range cycling from ambient to design temperature.

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